

BEACON EDUCATIONAL SERIES

Integrated Science for Basic 7

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 7 (JHS 1) Students and Teachers

Beacon Educational Consult
2026 Edition

Integrated Science for Basic 7
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 7 (JHS 1) students and teachers in their study of Integrated Science, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY AND MARKING SCHEMES

Paper 1: Agricultural Tools	
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Paper 1: Agriculture

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A simple agricultural tool is one that is
 - A. made entirely of plastic
 - B. driven by a diesel engine
 - C. operated by hand and needs no engine
 - D. used only on very large farms
2. The tool used mainly for clearing weeds and cutting down bushes on a farm is the
 - A. rake
 - B. garden trowel
 - C. watering can
 - D. cutlass
3. The farm tool used for making ridges, mounds and weeding is the
 - A. wheelbarrow
 - B. hoe
 - C. secateurs
 - D. measuring tape
4. Seedlings in a nursery are watered gently with a
 - A. watering can
 - B. spraying machine for weedicide
 - C. shovel
 - D. bucket of ice
5. The tool used for gathering dry leaves and levelling the surface of a seedbed is the
 - A. knapsack sprayer
 - B. rake
 - C. axe
 - D. cutlass

- 6.** A tool used for digging holes for planting tree seedlings is the
- A. rake
 - B. pruning saw
 - C. watering can
 - D. pickaxe or mattock
- 7.** Secateurs are used on a farm mainly for
- A. pruning or cutting small branches of plants
 - B. carrying farm produce
 - C. watering young plants
 - D. ploughing the land
- 8.** Which of the following tools is used on an ANIMAL farm?
- A. A garden rake
 - B. A seed drill
 - C. A feeding trough brush
 - D. A pair of secateurs
- 9.** A wheelbarrow is used on a farm mainly for
- A. spraying insecticide
 - B. carrying manure, soil and harvested crops
 - C. cutting grass
 - D. measuring land
- 10.** Which of the following is used to protect the hands while working with farm tools?
- A. A watering can
 - B. Hand gloves
 - C. A cutlass sheath
 - D. A head pan
- 11.** The handling of an agricultural tool means the way the tool is
- A. shared among the farmers
 - B. painted at the factory
 - C. bought from the market
 - D. held and used during work
- 12.** The maintenance of agricultural tools means
- A. keeping the tools in a locked box for ever
 - B. buying new tools every season
 - C. selling old tools to other farmers
 - D. keeping the tools in good working condition

- 13.** Which of the following is NOT a reason for maintaining farm tools?
- A. To prevent accidents during work
 - B. To make the tools last longer
 - C. To make the tools heavier to use
 - D. To save the money spent on new tools
- 14.** Well-maintained farm tools help the farmer to
- A. harvest a poorer crop
 - B. spend more time on the farm
 - C. use more fuel
 - D. work faster and with less effort
- 15.** A blunt cutlass is dangerous to use because it
- A. becomes too light to hold
 - B. cuts the grass too quickly
 - C. may slip off the stem and injure the user
 - D. makes no sound when used
- 16.** The main reason for washing a hoe after use is to
- A. add weight to the blade
 - B. make the hoe look new for sale
 - C. soften the wooden handle
 - D. remove soil that would hold moisture and cause rusting
- 17.** Farm tools that are poorly maintained will MOST likely
- A. wear out quickly and need early replacement
 - B. become sharper on their own
 - C. work better with time
 - D. increase the farmer's harvest
- 18.** When walking with a cutlass on a farm, the blade should be
- A. held in front of the face
 - B. swung freely while walking
 - C. pointed downwards and away from the body
 - D. carried on the shoulder with the edge up
- 19.** A cutlass should be handed to another person by
- A. holding the blade and offering the handle first
 - B. offering the sharp edge first
 - C. dropping it at the person's feet
 - D. throwing it gently to the person

- 20.** Before using any farm tool, the user should FIRST
- A. check that the tool is in good condition and the handle is firm
 - B. paint the blade of the tool
 - C. sharpen the handle of the tool
 - D. wash the tool in the stream
- 21.** Farm tools should be used only for
- A. digging out stones with the blade of a cutlass
 - B. any work at all, to save time
 - C. games during break
 - D. the work for which they are made
- 22.** While weeding with a hoe, pupils should stand
- A. directly behind one another
 - B. very close together to finish quickly
 - C. a safe distance apart
 - D. in a single line facing each other
- 23.** The correct protective clothing for working with farm tools includes
- A. sandals and short sleeves
 - B. open slippers
 - C. a school uniform only
 - D. boots and hand gloves
- 24.** A tool with a loose handle should be
- A. repaired or the handle refitted before use
 - B. thrown into the bush
 - C. used quickly before it falls off
 - D. used only by the strongest pupil
- 25.** When a tool is not in use during work on the farm, it should be
- A. placed across the farm path
 - B. kept in a safe place with the sharp edge down or covered
 - C. hung on a growing plant
 - D. left lying in the grass with the edge up
- 26.** The maintenance practice that prevents the metal parts of farm tools from rusting is
- A. burying them in wet soil
 - B. leaving them in the sun for days
 - C. soaking them in water overnight
 - D. cleaning and oiling them after use

- 27.** A cutlass is kept in good cutting condition by
- A. rubbing it with wet sand only
 - B. sharpening it with a file or whetstone
 - C. leaving it in the rain to soften
 - D. heating it in a fire until it is red
- 28.** Wooden handles of farm tools are rubbed with oil or kept dry mainly to
- A. make them slippery to hold
 - B. make them heavier
 - C. change their colour
 - D. prevent cracking and attack by termites
- 29.** The BEST place to keep farm tools after work is
- A. in a pool of water to keep them cool
 - B. on the bare ground in the open
 - C. a dry, well-ventilated store, hung on racks
 - D. buried in the soil of the farm
- 30.** Rust on the blade of a hoe is BEST removed by
- A. painting over the rust
 - B. scraping or filing it off and then oiling the blade
 - C. heating the blade in a fire
 - D. soaking the blade in water
- 31.** Moving parts of a knapsack sprayer are kept working smoothly by
- A. tightening them until they cannot move
 - B. washing them with hot pepper water
 - C. oiling or greasing them regularly
 - D. leaving them in the sun
- 32.** A knapsack sprayer should be washed out thoroughly after use mainly because
- A. chemicals left inside will block the nozzle and corrode the parts
 - B. the chemicals will become more effective
 - C. washing makes the sprayer lighter to carry
 - D. the sprayer must be refilled with water for storage
- 33.** Which maintenance practice is MOST suitable for a wheelbarrow?
- A. Painting the wheel so that it cannot turn
 - B. Sharpening the tray with a file
 - C. Oiling the wheel and tightening the bolts
 - D. Soaking the whole barrow in water

- 34.** Tools are hung on racks or nails in the store rather than left on the floor so that they
- A. do not take up moisture from the floor and are not stepped on
 - B. will become lighter in weight
 - C. can be seen by visitors to the farm
 - D. can dry out and crack quickly
- 35.** During a visit to a farm in the community, the BEST way for pupils to record what they observe about tool maintenance is to
- A. argue with the farmer about his methods
 - B. take notes and write a report afterwards
 - C. remember everything without writing
 - D. borrow the farmer's tools and take them home
- 36.** Before pupils handle tools on a community farm, they should
- A. race to see who finishes first
 - B. be shown the correct way to use each tool by the teacher or farmer
 - C. remove their boots for comfort
 - D. share the tools among themselves and begin at once
- 37.** A set of written statements that tell workers how a tool should be used safely is called the
- A. harvest record of the farm
 - B. crop rotation plan
 - C. operational rules for handling the tool
 - D. price list of the tool
- 38.** A farmer in the community keeps his cutlass sharp, clean and hung in a dry store. The MOST likely result is that his cutlass will
- A. last for many years and work efficiently
 - B. rust within a few weeks
 - C. become blunt after one season
 - D. be rejected by other farmers
- 39.** A school farm club found that most tools in the store were rusty. The FIRST step they should take is to
- A. lock the store and stop farming
 - B. clean and oil the tools and repair the loose handles
 - C. throw all the tools away
 - D. buy a complete new set of tools
- 40.** The club then drew up a list of maintenance rules and put it on the store wall. This is useful mainly because it
- A. reminds every user of what to do after work with the tools
 - B. shows the price of each tool
 - C. decorates the wall of the store

D. stops pupils from entering the store

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 2: Animal Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An animal that is tamed and kept by people in the home or on a farm is called
 - A. a game animal
 - B. a wild animal
 - C. a domestic animal
 - D. a pest
2. Which of the following is NOT a domestic animal?
 - A. The pig
 - B. The antelope
 - C. The goat
 - D. The fowl
3. A group of animals of the same kind having the same features that they pass on to their young is called
 - A. a litter
 - B. a flock
 - C. a breed
 - D. a herd
4. Sanga, N'dama and White Fulani are breeds of
 - A. goats
 - B. cattle
 - C. sheep
 - D. fowls
5. West African Dwarf is a breed commonly used in Ghana for
 - A. goats
 - B. rabbits
 - C. turkeys
 - D. cattle only

- 6.** A farmer at Walewale says he keeps a 'flock'. The animals he is most likely talking about are
- A. pigs
 - B. rabbits
 - C. sheep or fowls
 - D. cattle
- 7.** An animal with a stomach divided into four chambers is described as
- A. a monogastric animal
 - B. a scavenger
 - C. a carnivore
 - D. a ruminant
- 8.** Which of the following animals is a ruminant?
- A. Fowl
 - B. Sheep
 - C. Pig
 - D. Rabbit
- 9.** The habit of bringing swallowed food back to the mouth to chew it again is called
- A. scavenging
 - B. grazing
 - C. browsing
 - D. chewing the cud
- 10.** Ruminants are able to feed mainly on grass because
- A. grass contains no fibre at all
 - B. they swallow stones to grind the grass
 - C. the micro-organisms in their stomach chambers break down the grass
 - D. they have very sharp cutting teeth in the upper jaw
- 11.** Which of the following groups contains ONLY ruminants?
- A. Cattle, sheep and goats
 - B. Sheep, fowls and goats
 - C. Cattle, pigs and goats
 - D. Goats, rabbits and sheep
- 12.** A goat and a pig are given the same heap of fresh grass. The goat does better on it because the goat
- A. eats much faster than the pig
 - B. can digest the fibre in the grass with the help of its stomach micro-organisms
 - C. has a longer mouth than the pig
 - D. does not need any other feed at all

- 13.** The four chambers of the stomach of a ruminant include the
- A. gizzard
 - B. crop
 - C. rumen
 - D. caecum only
- 14.** An animal with a simple stomach of one chamber is described as
- A. a cud chewer
 - B. a monogastric animal
 - C. a ruminant
 - D. a herbivore
- 15.** Which of the following is a monogastric animal?
- A. The sheep
 - B. The cow
 - C. The pig
 - D. The goat
- 16.** Birds kept on a farm for their meat and eggs are known as
- A. draught animals
 - B. livestock feed
 - C. ruminants
 - D. poultry
- 17.** Which of the following is NOT poultry?
- A. The rabbit
 - B. The turkey
 - C. The guinea fowl
 - D. The duck
- 18.** The part of the digestive system of a fowl that grinds its food is the
- A. gizzard
 - B. rumen
 - C. crop only
 - D. udder
- 19.** A pig is usually fed on kitchen waste, maize bran and other concentrates rather than on grass alone because the pig
- A. refuses to eat any plant material
 - B. has a simple stomach and cannot digest much fibre
 - C. is too large to walk to the grass
 - D. has no teeth for cutting grass

- 20.** A rabbit is a monogastric animal, yet it feeds mainly on leaves and grass. This shows that
- A. leaves contain no fibre
 - B. the rabbit is a ruminant
 - C. all monogastric animals eat only meat
 - D. monogastric animals can also be herbivores
- 21.** Ruminants, monogastric animals and poultry are alike in that all of them
- A. have four stomach chambers
 - B. lay eggs
 - C. are kept only for their milk
 - D. feed, breathe, grow and produce young ones
- 22.** Which of the following is a difference between a fowl and a goat?
- A. The goat lays eggs while the fowl does not
 - B. The fowl has feathers and two legs while the goat has hair and four legs
 - C. The fowl is a ruminant while the goat is not
 - D. The goat has a gizzard while the fowl has a rumen
- 23.** A pupil wrote: 'Cattle and pigs are both ruminants because both are kept on farms.' The statement is wrong because
- A. only wild animals are ruminants
 - B. pigs have four stomach chambers
 - C. being kept on a farm does not determine the type of stomach an animal has
 - D. cattle are not kept on farms
- 24.** A farmer who has very little money for feed but plenty of grass on his land should best keep
- A. broiler fowls only
 - B. rabbits only
 - C. pigs only
 - D. sheep and goats
- 25.** Which observation would BEST help a pupil to decide whether an animal is a ruminant?
- A. Watching whether the animal brings food back to its mouth to chew again while resting
 - B. Noting the colour of its hair
 - C. Measuring the length of its tail
 - D. Counting the number of its legs

- 26.** Poultry are kept in large numbers on a small piece of land more easily than cattle mainly because poultry
- A. produce no waste at all
 - B. are small, grow quickly and need less space
 - C. cannot move about
 - D. do not need any feed
- 27.** The use of an animal to meet the needs of the family at home is called
- A. industrial use
 - B. wild use
 - C. commercial use
 - D. domestic use
- 28.** Which of the following is a domestic use of a fowl?
- A. Providing eggs and meat for the family
 - B. Exporting frozen chicken
 - C. Supplying a hotel with eggs every week
 - D. Selling day-old chicks in the market
- 29.** A farmer uses his bull to pull a plough on his farm. This use of the animal is called
- A. draught (work) use
 - B. commercial sale
 - C. a breed
 - D. a by-product
- 30.** Cattle dung spread on a food crop farm is useful because it
- A. kills all insects on the farm
 - B. prevents rain from falling
 - C. makes the soil hard and dry
 - D. adds nutrients and humus to the soil
- 31.** A family keeps goats and sells two of them to pay school fees. The goats have been used
- A. as draught animals
 - B. commercially
 - C. domestically only
 - D. as poultry
- 32.** Which of the following best shows the domestic use of a cow in a home?
- A. Renting the cow to another farmer
 - B. Supplying milk to a factory
 - C. Taking milk from the cow for the family to drink
 - D. Selling the cow at the market

- 33.** Animals are also kept for social and cultural purposes in Ghana, such as
- A. providing sheep and fowls for naming ceremonies and festivals
 - B. guarding the national borders
 - C. producing electricity for the home
 - D. pulling vehicles in the city
- 34.** Keeping animals mainly in order to sell them or their products for money is called
- A. cultural use
 - B. domestic use
 - C. subsistence use
 - D. commercial use
- 35.** A useful material obtained from an animal apart from its meat is called
- A. a by-product
 - B. a carcass only
 - C. a ration
 - D. a breed
- 36.** Which of the following is an animal by-product?
- A. Maize bran fed to the animal
 - B. Water given to the animal
 - C. Grass grown on the farm
 - D. Hide used for making leather
- 37.** The skin of a goat is used in many Ghanaian communities to
- A. feed other animals
 - B. replace roofing sheets
 - C. make the heads of drums and leather goods
 - D. build the walls of houses
- 38.** Animal droppings from a poultry house are best described as
- A. a by-product that can be used as manure
 - B. a type of animal feed for cattle
 - C. a disease of poultry
 - D. a waste of no use at all
- 39.** A farmer near Kumasi keeps 500 layers and sells the eggs to shops in the town every week. The main purpose of his farm is
- A. cultural
 - B. experimental
 - C. commercial
 - D. domestic

- 40.** A young farmer has a small plot behind her house and wants a quick return on the little money she has. The BEST advice is to
- A. start with a herd of cattle, because cattle are the largest animals
 - B. start with a small flock of fowls, because they grow fast and produce eggs and meat within a short time
 - C. keep one pig and wait for it to produce manure only
 - D. keep no animals until she can afford a large farm

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 3: Climate Change and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A sustainable energy source is one that
 - A. is found only outside Ghana
 - B. is always the cheapest to buy
 - C. can be used again and again without being used up
 - D. burns with the brightest flame
2. Which of the following is a renewable source of energy?
 - A. petrol
 - B. sunlight
 - C. coal
 - D. diesel
3. The energy of moving air is called
 - A. tidal energy
 - B. wind energy
 - C. chemical energy
 - D. nuclear energy
4. Energy obtained from the heat and light of the Sun is
 - A. geothermal energy
 - B. sound energy
 - C. solar energy
 - D. fossil energy
5. Electricity produced at the Akosombo Dam is obtained from
 - A. burning petrol
 - B. burning charcoal
 - C. the wind
 - D. falling water
6. Which of the following is NOT a fossil fuel?
 - A. coal
 - B. natural gas
 - C. firewood
 - D. crude oil

- 7.** Fossil fuels are described as non-renewable because they
- A. cannot burn in air
 - B. take millions of years to form and are being used up quickly
 - C. are found in only one country
 - D. are made in factories every day
- 8.** The gas produced when organic waste rots in a closed tank without air is
- A. biogas
 - B. oxygen
 - C. nitrogen
 - D. chlorine
- 9.** Planting trees on bare land to protect the environment is called
- A. afforestation
 - B. urbanisation
 - C. deforestation
 - D. desertification
- 10.** A pupil who switches off the lights and the fan before leaving the classroom is practising
- A. energy wastage
 - B. energy production
 - C. energy conservation
 - D. energy transformation
- 11.** Replacing filament bulbs with LED bulbs in a school helps the environment MAINLY because LED bulbs
- A. are made of glass only
 - B. are brighter than the Sun
 - C. never need to be switched off
 - D. use less electricity for the same amount of light
- 12.** Which of these practices adds the LEAST pollution to the air?
- A. drying clothes in the Sun
 - B. running a petrol generator all day
 - C. burning rubbish in the yard
 - D. burning old tyres
- 13.** The use of an improved cooking stove that needs less firewood helps the environment because it
- A. costs more money to buy
 - B. increases the heat lost to the air
 - C. makes the food taste better
 - D. reduces the number of trees cut down and the smoke produced

- 14.** Walking or cycling to school instead of being driven helps the environment because it
- A. reduces the exhaust gases released into the air
 - B. increases the number of vehicles on the road
 - C. makes the roads wider
 - D. uses more fuel
- 15.** Which of the following BEST describes a green economy?
- A. an economy based only on farming
 - B. an economy that creates wealth while protecting the environment
 - C. an economy that uses no energy at all
 - D. an economy in which only green goods are sold
- 16.** Sorting household waste so that bottles and tins are collected separately supports
- A. recycling
 - B. open burning
 - C. littering
 - D. land filling of everything
- 17.** The greenhouse effect is the
- A. reflection of all sunlight back into space
 - B. growing of green plants inside a glass house
 - C. cooling of the Earth by rain clouds
 - D. trapping of heat near the Earth's surface by gases in the atmosphere
- 18.** Which of the following is a greenhouse gas?
- A. helium
 - B. nitrogen
 - C. oxygen
 - D. carbon dioxide
- 19.** Besides carbon dioxide, another important greenhouse gas is
- A. hydrogen
 - B. methane
 - C. neon
 - D. argon
- 20.** Carbon dioxide is added to the atmosphere MAINLY by
- A. the burning of fuels such as petrol, coal and firewood
 - B. the freezing of water
 - C. the evaporation of sea water
 - D. the rusting of iron

- 21.** Green plants help to reduce the amount of carbon dioxide in the air because they
- A. take in carbon dioxide during photosynthesis
 - B. give out carbon dioxide at night only
 - C. turn carbon dioxide into soil
 - D. store carbon dioxide in their roots as salt
- 22.** A steady rise in the average temperature of the Earth is called
- A. condensation
 - B. global dimming
 - C. evaporation
 - D. global warming
- 23.** Which of the following is a likely effect of global warming?
- A. a permanent end to the rainy season everywhere
 - B. melting of ice at the poles and a rise in sea level
 - C. an increase in the number of planets
 - D. a fall in the amount of carbon dioxide in the air
- 24.** Cutting down large areas of forest increases the greenhouse effect because it
- A. cools the soil too quickly
 - B. adds oxygen to the air
 - C. stops the wind from blowing
 - D. removes the trees that would have taken in carbon dioxide
- 25.** The layer of gases that surrounds the Earth and holds in some of its heat is the
- A. biosphere
 - B. atmosphere
 - C. lithosphere
 - D. hydrosphere
- 26.** Climate change is BEST described as
- A. the change of weather from morning to afternoon
 - B. a long-term change in the usual pattern of weather of a place
 - C. the movement of clouds across the sky
 - D. the change of seasons within a single week
- 27.** Which of the following is an effect of climate change on farming in Ghana?
- A. crops no longer need water
 - B. all soils become more fertile
 - C. rains become unreliable, so planting times are harder to predict
 - D. weeds stop growing on farms

- 28.** Flooding in coastal towns is made worse by climate change MAINLY because
- A. rivers flow backwards during the night
 - B. the Moon moves nearer to the Earth
 - C. the level of the sea rises and storms become stronger
 - D. the soil becomes lighter in colour
- 29.** A community that wants to reduce the greenhouse gases it produces should
- A. cut down trees to make more charcoal
 - B. use solar lamps and plant trees instead of burning more fuel
 - C. burn its rubbish in open heaps every evening
 - D. buy more petrol generators for every house
- 30.** The BEST reason for using public transport, or sharing a vehicle, is that it
- A. removes the need for roads
 - B. makes journeys take longer
 - C. carries many people while releasing less exhaust gas per person
 - D. allows drivers to use more fuel
- 31.** Burning farm residues in the open field after harvest is discouraged because it
- A. increases the number of earthworms
 - B. adds water to the soil
 - C. releases carbon dioxide and smoke and destroys soil organisms
 - D. makes the field cooler
- 32.** Which pair of actions would MOST reduce a school's contribution to climate change?
- A. switching off unused appliances and planting trees in the compound
 - B. clearing the school garden and paving all the ground
 - C. burning waste paper daily and using more generators
 - D. leaving taps running and buying more bulbs
- 33.** A country is said to use clean energy when its electricity comes mainly from
- A. firewood and charcoal
 - B. old tyres and plastics
 - C. coal and crude oil
 - D. the Sun, wind and water
- 34.** A simple solar dryer used to dry pepper and fish works by
- A. burning charcoal under the trays
 - B. trapping the Sun's heat so that the warm air removes moisture
 - C. adding salt to the food in a tank
 - D. cooling the food with ice

- 35.** In a biogas project, the material fed into the digester is
- A. animal dung and other organic waste
 - B. plastic bags and metal
 - C. sand and gravel
 - D. broken bottles and tins
- 36.** Apart from gas for cooking, a biogas plant also provides
- A. slurry that can be used as manure on the farm
 - B. pure drinking water
 - C. cement for building
 - D. petrol for vehicles
- 37.** A school wishing to light its classrooms during a power cut, without polluting the air, should install
- A. a petrol generator
 - B. a diesel generator
 - C. a charcoal stove
 - D. solar panels with a storage battery
- 38.** Before building a small energy project, the FIRST step a class should take is to
- A. paint the school wall
 - B. call the whole town for a party
 - C. plan and find out which source of energy is available in the area
 - D. buy all the materials at once
- 39.** Which of the following would BEST show that a class project on sustainable energy has succeeded?
- A. the device uses more fuel than before
 - B. the device is the most expensive in the district
 - C. the device works and reduces the fuel or electricity the school uses
 - D. the device is kept in a locked store
- 40.** Green jobs are jobs that
- A. require workers to wear green uniforms
 - B. are done only on farms
 - C. are done only during the rainy season
 - D. help to protect the environment while providing a living

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 4: Conversion and Conservation

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The law of conservation of energy states that energy can be
 - A. created whenever it is needed
 - B. destroyed after it has been used
 - C. stored only in dry cells
 - D. changed from one form to another but cannot be created or destroyed
2. The unit in which energy is measured is the
 - A. joule
 - B. newton
 - C. ampere
 - D. volt
3. A system into which no energy enters and from which no energy escapes is described as a
 - A. moving system
 - B. closed system
 - C. heated system
 - D. wasted system
4. In a closed system, the total amount of energy at the end of a change is
 - A. always less than at the beginning
 - B. the same as the total amount at the beginning
 - C. equal to zero
 - D. always more than at the beginning
5. A dry cell supplies 60 J of energy to a torch bulb. The bulb gives out 18 J as light. The energy given out as heat is
 - A. 60 J
 - B. 78 J
 - C. 18 J
 - D. 42 J

- 6.** Which of the following statements about energy changes is TRUE?
- A. Energy is only changed from one useful form into other forms
 - B. Energy disappears when a machine stops working
 - C. New energy is made inside a dry cell
 - D. Energy can be weighed on a balance
- 7.** A pupil said that the electricity used by a fan is 'lost' when the fan is switched off. The correct statement is that the energy has been
- A. changed into kinetic energy, sound and heat
 - B. destroyed by the motor
 - C. stored inside the blades
 - D. returned unchanged to the socket
- 8.** A machine is said to waste energy when some of the energy supplied to it is
- A. destroyed completely inside the machine
 - B. changed into forms that are not useful for the job
 - C. weighed and thrown away
 - D. returned to the source of the energy
- 9.** The energy change that takes place in a dry-cell torch when it is switched on is
- A. electrical to chemical to sound
 - B. light to electrical to chemical
 - C. chemical to electrical to light and heat
 - D. heat to chemical to light
- 10.** A solar panel on the roof of a clinic changes
- A. solar energy into electrical energy
 - B. sound energy into electrical energy
 - C. electrical energy into solar energy
 - D. heat energy into chemical energy
- 11.** An electric pressing iron changes electrical energy mainly into
- A. heat energy
 - B. chemical energy
 - C. sound energy
 - D. light energy
- 12.** The main energy change in a loudspeaker of a radio set is
- A. chemical energy to light energy
 - B. sound energy to electrical energy
 - C. heat energy to kinetic energy
 - D. electrical energy to sound energy

- 13.** An electric fan changes electrical energy mainly into
- A. kinetic energy
 - B. chemical energy
 - C. light energy
 - D. nuclear energy
- 14.** A microphone used at a school gathering changes
- A. heat energy into sound energy
 - B. sound energy into electrical energy
 - C. electrical energy into sound energy
 - D. light energy into sound energy
- 15.** When a mobile phone is being charged, the energy change is
- A. sound energy to heat energy
 - B. light energy to electrical energy
 - C. chemical energy to electrical energy
 - D. electrical energy to chemical energy
- 16.** Green plants carry out an energy conversion in which
- A. chemical energy is changed into light energy
 - B. heat energy is changed into sound energy
 - C. light energy is changed into chemical energy in food
 - D. kinetic energy is changed into light energy
- 17.** Which device changes kinetic energy into electrical energy?
- A. A generator
 - B. An electric motor
 - C. A pressing iron
 - D. A dry cell
- 18.** In a hydroelectric station such as Akosombo, the energy of the stored water is changed in the order
- A. chemical to heat to electrical
 - B. kinetic to chemical to electrical
 - C. potential to kinetic to electrical
 - D. electrical to kinetic to potential
- 19.** A boy winds up a toy car and releases it. The energy change is
- A. heat energy to potential energy
 - B. sound energy to kinetic energy
 - C. kinetic energy to chemical energy
 - D. elastic potential energy to kinetic energy

- 20.** A candle burning in a dark room changes chemical energy into
- A. electrical and nuclear energy
 - B. potential and elastic energy
 - C. light and heat energy
 - D. sound and kinetic energy
- 21.** A motor is supplied with 100 J of electrical energy. It gives 70 J of kinetic energy. The energy that appears as heat and sound is
- A. 100 J
 - B. 30 J
 - C. 70 J
 - D. 170 J
- 22.** A lamp supplied with 50 J of electrical energy gives out 10 J of light. The fraction of the energy that is useful is
- A. the whole
 - B. four-fifths
 - C. one-half
 - D. one-fifth
- 23.** In every energy conversion, some of the energy is usually changed into
- A. solar energy
 - B. nuclear energy
 - C. chemical energy
 - D. heat energy
- 24.** A pupil lists the energy chain of a petrol lorry as: chemical to to kinetic. The missing form is
- A. light energy
 - B. nuclear energy
 - C. heat energy
 - D. sound energy
- 25.** Which of the following energy chains is CORRECT for an electric kettle boiling water?
- A. kinetic energy to light energy
 - B. heat energy to electrical energy
 - C. electrical energy to heat energy
 - D. chemical energy to sound energy
- 26.** A filament bulb gives out far more heat than light. This means that most of the energy supplied to it is
- A. destroyed
 - B. stored
 - C. returned to the mains

D. wasted

- 27.** An LED lamp is preferred to a filament bulb for lighting a classroom because the LED lamp
- A. gives out more heat
 - B. uses a larger current
 - C. changes a greater part of the electrical energy into light
 - D. needs no switch to control it
- 28.** Oiling the moving parts of a machine saves energy because it reduces the energy wasted as
- A. light given out by the metal
 - B. heat and sound caused by friction
 - C. electrical energy in the wires
 - D. chemical energy in the oil
- 29.** Of the energy supplied to it, the useful energy given out by a machine is always
- A. equal to the energy supplied
 - B. less than the energy supplied
 - C. equal to zero
 - D. more than the energy supplied
- 30.** Heat lost through the walls of an electric oven is an example of
- A. destroyed energy
 - B. created energy
 - C. wasted energy
 - D. stored energy
- 31.** A pupil claimed to have built a machine that works for ever without any supply of energy. The claim must be false because
- A. machines are always made of metal
 - B. no machine can give out heat
 - C. energy cannot be created; a machine must be supplied with energy
 - D. every machine must have an electric motor
- 32.** Energy conservation means
- A. using energy carefully so that less of it is wasted
 - B. making new energy in a laboratory
 - C. destroying energy that is not needed
 - D. using as much energy as possible

- 33.** Which of the following is the BEST way of conserving electrical energy at home?
- A. Using filament bulbs in every room
 - B. Switching off lights and fans in rooms that are not in use
 - C. Leaving the television on all night
 - D. Opening the fridge door often
- 34.** Drying clothes in the open air instead of using a drying machine conserves energy because it
- A. increases the cost of electricity
 - B. uses energy from the Sun, which is free and renewable
 - C. destroys the electrical energy saved
 - D. makes the clothes heavier
- 35.** Ironing all the family's clothes at one time rather than one garment at a time conserves energy because
- A. the iron gives out no heat
 - B. the clothes need less water
 - C. the iron is heated once instead of many times
 - D. the iron works at a lower voltage
- 36.** A sign of energy being wasted in a school is
- A. an LED lamp used in the library
 - B. a kettle switched off after the water has boiled
 - C. a classroom with the windows opened for air
 - D. a classroom with the lights and fans left on at break time
- 37.** Which of the following sources of energy will NOT run out if it is used wisely?
- A. Sunlight
 - B. Crude oil
 - C. Coal
 - D. Natural gas
- 38.** A reason for conserving energy in Ghana is that it
- A. increases the number of power cuts
 - B. raises the cost of electricity for everybody
 - C. makes dry cells last a shorter time
 - D. reduces the fuel burnt and so protects the environment
- 39.** Putting a lid on a cooking pot conserves energy because it
- A. changes the heat into light
 - B. keeps the heat in the pot so that the food cooks faster
 - C. adds more heat to the food
 - D. makes the fire burn more fuel

- 40.** A school wishes to teach the community about saving energy. The MOST suitable activity is to
- A. make posters and leaflets on energy saving and display them in the town
 - B. ask every home to stop using electricity
 - C. switch off the school water pump for a term
 - D. sell dry cells to the community

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram, an energy-flow diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 5: Crop Production

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Substances that plants take in from the soil for healthy growth are called
 - A. plant hormones
 - B. plant nutrients
 - C. plant cells
 - D. plant fibres
2. The part of the plant through which most nutrients are taken in from the soil is the
 - A. root
 - B. flower
 - C. fruit
 - D. leaf
3. Plant nutrients are taken up by the roots
 - A. as dry solid grains
 - B. only when the soil is completely dry
 - C. as a gas from the air
 - D. dissolved in soil water
4. Any substance added to the soil to supply nutrients to crops is known as
 - A. a fertilizer
 - B. a fungicide
 - C. a herbicide
 - D. a pesticide
5. Maize plants on a school farm are short, with pale yellow leaves, while those on the next plot are tall and dark green. The most likely cause is that the soil of the first plot
 - A. lacks enough plant nutrients
 - B. is in a cooler part of the school
 - C. has too many stones
 - D. contains too much air

- 6.** Which of the following is the chief reason for adding plant nutrients to a school garden year after year?
- A. Plants cannot take in water without fertilizer
 - B. The crops harvested each year remove nutrients from the soil
 - C. Fertilizers make the soil heavier
 - D. Nutrients make the soil change colour
- 7.** A source of plant nutrients obtained from the remains of plants and animals is described as
- A. mineral
 - B. artificial
 - C. organic
 - D. inorganic
- 8.** Which of the following is an organic source of plant nutrients?
- A. Compost
 - B. Urea
 - C. Ammonium sulphate
 - D. Muriate of potash
- 9.** Rotted plant remains, kitchen waste and animal dung heaped together and allowed to decay form
- A. gravel
 - B. cement
 - C. charcoal
 - D. compost
- 10.** Dung and urine of farm animals mixed with their bedding is known as
- A. wood ash
 - B. farmyard manure
 - C. green manure
 - D. liquid fertilizer
- 11.** A farmer ploughs a young leguminous crop into the soil while it is still fresh and green. This practice provides
- A. green manure
 - B. inorganic fertilizer
 - C. poultry litter only
 - D. farmyard manure
- 12.** Which of the following organic materials is likely to be collected from a poultry house?
- A. Granules of NPK fertilizer
 - B. Bags of lime
 - C. Droppings mixed with wood shavings

- D. Crystals of ammonium sulphate
- 13.** Organic sources of plant nutrients improve the soil in a way that inorganic fertilizers do not, because they
- A. supply nutrients instantly to the crop
 - B. make the soil completely free of germs
 - C. add humus, which improves the structure of the soil and its ability to hold water
 - D. stop all weeds from growing
- 14.** A source of plant nutrients that is manufactured in a factory from mineral substances is described as
- A. natural
 - B. decayed
 - C. organic
 - D. inorganic
- 15.** Which of the following is an inorganic fertilizer?
- A. Compost
 - B. NPK 15-15-15
 - C. Green manure
 - D. Poultry manure
- 16.** The letters N, P and K on a bag of fertilizer stand for
- A. nitrogen, phosphorus and potassium
 - B. nickel, phosphorus and potassium
 - C. nitrogen, phosphorus and carbon
 - D. nitrogen, potassium and calcium
- 17.** Which of the following is supplied mainly by sulphate of ammonia and urea?
- A. Calcium
 - B. Phosphorus
 - C. Nitrogen
 - D. Potassium
- 18.** Inorganic fertilizers act faster on a crop than organic manures because they
- A. are always applied in larger quantities
 - B. are heavier than manure
 - C. contain living organisms
 - D. dissolve quickly in soil water and are taken up at once
- 19.** A danger of using too much inorganic fertilizer on a bed of vegetables is that it may
- A. scorch (burn) the plants and harm the soil
 - B. prevent rain from falling on the farm
 - C. turn the soil into stone

D. make the plants grow too slowly to harvest

- 20.** Which of the following statements about inorganic fertilizers is correct?
- A. They are always cheaper than organic manures
 - B. They supply nutrients in a concentrated form, so only small quantities are needed
 - C. They are made by heaping plant remains to decay
 - D. They add humus to the soil
- 21.** Which of the following pairs contains ONLY organic sources of plant nutrients?
- A. Compost and farmyard manure
 - B. Urea and poultry manure
 - C. Superphosphate and urea
 - D. NPK and green manure
- 22.** Compared with inorganic fertilizers, organic manures are
- A. quicker in their action on the crop
 - B. light, so very small quantities are enough
 - C. bulky, so large quantities must be applied
 - D. always sold in sealed sacks
- 23.** A farmer needs 2 kg of NPK fertilizer to supply the nutrients for one bed of vegetables. To supply the same nutrients with poultry manure he would need
- A. exactly 2 kg as well
 - B. less than 1 kg
 - C. no manure at all, since manure has no nutrients
 - D. a much larger quantity, measured in many kilograms
- 24.** One advantage of organic manure over inorganic fertilizer is that organic manure
- A. is sold in small light bags
 - B. acts within one or two days
 - C. contains only one nutrient
 - D. releases its nutrients slowly over a long period
- 25.** A school has little money but keeps goats and sweeps up plenty of leaves and kitchen waste. The best source of nutrients for its garden is
- A. no nutrients at all until money is available
 - B. bags of imported NPK fertilizer
 - C. compost and goat manure made on the school compound
 - D. liquid fertilizer bought from the city

- 26.** A farmer who uses only inorganic fertilizer year after year without adding any organic matter will most likely find that
- A. the soil becomes richer in humus each year
 - B. the soil loses its structure and holds less water
 - C. weeds disappear from the farm completely
 - D. the crop no longer needs water
- 27.** Well-made compost is usually
- A. clear, colourless and watery
 - B. yellow, dry and powdery
 - C. white, hard and shiny
 - D. dark brown to black, crumbly and earthy-smelling
- 28.** NPK fertilizer is commonly supplied as
- A. a dark crumbly heap
 - B. a wet, smelly paste
 - C. small hard granules in a sealed sack
 - D. a block of pressed leaves
- 29.** Which of the following nutrient sources is a liquid?
- A. Farmyard manure
 - B. Liquid fertilizer mixed with water for spraying
 - C. Dry poultry litter
 - D. Muriate of potash crystals
- 30.** Fresh poultry droppings differ from dried poultry manure in that the fresh droppings are
- A. dry, light and odourless
 - B. hard, white crystals
 - C. clear and colourless
 - D. wet, sticky and strong-smelling
- 31.** Inorganic fertilizer is kept in a sealed bag in a dry store mainly because it
- A. turns into compost when it is left open
 - B. is food for farm animals
 - C. absorbs moisture from the air and cakes into lumps
 - D. gives off a pleasant smell
- 32.** Why is a bulky organic manure usually worked into the soil rather than left on the surface?
- A. So that it can be seen by the farmer
 - B. So that the soil can become harder
 - C. So that it decays in the soil and releases its nutrients where the roots are
 - D. So that birds can feed on it easily

- 33.** The physical form of a nutrient source affects how it is applied. A fine powder is harder to spread by hand on a windy day than granules because the powder
- A. dissolves in the air
 - B. has no nutrients in it
 - C. is heavier than the granules
 - D. is light and is blown away easily
- 34.** Spreading fertilizer evenly over the whole surface of a bed is called
- A. broadcasting
 - B. ring application
 - C. foliar spraying
 - D. drilling
- 35.** Placing fertilizer in a circle round a plant, a short distance from the stem, is known as
- A. top dressing of the leaves
 - B. ring application
 - C. broadcasting
 - D. deep ploughing
- 36.** Fertilizer is placed a short distance away from the stem of a plant rather than touching it in order to
- A. allow the wind to blow it away
 - B. avoid scorching the stem and roots while still feeding the plant
 - C. make the plant grow in a circle
 - D. keep the fertilizer away from the roots
- 37.** Liquid fertilizer that is sprayed onto the leaves of a crop is said to be applied as
- A. a foliar spray
 - B. a basal dressing
 - C. a ring application
 - D. a compost heap
- 38.** Compost is best applied to a vegetable bed
- A. sprinkled on the leaves of the growing crop
 - B. during land preparation, worked into the soil before planting
 - C. on the day of harvesting
 - D. only after the crop has been uprooted

- 39.** A pupil applies fertilizer to a bed in the middle of a hot, dry afternoon and the crop later shows burnt leaf edges. The best advice to the pupil is to
- A. apply twice as much fertilizer next time
 - B. stop using any fertilizer on the bed again
 - C. apply fertilizer when the soil is moist, such as after a light rain or watering
 - D. apply the fertilizer directly on the leaves instead
- 40.** A school garden has one wheelbarrow of poultry manure and one small bag of NPK for four vegetable beds. The BEST plan is to
- A. spray both the manure and the NPK onto the leaves of the crop
 - B. mix the manure and the NPK together and heap them in one corner
 - C. pour all the NPK onto one bed and leave the manure unused
 - D. work the manure into all four beds during land preparation and side-dress the growing crop with small measured amounts of NPK

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 6: Ecosystem

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. An ecosystem is best described as
 - A. a place where only animals are found
 - B. a group of animals of the same kind
 - C. a unit in which living things and their non-living surroundings interact
 - D. the soil in which plants grow
2. An ecosystem is said to be self-sustaining because
 - A. no living thing ever dies in it
 - B. it never changes from year to year
 - C. its parts supply one another with what they need to continue
 - D. it is kept going by human beings
3. Which of the following is an example of an ecosystem?
 - A. A piece of iron sheet
 - B. A pond
 - C. A bucket of sand
 - D. A single mango fruit
4. The place where a plant or an animal lives in an ecosystem is called its
 - A. nutrient
 - B. community
 - C. population
 - D. habitat
5. All the living things of different kinds living together in one place form
 - A. a community
 - B. a habitat
 - C. an organ
 - D. a population
6. Which of the following best explains why the Owabi forest can be called an ecosystem?
 - A. It is found in the Ashanti Region
 - B. It contains very many trees
 - C. Its plants, animals, soil, water and air act on one another as one unit
 - D. It is protected by the government

- 7.** The study of ecosystems is important mainly because it
- A. helps us to use and protect our surroundings wisely
 - B. shows us how to build better houses
 - C. helps us to kill dangerous animals
 - D. teaches us the names of all plants
- 8.** An ecosystem found on land is described as
- A. artificial
 - B. terrestrial
 - C. aquatic
 - D. arboreal
- 9.** An ecosystem found in water is described as
- A. arid
 - B. terrestrial
 - C. arboreal
 - D. aquatic
- 10.** An ecosystem made up of the life found on trees is described as
- A. terrestrial
 - B. arboreal
 - C. marine
 - D. aquatic
- 11.** Which of the following is an aquatic ecosystem?
- A. The Kakum forest
 - B. The Volta Lake
 - C. The Accra plains
 - D. A baobab tree
- 12.** A grassland in the Northern Region of Ghana is an example of
- A. a terrestrial ecosystem
 - B. a man-made lake
 - C. an arboreal ecosystem
 - D. an aquatic ecosystem
- 13.** Monkeys, birds, orchids and insects living in the branches of a tall forest tree make up
- A. a desert ecosystem
 - B. an arboreal ecosystem
 - C. an aquatic ecosystem
 - D. a soil ecosystem

- 14.** Which of the following pairs are BOTH aquatic ecosystems?
- A. The sea and a grassland
 - B. A mangrove tree and a rock
 - C. A pond and a forest
 - D. A river and the sea
- 15.** The living parts of an ecosystem are called
- A. mineral components
 - B. physical components
 - C. abiotic components
 - D. biotic components
- 16.** Which of the following is a biotic component of a pond?
- A. Sunlight
 - B. Tilapia
 - C. Mud
 - D. Water
- 17.** Green plants in an ecosystem are called producers because they
- A. produce the water of the ecosystem
 - B. break down dead bodies
 - C. are eaten by all other organisms
 - D. make their own food using sunlight
- 18.** Animals in an ecosystem are called consumers because they
- A. produce oxygen for other organisms
 - B. use sunlight to make food
 - C. depend on other organisms for their food
 - D. live only in water
- 19.** Bacteria and fungi in an ecosystem are important as
- A. decomposers that break down dead plants and animals
 - B. producers that make food from sunlight
 - C. abiotic components of the ecosystem
 - D. animals that feed on living plants
- 20.** If all the decomposers in a forest were removed, the MOST likely result would be that
- A. more rain would fall in the forest
 - B. all the animals would become plants
 - C. the forest would grow faster
 - D. dead plants and animals would pile up and nutrients would not return to the soil

- 21.** The non-living parts of an ecosystem are called
- A. biotic components
 - B. decomposers
 - C. abiotic components
 - D. consumers
- 22.** Which of the following is an abiotic component of a forest ecosystem?
- A. Earthworm
 - B. Climbing plant
 - C. Sunlight
 - D. Mushroom
- 23.** Which group contains ONLY abiotic components?
- A. Water, fish, soil, air
 - B. Water, air, soil, temperature
 - C. Air, water, bacteria, rock
 - D. Soil, grass, light, air
- 24.** Sunlight is an important abiotic component of an ecosystem chiefly because
- A. it keeps animals awake
 - B. it makes water salty
 - C. it helps the wind to blow
 - D. green plants need it to make food
- 25.** Water is important in an ecosystem because
- A. it prevents plants from growing too tall
 - B. it supplies the soil with sunlight
 - C. it is a living part of the ecosystem
 - D. all living things need it for their life processes
- 26.** A pond dries up completely during a long dry season. The fish die mainly because an abiotic component
- A. needed by the fish has been removed
 - B. has turned into a biotic component
 - C. has eaten the fish
 - D. has multiplied too fast
- 27.** Fish are suited to life in water because they
- A. have lungs and legs
 - B. have roots that take in water
 - C. have gills for breathing in water and fins for swimming
 - D. have wings and feathers

- 28.** Which structure helps a monkey to live in the branches of trees?
- A. A hard shell for protection
 - B. Long limbs and a gripping tail for holding on to branches
 - C. Gills for taking in air
 - D. Fins for moving about
- 29.** Water lilies are suited to life in a pond because they have
- A. deep woody roots
 - B. leaves reduced to spines
 - C. broad floating leaves that stay on the surface
 - D. thick bark on the stem
- 30.** Grasses of the northern savanna survive the long dry season mainly because they
- A. take in food from the air
 - B. can swim to wetter places
 - C. grow best without any water
 - D. have underground parts that sprout again when the rains come
- 31.** Which of the following animals is found in a terrestrial ecosystem?
- A. Grasscutter
 - B. Crab in a rock pool
 - C. Tilapia
 - D. Shrimp
- 32.** An earthworm taken from the soil and placed in a bucket of water soon dies. This shows that
- A. an organism survives best in the ecosystem to which it is suited
 - B. earthworms are not living things
 - C. every animal can live anywhere
 - D. water is harmful to all animals
- 33.** The MAIN difference between the organisms of a pond and those of a grassland is that
- A. pond organisms are all larger
 - B. each group has features suited to the conditions of its own surroundings
 - C. grassland organisms do not need water
 - D. pond organisms do not need food
- 34.** The feeding relationship in which a hawk feeds on a chick is an example of
- A. predation
 - B. competition
 - C. decomposition
 - D. pollination

- 35.** Two mango trees growing very close together struggle for light, water and soil nutrients. This is an example of
- A. competition
 - B. predation
 - C. decomposition
 - D. pollination
- 36.** Which of the following shows a dependence of a plant on an animal?
- A. A plant taking in water through its roots
 - B. A plant using sunlight to make food
 - C. A plant losing water from its leaves
 - D. A bee carrying pollen from one flower to another
- 37.** Green plants in an ecosystem supply animals with
- A. wind and rain
 - B. soil and rock
 - C. food and oxygen
 - D. light and heat
- 38.** Which of the following best shows how a biotic component depends on an abiotic component?
- A. A tree takes up water from the soil
 - B. A bird builds a nest with twigs
 - C. A fungus grows on a dead log
 - D. A goat feeds on grass
- 39.** A factory pours waste into a river and most of the fish die. The farmers who catch the fish also suffer. This shows that
- A. fish are not part of the ecosystem
 - B. rivers can clean themselves at once
 - C. the parts of an ecosystem are linked, so a change in one affects the others
 - D. only the fish are affected by pollution
- 40.** Which of the following human activities would MOST help to protect a wetland ecosystem near a town?
- A. Catching all the fish before the rains
 - B. Stopping the dumping of refuse in the wetland and replanting the plants around it
 - C. Filling the wetland with sand for building
 - D. Burning the plants at the edge of the water

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 7: Electricity and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Electricity produced by burning fuels such as natural gas, crude oil or coal is called
 - A. wind electricity
 - B. nuclear electricity
 - C. solar electricity
 - D. thermal electricity
2. In a thermal power station, the heat from the burning fuel is used to
 - A. cool the generator
 - B. melt the metal of the turbine
 - C. boil water and produce steam
 - D. charge large dry cells directly
3. In both thermal and nuclear power stations, the part that is turned by steam is the
 - A. pylon
 - B. transformer
 - C. turbine
 - D. chimney
4. The machine that changes the movement of the turbine into electricity is the
 - A. boiler
 - B. resistor
 - C. switch
 - D. generator
5. Nuclear electricity is produced from the heat released when
 - A. water falls from a high dam
 - B. coal is burnt in a large furnace
 - C. the nuclei of atoms of a fuel such as uranium are split
 - D. the Sun shines on a panel

- 6.** Which of the following is the fuel used in a nuclear power station?
- A. Uranium
 - B. Crude oil
 - C. Natural gas
 - D. Charcoal
- 7.** The energy change that takes place in the generator of a power station is
- A. chemical energy to heat energy
 - B. kinetic energy to electrical energy
 - C. electrical energy to kinetic energy
 - D. light energy to electrical energy
- 8.** Which statement about thermal and nuclear power stations is correct?
- A. Both burn fuel in a furnace to obtain heat
 - B. Both use steam to turn a turbine, but they obtain their heat in different ways
 - C. Both of them depend on falling water
 - D. Neither of them uses a generator
- 9.** The gas given off in the largest amount when fossil fuels are burnt in a thermal power station is
- A. nitrogen
 - B. oxygen
 - C. carbon dioxide
 - D. helium
- 10.** Carbon dioxide released from thermal power stations is of concern because it
- A. adds to the greenhouse gases that cause global warming
 - B. turns rain water into fresh water
 - C. makes the soil more fertile
 - D. destroys the ozone layer at once
- 11.** Smoke and soot from a thermal power station affect people living nearby mainly by
- A. improving the quality of their drinking water
 - B. causing breathing problems such as coughs and asthma
 - C. increasing the amount of oxygen in the air
 - D. making their houses stronger
- 12.** Sulphur dioxide released when some fuels are burnt may cause
- A. acid rain
 - B. a water cycle
 - C. an earthquake
 - D. a rainbow

- 13.** The MAIN environmental danger of a nuclear power station is
- A. the smoke from its chimney
 - B. the radioactive waste it produces and the risk of leakage
 - C. the carbon dioxide it releases
 - D. the noise of its boiler
- 14.** Which of the following would reduce the harmful effects of electricity generation on the environment?
- A. Building more chimneys for the same station
 - B. Pouring hot water from the station into a nearby river
 - C. Burning more coal instead of natural gas
 - D. Using more solar and hydro-electric power and fitting filters to chimneys
- 15.** Hot water released from a power station into a river may harm the river because it
- A. makes the river flow more quickly
 - B. raises the temperature of the water and kills fish and other organisms
 - C. removes all the salt from the water
 - D. adds useful nutrients to the water
- 16.** The letters LED stand for
- A. light energy detector
 - B. light-emitting diode
 - C. long electric drive
 - D. low energy device
- 17.** The component that gives out light when current passes through it in the correct direction is the
- A. LED
 - B. resistor
 - C. inductor
 - D. capacitor
- 18.** The MAIN function of a resistor in an electronic circuit is to
- A. reduce (limit) the current flowing in the circuit
 - B. change the current into heat only
 - C. give out light
 - D. store electrical charge
- 19.** The value of a small resistor is shown on its body by
- A. its length
 - B. the number of legs it has
 - C. its weight
 - D. coloured bands

- 20.** The component that stores electrical charge and releases it later is the
- A. switch
 - B. resistor
 - C. capacitor
 - D. diode
- 21.** A coil of wire that opposes a change in the current flowing through it is the
- A. inductor
 - B. LED
 - C. capacitor
 - D. resistor
- 22.** Which component allows current to pass in one direction only?
- A. A resistor
 - B. A connecting wire
 - C. A diode
 - D. A capacitor
- 23.** A source of electrical energy is needed in every working circuit. In the circuit built in class, this is provided by the
- A. resistor
 - B. LED
 - C. switch
 - D. dry cells
- 24.** Two dry cells of 1.5 V each are joined in series. The voltage of the battery formed is
- A. 3 V
 - B. 0.75 V
 - C. 6 V
 - D. 1.5 V
- 25.** A P-N junction diode is made by joining
- A. two pieces of copper wire
 - B. a resistor and a capacitor
 - C. a P-type region and an N-type region of semiconductor material
 - D. two dry cells in series
- 26.** In a P-N junction diode, the P region is the
- A. region that stores charge
 - B. positive region, which must be joined to the positive side of the battery for current to flow
 - C. region that gives off light
 - D. negative region, which must be joined to the positive side of the battery

- 27.** A diode connected so that current flows through it is said to be
- A. short circuited
 - B. reverse biased
 - C. earthed
 - D. forward biased
- 28.** An LED in a circuit does not light when the cells are connected one way round, but lights when the cells are turned round. This is because the LED
- A. needs a larger resistor
 - B. stores charge before lighting
 - C. allows current to pass in one direction only
 - D. is partly spoilt
- 29.** The longer leg of a new LED is the
- A. leg that must be cut off before use
 - B. positive leg (anode), which is joined to the positive side of the battery
 - C. negative leg, which is joined to the positive side of the battery
 - D. leg that carries no current
- 30.** When the switch in a circuit is closed,
- A. the circuit is broken and no current flows
 - B. the LED stores charge
 - C. the circuit is complete and current flows
 - D. the cells are destroyed
- 31.** When the switch in a circuit is opened, the LED goes out because
- A. the cells have run down at once
 - B. the path of the current is broken
 - C. the resistor has melted
 - D. the LED has been spoilt by the current
- 32.** In a circuit diagram, a long thin line beside a short thick line is the symbol for
- A. a cell
 - B. a resistor
 - C. a switch
 - D. a lamp
- 33.** Before joining an LED to a 3 V battery, it is wise to join a resistor in the circuit in order to
- A. prevent too large a current from spoiling the LED
 - B. store charge for the LED
 - C. make the LED brighter than normal
 - D. change the direction of the current

- 34.** A pupil built a circuit with two cells, a switch, a resistor and an LED, but the LED did not light even when the switch was closed. The FIRST thing she should check is
- A. whether the LED is warm
 - B. whether the wires are long enough
 - C. whether the room is dark enough
 - D. whether all the connections are tight and the cells are the right way round
- 35.** In a simple series circuit, the components are joined
- A. with the LED outside the circuit
 - B. in no particular order, since current finds its own way
 - C. side by side, so that each carries a different current
 - D. one after another, so that the same current passes through each of them
- 36.** When a second resistor is added in series to a circuit containing an LED, the LED becomes
- A. brighter, because the resistance is reduced
 - B. dimmer, because the voltage of the cells increases
 - C. dimmer, because the current is reduced
 - D. brighter, because the current is increased
- 37.** If the resistor in an LED circuit is replaced by one of much smaller resistance, the LED will
- A. go out completely
 - B. store more charge
 - C. change the direction of the current
 - D. shine more brightly and may be damaged by the larger current
- 38.** A diode joined into an LED circuit the wrong way round will make the LED
- A. shine more brightly
 - B. flash on and off
 - C. shine with a different colour
 - D. stay off, because the diode blocks the current
- 39.** The components of an electronic circuit are said to be interdependent because
- A. the way one component behaves affects the working of the others
 - B. each component works on its own
 - C. they are all made of the same material
 - D. they are all the same size

- 40.** A radio set, a mobile phone and a television all contain resistors, diodes and capacitors. This shows that
- A. all these gadgets do the same work
 - B. the same basic components are used in many different electronic gadgets
 - C. electronic gadgets need no source of energy
 - D. the components work only in radios

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a circuit diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 8: Energy

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Energy is best described as
 - A. the speed of a moving body
 - B. the space a body occupies
 - C. the mass of a body
 - D. the ability to do work
2. The unit in which energy is measured is the
 - A. newton
 - B. joule
 - C. kilogram
 - D. watt
3. The energy possessed by a body because of its position above the ground is
 - A. sound energy
 - B. potential energy
 - C. kinetic energy
 - D. nuclear energy
4. The energy possessed by a body because of its motion is
 - A. light energy
 - B. potential energy
 - C. chemical energy
 - D. kinetic energy
5. The energy stored in food, fuel and a dry cell is
 - A. solar energy
 - B. kinetic energy
 - C. sound energy
 - D. chemical energy
6. The energy that reaches the Earth directly from the Sun is called
 - A. solar energy
 - B. chemical energy
 - C. potential energy
 - D. nuclear energy

- 7.** Which form of energy reaches our ears when a drum is beaten?
- A. Light energy
 - B. Sound energy
 - C. Chemical energy
 - D. Nuclear energy
- 8.** Which of the following is NOT a form of energy?
- A. Heat
 - B. Light
 - C. Sound
 - D. Mass
- 9.** The sum of the potential energy and the kinetic energy of a body is called its
- A. nuclear energy
 - B. electrical energy
 - C. chemical energy
 - D. mechanical energy
- 10.** A coconut hanging on a tree possesses
- A. potential energy
 - B. sound energy
 - C. no energy at all
 - D. kinetic energy only
- 11.** As a mango falls from a tree to the ground, its
- A. potential energy increases while its kinetic energy decreases
 - B. potential and kinetic energy both increase
 - C. potential energy decreases while its kinetic energy increases
 - D. potential and kinetic energy both decrease
- 12.** A stone thrown straight upwards has its greatest potential energy
- A. half-way up
 - B. just before it hits the ground
 - C. at the highest point of its flight
 - D. at the moment it leaves the hand
- 13.** At the highest point of its flight, a ball thrown straight upwards has
- A. maximum potential energy and least kinetic energy
 - B. equal potential and kinetic energy
 - C. neither potential nor kinetic energy
 - D. maximum kinetic energy and least potential energy

- 14.** A pendulum bob swings from side to side. At the lowest point of the swing the bob has
- A. more potential than kinetic energy
 - B. no energy at all
 - C. the greatest kinetic energy
 - D. the greatest potential energy
- 15.** When a mango falls from a tree, the potential energy it loses is changed mainly into
- A. kinetic energy
 - B. nuclear energy
 - C. electrical energy
 - D. chemical energy
- 16.** A torch (flashlight) changes
- A. potential energy into sound energy
 - B. light energy into chemical energy
 - C. chemical energy into light energy
 - D. sound energy into heat energy
- 17.** An electric iron changes electrical energy into
- A. heat energy
 - B. chemical energy
 - C. nuclear energy
 - D. sound energy
- 18.** A solar panel on the roof of a clinic changes
- A. solar energy into electrical energy
 - B. heat energy into sound energy
 - C. electrical energy into solar energy
 - D. chemical energy into light energy
- 19.** A radio set changes electrical energy mainly into
- A. potential energy
 - B. chemical energy
 - C. sound energy
 - D. nuclear energy
- 20.** Which appliance changes electrical energy into kinetic energy?
- A. An electric fan
 - B. A candle
 - C. An electric bulb
 - D. A dry cell

- 21.** Water stored in the Akosombo dam high above the turbines possesses
- A. chemical energy, which is changed into heat energy
 - B. potential energy, which is changed into electrical energy at the power station
 - C. sound energy, which is changed into light energy
 - D. nuclear energy, which is changed into kinetic energy
- 22.** A burning candle changes chemical energy into
- A. sound and nuclear energy
 - B. heat and light energy
 - C. potential and sound energy
 - D. electrical and chemical energy
- 23.** The potential energy of a body is calculated using the expression
- A. $PE = mgh$
 - B. $PE = mv$
 - C. $PE = mg/h$
 - D. $PE = 1/2 mv^2$
- 24.** The kinetic energy of a body is calculated using the expression
- A. $KE = 2mv$
 - B. $KE = mgh$
 - C. $KE = mv$
 - D. $KE = 1/2 mv^2$
- 25.** A bag of mass 2 kg is lifted onto a shelf 5 m above the ground. Its potential energy is ($g = 10 \text{ m/s}^2$)
- A. 25 J
 - B. 1 000 J
 - C. 100 J
 - D. 10 J
- 26.** A boy of mass 40 kg runs at a speed of 3 m/s. His kinetic energy is
- A. 120 J
 - B. 180 J
 - C. 360 J
 - D. 60 J
- 27.** Two stones are held at the same height. The stone with the greater mass has
- A. no potential energy
 - B. the smaller potential energy
 - C. the same potential energy as the lighter stone
 - D. the greater potential energy

- 28.** If the speed of a moving car is doubled while its mass stays the same, its kinetic energy becomes
- A. half as large
 - B. the same
 - C. twice as large
 - D. four times as large
- 29.** The transfer of heat through a solid from the hot end to the cold end is called
- A. evaporation
 - B. conduction
 - C. radiation
 - D. convection
- 30.** Heat is transferred through liquids and gases mainly by
- A. radiation
 - B. convection
 - C. reflection
 - D. conduction
- 31.** The transfer of heat from the Sun to the Earth takes place by
- A. condensation
 - B. conduction
 - C. radiation
 - D. convection
- 32.** Which of the following is the BEST conductor of heat?
- A. Rubber
 - B. Aluminium
 - C. Plastic
 - D. Wood
- 33.** The handles of cooking pots are often made of wood or plastic because these materials are
- A. heavier than metal
 - B. good conductors of heat
 - C. poor conductors of heat
 - D. able to store heat for long
- 34.** Warm air above a hot road rises. This is because warm air
- A. has no weight at all
 - B. is pushed up by sound waves
 - C. is lighter (less dense) than the cooler air around it
 - D. is heavier than the cooler air

- 35.** The formation of a sharp shadow behind an object placed in sunlight shows that light
- A. is made up of sound waves
 - B. travels in a straight line
 - C. travels only in water
 - D. bends round objects
- 36.** Three cardboard sheets, each with a small hole, are arranged so that a candle flame can be seen through them. The flame can be seen only when the three holes are
- A. covered with glass
 - B. placed at different heights
 - C. of different sizes
 - D. in a straight line
- 37.** The bouncing back of light when it strikes a smooth, shiny surface is called
- A. refraction
 - B. dispersion
 - C. absorption
 - D. reflection
- 38.** A pencil standing in a glass of water appears bent at the surface of the water. This is due to
- A. refraction of light
 - B. absorption of light
 - C. reflection of light
 - D. dispersion of light
- 39.** The splitting of white light into its colours as it passes through a glass prism is called
- A. dispersion
 - B. conduction
 - C. reflection
 - D. diffusion
- 40.** A rainbow seen in the sky after a shower of rain is formed because the raindrops
- A. absorb all the colours of sunlight
 - B. reflect sound back to the Earth
 - C. split the sunlight into its different colours
 - D. give out light of their own

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps of your working in every calculation. Take the acceleration due to gravity, g , as 10 m/s^2 .

Question 7

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 9: Farming Systems

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A farming system is best described as
 - A. the way a farmer arranges and manages the growing of crops and the keeping of animals
 - B. the tools used to clear a piece of land
 - C. the number of workers on a farm
 - D. the type of soil found on a farm
2. Which of the following is NOT one of the farming systems studied at this level?
 - A. Cold storage farming
 - B. Organic farming
 - C. Crop rotation
 - D. Mixed farming
3. A farmer chooses a farming system mainly according to
 - A. the day of the week on which he starts
 - B. the number of children in his house
 - C. the colour of his farm tools
 - D. the land, the crops, the animals and the money available to him
4. Growing crops and keeping animals on the same farm at the same time is called
 - A. land rotation
 - B. mixed farming
 - C. mixed cropping
 - D. crop rotation
5. Growing two or more crops together on the same piece of land at the same time is called
 - A. mixed cropping
 - B. mixed farming
 - C. crop rotation
 - D. monocropping

- 6.** Growing only one type of crop on a piece of land season after season is called
- A. mixed farming
 - B. organic farming
 - C. mixed cropping
 - D. monocropping (sole cropping)
- 7.** In land rotation, the farmer
- A. uses only manure and no chemicals
 - B. keeps animals and crops on the same land
 - C. grows different crops on the same land each year
 - D. cultivates one piece of land for a few years, then leaves it and moves to another piece
- 8.** Land rotation is also known as
- A. crop rotation
 - B. shifting cultivation
 - C. plantation farming
 - D. mixed farming
- 9.** Land that has been left uncultivated so that it can regain its fertility is said to be under
- A. rotation of crops
 - B. fallow
 - C. mulch
 - D. irrigation
- 10.** During the fallow period the fertility of the land is restored mainly by
- A. the fertilizer spread before the land is left
 - B. the heat of the Sun on the bare soil
 - C. the ploughing done by the farmer
 - D. the natural vegetation that grows and decays on it
- 11.** Land rotation is practised mostly in areas where
- A. land is scarce and the population is large
 - B. the soil is already very rich
 - C. there are many tractors and fertilizers
 - D. land is plentiful and the population is small
- 12.** Which of the following is a disadvantage of land rotation?
- A. It gives the land no time to rest
 - B. It needs a large area of land and the burning of bush destroys soil organisms
 - C. It cannot be practised without a tractor
 - D. It makes the farmer grow only one crop

- 13.** Land rotation is becoming less common in many parts of Ghana today mainly because
- A. the government has banned all farming
 - B. the soil has become too rich for it
 - C. the growing population has reduced the amount of land available
 - D. farmers no longer know how to practise it
- 14.** In crop rotation, the farmer
- A. grows different crops on the same piece of land in a planned order, season after season
 - B. keeps animals instead of growing crops
 - C. moves to a new piece of land each year
 - D. grows the same crop on the land every year
- 15.** The MAIN reason for including a legume such as groundnut in a crop rotation is that the legume
- A. takes all the pests away
 - B. adds nitrogen to the soil
 - C. dries the soil quickly
 - D. gives the largest harvest
- 16.** The nitrogen added to the soil by a groundnut crop comes from
- A. the leaves falling on the soil
 - B. bacteria living in nodules on its roots
 - C. the water the plant takes in
 - D. the sunlight trapped by the plant
- 17.** Crop rotation helps to control pests and diseases because
- A. the pests of one crop do not find their food plant in the following season
 - B. the crops are sprayed with more chemicals
 - C. the pests are frightened by the new crops
 - D. the land is burnt before every planting
- 18.** In a good crop rotation plan, a deep-rooted crop is followed by a shallow-rooted one so that
- A. the soil becomes harder
 - B. only one type of nutrient is used up
 - C. the plants draw nutrients from different depths of the soil
 - D. the farmer can harvest at the same time
- 19.** Which of the following is the MAIN difference between crop rotation and land rotation?
- A. In land rotation the crops are changed each season
 - B. In crop rotation the same land is used and the crops are changed; in land rotation the land itself is changed
 - C. In crop rotation only one crop is grown

D. Crop rotation can be practised only with animals

- 20.** A farmer at Wenchi grows maize, then beans, then cassava and then rests the land on the same plot year after year. This farmer is practising
- A. crop rotation
 - B. monocropping
 - C. mixed cropping
 - D. land rotation
- 21.** Which of the following is an example of mixed cropping?
- A. Maize, cassava and cowpea growing together in the same field
 - B. Maize this year and cassava on the same field next year
 - C. Maize grown alone on a large field
 - D. Maize grown on one field and goats kept on another
- 22.** One advantage of mixed cropping is that
- A. harvesting is always done by machine
 - B. the crops never compete for anything
 - C. the farmer still has something to harvest if one crop fails
 - D. only one type of nutrient is removed from the soil
- 23.** Mixed cropping helps to protect the soil because
- A. the different crops cover the soil and reduce erosion
 - B. the soil is left bare between the plants
 - C. the farmer ploughs the soil more often
 - D. the crops are burnt after harvesting
- 24.** A disadvantage of mixed cropping is that
- A. the soil cannot hold water at all
 - B. no crop can be grown with another
 - C. the crops may compete with one another for light, water and nutrients
 - D. the farmer can harvest only once in ten years
- 25.** Mixed cropping is difficult to carry out with machines mainly because
- A. the crops are planted too deep
 - B. the farmer plants only in the dry season
 - C. machines cannot move on a farm
 - D. the crops are of different kinds and ripen at different times
- 26.** A farmer plants cowpea between his rows of maize. Apart from extra food, the cowpea also helps by
- A. covering the ground and adding nitrogen to the soil
 - B. making the maize grow taller than usual
 - C. preventing rain from reaching the soil
 - D. driving away all the birds

- 27.** Which of the following describes a farm practising mixed farming?
- A. A farm where only cattle are kept
 - B. A farm where the land is left to rest for five years
 - C. A farm at Ejura where maize is grown and cattle are also kept
 - D. A farm where only maize and cassava are grown together
- 28.** In mixed farming, the animals are useful to the crops mainly because they
- A. shade the young crops from the Sun
 - B. supply manure that improves the fertility of the soil
 - C. carry the seeds to the field
 - D. eat the pests of the crops at night
- 29.** In mixed farming, the crops are useful to the animals mainly because they
- A. provide medicine for the animals
 - B. supply water to the animals
 - C. keep the animals warm at night
 - D. supply feed such as crop residues and grain
- 30.** One financial advantage of mixed farming is that the farmer
- A. is paid by the government each month
 - B. earns income from both crops and animals during the year
 - C. does not need to buy any tools
 - D. pays no money for labour
- 31.** A disadvantage of mixed farming is that it
- A. gives the farmer only one source of income
 - B. needs more capital, more labour and more skill than a single enterprise
 - C. cannot be practised in Ghana
 - D. always destroys the soil
- 32.** Mixed farming is said to make good use of a farm's resources because
- A. the farmer buys everything from the market
 - B. nothing is ever harvested from the farm
 - C. the waste of one enterprise is used by the other
 - D. the animals are kept far away from the crops
- 33.** Organic farming is a system in which the farmer
- A. keeps animals in cages all year round
 - B. grows only one crop on a large farm
 - C. grows crops and keeps animals without using artificial fertilizers and chemical pesticides
 - D. uses only artificial fertilizers

- 34.** Which of the following would an organic farmer use to feed his crops?
- A. NPK fertilizer
 - B. Sulphate of ammonia
 - C. Weedicide
 - D. Compost and animal manure
- 35.** An organic farmer controls insect pests mainly by
- A. leaving the pests to multiply freely
 - B. burning the whole farm every month
 - C. using natural methods such as neem extract, hand picking and natural enemies
 - D. spraying strong chemical insecticides
- 36.** One advantage of organic farming is that
- A. it works only on very poor soils
 - B. it needs very little labour
 - C. the yields are always the highest possible
 - D. the food produced is free from chemical residues and the soil organisms are protected
- 37.** A disadvantage of organic farming is that
- A. yields are often lower and more labour is needed
 - B. the food produced cannot be eaten
 - C. it cannot be practised without machines
 - D. it destroys the structure of the soil
- 38.** Organic produce often sells at a higher price than other produce because
- A. the government fixes a higher price for it
 - B. it can be stored for ten years
 - C. many buyers are willing to pay more for food grown without chemicals
 - D. it is always larger in size
- 39.** A farmer with a small piece of land near Kumasi, little money and a family to feed all year would be BEST advised to practise
- A. leaving the land to fallow for five years
 - B. land rotation on several large plots
 - C. mixed cropping with manure from his own animals
 - D. monocropping of one crop on the whole land
- 40.** Which statement about the farming systems is correct?
- A. Organic farming cannot be combined with crop rotation
 - B. Only one system may be used on a farm
 - C. Mixed farming cannot include any crops
 - D. More than one system can be used on the same farm at the same time

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 10: Force and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A force is best described as
 - A. the speed of a moving body
 - B. the mass of a body at rest
 - C. a push or a pull acting on a body
 - D. the energy stored in a body
2. The unit in which force is measured is the
 - A. newton
 - B. kilogram
 - C. watt
 - D. joule
3. Two forces acting on a body are said to be balanced when they are
 - A. applied at different times
 - B. equal in size and act in the same direction
 - C. unequal and act in opposite directions
 - D. equal in size and act in opposite directions
4. When the forces acting on a body at rest are balanced, the body will
 - A. move with increasing speed
 - B. begin to move slowly
 - C. turn round and round
 - D. remain at rest
5. A body begins to move only when the forces acting on it are
 - A. unbalanced
 - B. very small
 - C. equal and opposite
 - D. balanced
6. Two pupils pull a box in opposite directions with forces of 30 N and 50 N. The box will move
 - A. in neither direction
 - B. in the direction of the 50 N force
 - C. at right angles to both forces

- D. in the direction of the 30 N force
- 7.** The tendency of a body to resist a change in its state of rest or of motion is called
- A. gravity
 - B. friction
 - C. inertia
 - D. pressure
- 8.** Of the following bodies, the one with the greatest inertia is a
- A. bicycle
 - B. football
 - C. wheelbarrow
 - D. loaded lorry
- 9.** Newton's first law of motion states that a body continues in its state of rest or of uniform motion in a straight line unless
- A. an unbalanced force acts on it
 - B. it is made of metal
 - C. it is placed on a smooth surface
 - D. its mass is increased
- 10.** Newton's first law of motion is also known as the law of
- A. conservation of energy
 - B. inertia
 - C. gravity
 - D. friction
- 11.** A metal ball rolling on a smooth floor finally stops. The force that stops it is
- A. the push that started it
 - B. the inertia of the ball
 - C. the weight of the ball
 - D. friction between the ball and the floor
- 12.** According to Newton's first law, a ball rolling on a perfectly smooth surface with no opposing force would
- A. move in a circle
 - B. keep moving in a straight line for ever
 - C. rise into the air
 - D. stop after a short distance
- 13.** A body moving with uniform velocity has
- A. a force acting on it in the direction of motion only
 - B. a large unbalanced force acting on it
 - C. no unbalanced force acting on it
 - D. no mass

- 14.** Which of the following is the BEST example of inertia of rest?
- A. A ball that bounces off a wall
 - B. A lorry that is difficult to stop when moving
 - C. A heavy stone that is difficult to start moving
 - D. A magnet that attracts a pin
- 15.** The quantity that measures the inertia of a body is its
- A. weight
 - B. mass
 - C. volume
 - D. speed
- 16.** Passengers in a trolley jerk forward when the driver brakes suddenly because their bodies
- A. tend to continue moving forward
 - B. lose their inertia
 - C. are pushed forward by the brakes
 - D. become heavier at that moment
- 17.** Seat belts are worn in vehicles mainly to
- A. hold the passenger back when the vehicle stops suddenly
 - B. make the seat comfortable
 - C. reduce the speed of the vehicle
 - D. keep the passenger warm
- 18.** When a vehicle at rest starts off suddenly, the passengers are thrown backwards because the
- A. seats push the passengers backwards
 - B. engine pulls the passengers backwards
 - C. weight of the passengers increases
 - D. lower part of the body moves with the vehicle while the upper part stays at rest
- 19.** Knocking the handle of a hoe against a hard surface tightens the blade. This is because the
- A. handle pushes the blade upwards
 - B. blade continues to move downwards when the handle stops
 - C. blade becomes heavier on hitting the surface
 - D. blade loses its inertia
- 20.** A tablecloth can be pulled quickly from under a glass without the glass falling because the glass
- A. tends to stay at rest
 - B. has no mass
 - C. is firmly stuck to the cloth

- D. is pushed upwards by the cloth
- 21.** Dust is removed from a mat by beating it because, when the mat is struck, the
- A. mat attracts the dust strongly
 - B. dust is burnt by the stick
 - C. dust gains weight and falls
 - D. mat moves away while the dust particles stay at rest and are left behind
- 22.** A magnet is a material that can
- A. attract all materials
 - B. attract iron and certain other materials
 - C. produce electricity by itself
 - D. attract all metals
- 23.** Which of the following materials will be attracted by a magnet?
- A. A copper wire
 - B. A wooden pencil
 - C. A plastic ruler
 - D. An iron nail
- 24.** A magnet that keeps its magnetism for a long time is described as a
- A. soft magnet
 - B. temporary magnet
 - C. permanent magnet
 - D. weak magnet
- 25.** The poles of a magnet are the parts where the magnetic force is
- A. weakest
 - B. completely absent
 - C. strongest
 - D. the same as at the middle
- 26.** When the north pole of one magnet is brought near the north pole of another magnet, the two magnets
- A. repel each other
 - B. attract each other
 - C. both lose their magnetism at once
 - D. stick together firmly
- 27.** The surest test that a bar of metal is a magnet is that it
- A. is made of steel
 - B. repels one end of a known magnet
 - C. attracts an iron nail
 - D. is shiny and hard

- 28.** A bar magnet hung freely by a thread finally comes to rest with its ends pointing
- A. in no fixed direction
 - B. east and west
 - C. north and south
 - D. upwards and downwards
- 29.** Which of the following is a use of a magnet in everyday life?
- A. Separating iron pieces from other waste at a scrap yard
 - B. Cutting metal sheets
 - C. Boiling water in a kettle
 - D. Measuring the mass of a body
- 30.** A simple machine is a device that
- A. creates energy for the user
 - B. makes work easier to do
 - C. works without any force
 - D. destroys the force applied to it
- 31.** Which of the following is NOT a simple machine?
- A. A pulley
 - B. An inclined plane
 - C. A wheel and axle
 - D. A dry cell
- 32.** The fixed point about which a lever turns is called the
- A. effort
 - B. load
 - C. fulcrum
 - D. pivot arm
- 33.** A lever in which the fulcrum lies between the effort and the load is a
- A. second-class lever
 - B. third-class lever
 - C. fourth-class lever
 - D. first-class lever
- 34.** A wheelbarrow is an example of a
- A. second-class lever
 - B. first-class lever
 - C. pulley
 - D. third-class lever

- 35.** In a pair of sugar tongs, the effort is applied between the fulcrum and the load. The tongs are therefore a
- A. second-class lever
 - B. wheel and axle
 - C. third-class lever
 - D. first-class lever
- 36.** The work done by the effort on a machine is called the
- A. work input
 - B. work output
 - C. efficiency
 - D. mechanical advantage
- 37.** The efficiency of a machine is the ratio of
- A. work input to work output, expressed as a percentage
 - B. load to effort, expressed in newtons
 - C. effort to distance, expressed in joules
 - D. work output to work input, expressed as a percentage
- 38.** A machine is supplied with 200 J of energy and does 150 J of useful work. Its efficiency is
- A. 133%
 - B. 75%
 - C. 35%
 - D. 50%
- 39.** No simple machine has an efficiency of 100% because some of the work input is used in
- A. increasing the mass of the machine
 - B. overcoming friction between the moving parts
 - C. creating new energy in the machine
 - D. destroying the load
- 40.** The efficiency of a machine can be improved by
- A. adding more load to the machine
 - B. removing the fulcrum
 - C. oiling the moving parts to reduce friction
 - D. making the parts rougher

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Show all the steps in any calculation, and give the unit with every answer.

Question 7

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 8

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 11: Human Body D

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Any substance taken into the body to provide energy, growth and health is called
 - A. waste
 - B. enzyme
 - C. food
 - D. mucus
2. Which of the following is NOT a food nutrient?
 - A. Vitamin
 - B. Sand
 - C. Carbohydrate
 - D. Protein
3. The nutrient that supplies the body with most of its energy is
 - A. protein
 - B. mineral salt
 - C. carbohydrate
 - D. vitamin
4. Foods such as beans, fish, meat and eggs are valuable in the diet mainly because they supply
 - A. energy only
 - B. protein for growth and repair of the body
 - C. roughage only
 - D. water only
5. A person who does not eat enough food over a long time will
 - A. store more energy in the body
 - B. grow taller than others
 - C. become stronger and more active
 - D. lose weight, become weak and fall ill easily

- 6.** Roughage (dietary fibre) is important in the diet because it
- A. helps the movement of food through the alimentary canal and prevents constipation
 - B. builds strong bones and teeth
 - C. is digested into amino acids
 - D. supplies most of the body's energy
- 7.** Two pupils of the same age are compared. One is active, alert and of normal weight; the other is thin, tired and often ill. The BEST conclusion is that the second pupil
- A. takes too much water
 - B. does too little homework
 - C. may not be getting enough of the right food
 - D. was born to be weak
- 8.** The long tube through which food passes from the mouth to the anus is called
- A. the alimentary canal
 - B. the windpipe
 - C. the blood vessel
 - D. the spinal cord
- 9.** The correct order of the parts of the alimentary canal is
- A. mouth, oesophagus, stomach, small intestine, large intestine, anus
 - B. mouth, oesophagus, small intestine, stomach, large intestine, anus
 - C. mouth, windpipe, stomach, small intestine, large intestine, anus
 - D. mouth, stomach, oesophagus, large intestine, small intestine, anus
- 10.** The tube that carries food from the mouth to the stomach is the
- A. duodenum
 - B. trachea
 - C. rectum
 - D. oesophagus
- 11.** Digestion begins in the
- A. large intestine
 - B. small intestine
 - C. mouth
 - D. stomach
- 12.** The part of the alimentary canal where most digested food is absorbed is the
- A. stomach
 - B. small intestine
 - C. large intestine
 - D. oesophagus

- 13.** The main work of the large intestine is to
- A. produce saliva
 - B. absorb amino acids
 - C. digest protein
 - D. absorb water from the undigested material
- 14.** Undigested material is passed out of the body through the
- A. lungs
 - B. anus
 - C. kidney
 - D. skin
- 15.** Food is pushed along the alimentary canal by
- A. the air we breathe in
 - B. the force of gravity alone
 - C. the beating of the heart
 - D. the squeezing movement of the muscles of its wall
- 16.** The teeth help in digestion by
- A. cutting and grinding the food into smaller pieces
 - B. adding enzymes to the food
 - C. turning starch into sugar
 - D. absorbing nutrients from the food
- 17.** Saliva is produced in the
- A. small intestine
 - B. stomach
 - C. liver
 - D. mouth
- 18.** The substance in saliva that acts on boiled yam or bread is an enzyme that begins the digestion of
- A. protein
 - B. fat
 - C. starch
 - D. vitamin
- 19.** The breaking down of food into smaller pieces by the teeth and the tongue, without any chemical change, is called
- A. physical (mechanical) digestion
 - B. absorption
 - C. assimilation
 - D. chemical digestion

- 20.** Boiled rice chewed for a long time begins to taste sweet. This is because
- A. the enzyme in saliva has changed some of the starch into sugar
 - B. the rice has started to go bad
 - C. sugar has been added by the tongue
 - D. the teeth produce sugar
- 21.** The digestion of protein, as in a piece of meat, begins in the
- A. rectum
 - B. large intestine
 - C. mouth
 - D. stomach
- 22.** The acid produced in the stomach helps digestion by
- A. absorbing the digested food
 - B. making the food sweeter
 - C. killing germs in the food and helping the enzymes to work
 - D. changing fats into sugar
- 23.** Bile, which is produced by the liver and helps in the digestion of palm oil, acts by
- A. absorbing the oil into the blood
 - B. changing the oil into starch
 - C. breaking the oil into tiny droplets so that enzymes can act on it
 - D. turning the oil into protein
- 24.** An orange eaten at breakfast supplies vitamin C. In the alimentary canal this vitamin is
- A. digested into amino acids
 - B. absorbed without being digested
 - C. changed into starch
 - D. passed out unchanged in the faeces
- 25.** The end product of the digestion of starch is
- A. glucose
 - B. amino acids
 - C. fatty acids
 - D. glycerol only
- 26.** Proteins are digested into
- A. simple sugars
 - B. amino acids
 - C. fatty acids and glycerol
 - D. minerals

- 27.** Oils and fats are digested into
- A. glucose
 - B. amino acids
 - C. fatty acids and glycerol
 - D. vitamins
- 28.** Which of the following pairs correctly matches a food with the end product of its digestion?
- A. Palm oil - glucose
 - B. Boiled yam - fatty acids
 - C. Egg - amino acids
 - D. Bread - amino acids
- 29.** A meal of kenkey and fried fish will produce which end products of digestion?
- A. Glucose and amino acids
 - B. Fatty acids only
 - C. Glucose only
 - D. Amino acids only
- 30.** The end products of digestion are described as 'simple' because they are
- A. sweet in taste
 - B. small enough to pass through the wall of the small intestine into the blood
 - C. produced only in the mouth
 - D. always liquid in form
- 31.** Which of the following statements about digestion is correct?
- A. All food is absorbed without any change
 - B. Large food substances are broken down into smaller ones that the body can absorb
 - C. Small food substances are joined into larger ones
 - D. Food is burnt into ash inside the stomach
- 32.** The passing of digested food through the wall of the small intestine into the blood is called
- A. ingestion
 - B. egestion
 - C. digestion
 - D. absorption
- 33.** The inner wall of the small intestine is folded and covered with millions of tiny finger-like structures. This arrangement
- A. makes the intestine shorter
 - B. increases the surface for absorbing digested food
 - C. helps to chew the food

D. produces the teeth of the body

34. Digested food is carried from the small intestine to all parts of the body by

- A. the air passages
- B. the blood
- C. the nerves
- D. the bones

35. Glucose carried to the cells of the body is used mainly to

- A. build the walls of cells
- B. make the hair grow dark
- C. form the enamel of teeth
- D. release energy during respiration

36. A patient whose small intestine has been damaged will suffer mainly because

- A. food cannot be chewed properly
- B. water cannot be drunk at all
- C. digested food cannot be properly absorbed into the blood
- D. saliva can no longer be produced

37. Iodine solution added to boiled yam turns

- A. blue-black
- B. purple
- C. colourless
- D. brick-red

38. Benedict's solution added to a glucose solution and heated turns

- A. purple
- B. milky white
- C. blue-black
- D. brick-red

39. Sodium hydroxide followed by copper sulphate solution added to egg white gives a

- A. purple colour
- B. blue-black colour
- C. green colour
- D. brick-red colour

40. A piece of paper rubbed with a food leaves a mark through which light passes. The food contains

- A. protein
- B. vitamin C
- C. starch
- D. oil or fat

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 12: Human Body

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The class of food that supplies the body with most of its energy is
 - A. carbohydrates
 - B. proteins
 - C. mineral salts
 - D. vitamins
2. Proteins are described as body-building foods because they are used for
 - A. cooling the blood
 - B. growth and the repair of worn-out tissues
 - C. supplying heat only
 - D. protecting the body against all diseases
3. Which of the following foods is the RICHEST source of protein?
 - A. Yam
 - B. Cassava
 - C. Rice
 - D. Beans
4. Fats and oils are important in the diet because they
 - A. supply energy and help to store it in the body
 - B. are digested in the mouth
 - C. supply water to the cells
 - D. cannot be stored in the body
5. Vitamins and mineral salts are needed in the diet mainly to
 - A. build the bones only
 - B. replace the blood lost daily
 - C. protect the body and keep it working properly
 - D. supply all the energy for work
6. Which nutrient is needed for the formation of strong bones and teeth?
 - A. Sodium chloride
 - B. Calcium
 - C. Starch
 - D. Iodine

- 7.** Iron is needed in the diet for the formation of
- A. bile in the liver
 - B. sweat in the skin
 - C. haemoglobin in red blood cells
 - D. enamel on the teeth
- 8.** A diet that contains all the classes of food in the correct proportions is described as a
- A. balanced diet
 - B. liquid diet
 - C. protein diet
 - D. staple diet
- 9.** A deficiency disease is a disease caused by
- A. the lack of a particular nutrient in the diet
 - B. eating too much of one food
 - C. a virus entering the body
 - D. drinking cold water
- 10.** Kwashiorkor in young children is caused by the lack of
- A. fibre in the diet
 - B. protein in the diet
 - C. carbohydrate in the diet
 - D. water in the diet
- 11.** Night blindness is caused by a lack of
- A. iron
 - B. vitamin C
 - C. vitamin D
 - D. vitamin A
- 12.** Scurvy, in which the gums bleed, is caused by a lack of
- A. vitamin K
 - B. vitamin C
 - C. vitamin A
 - D. calcium
- 13.** Rickets in children is caused by a lack of
- A. vitamin C and iron
 - B. iodine and sodium
 - C. protein and water
 - D. vitamin D and calcium

- 14.** Anaemia, in which a person is always weak and pale, is caused by a lack of
- A. vitamin C
 - B. fat
 - C. iron
 - D. iodine
- 15.** Goitre, a swelling in the neck, is caused by a lack of
- A. iodine
 - B. vitamin D
 - C. calcium
 - D. protein
- 16.** A child who eats only kenkey and sugar every day, with no fish, beans or eggs, is MOST likely to suffer from
- A. goitre
 - B. night blindness
 - C. scurvy
 - D. kwashiorkor
- 17.** A child with kwashiorkor usually shows
- A. a large swelling in front of the neck
 - B. bowed legs and a soft skull
 - C. a swollen belly and feet, with thin arms and reddish hair
 - D. bleeding gums and loose teeth
- 18.** Bowed legs and soft, weak bones in a young child are symptoms of
- A. scurvy
 - B. rickets
 - C. anaemia
 - D. night blindness
- 19.** A person who cannot see well in dim light is suffering from
- A. night blindness
 - B. goitre
 - C. rickets
 - D. scurvy
- 20.** Rickets is BEST prevented by eating foods rich in vitamin D and by
- A. bathing the child in warm water
 - B. giving the child plenty of sugar
 - C. allowing the child to play in the early morning sunshine
 - D. keeping the child indoors at all times

- 21.** The BEST way to prevent scurvy in a family is to
- A. boil all fruits before eating them
 - B. eat more fried yam
 - C. drink more sweetened drinks
 - D. include fresh fruits such as oranges and guava in the daily meals
- 22.** Which of these local foods would BEST help to prevent anaemia?
- A. Sugar-cane juice
 - B. Kontomire and liver
 - C. Plain white rice
 - D. Fried plantain only
- 23.** Deficiency diseases in a community are BEST prevented by
- A. educating families to eat balanced diets from locally available foods
 - B. giving injections to every child each week
 - C. boiling all drinking water twice
 - D. wearing nose masks at meal times
- 24.** A virus is BEST described as a
- A. worm that lives in the intestines
 - B. large single-celled animal found in water
 - C. type of bacterium that lives in the soil
 - D. very small organism that can multiply only inside the cells of a living host
- 25.** Which of the following diseases is caused by a virus?
- A. COVID-19
 - B. Ringworm
 - C. Cholera
 - D. Malaria
- 26.** Viral diseases cannot be cured by
- A. vaccination before infection
 - B. antibiotics
 - C. the body's own defence system
 - D. rest and good feeding
- 27.** COVID-19, Ebola and H1N1 are alike in that all three are
- A. spread only by mosquitoes
 - B. deficiency diseases of childhood
 - C. caused by viruses and spread from person to person
 - D. caused by bacteria in food

- 28.** COVID-19 is spread MAINLY through
- A. drinking water from a tap
 - B. droplets released when an infected person coughs, sneezes or talks
 - C. eating food that lacks protein
 - D. the bite of an infected mosquito
- 29.** Ebola is spread MAINLY through
- A. direct contact with the body fluids of an infected person
 - B. the bite of a housefly
 - C. breathing the air of a cold room
 - D. sharing a classroom desk
- 30.** A disease that spreads quickly from community to community and from country to country, affecting a very large number of people, is declared a
- A. deficiency disease
 - B. chronic disease
 - C. pandemic
 - D. hereditary disease
- 31.** Air travel helped COVID-19 to spread round the world quickly because
- A. aeroplanes produce the virus in their engines
 - B. the virus is carried in aviation fuel
 - C. the virus grows faster in an aeroplane
 - D. infected people moved from country to country before showing symptoms
- 32.** Common symptoms of COVID-19 include
- A. a swelling in front of the neck
 - B. bleeding gums and loose teeth
 - C. fever, dry cough, tiredness and loss of taste or smell
 - D. bowed legs and soft bones
- 33.** A person suspected of having a viral disease such as COVID-19 should FIRST
- A. visit many friends for advice
 - B. isolate himself and report to a health facility
 - C. buy antibiotics from a drug store
 - D. continue to attend school as usual
- 34.** Washing the hands with soap under running water helps to prevent COVID-19 because it
- A. removes and destroys virus particles picked up on the hands
 - B. makes the hands too cold for the virus
 - C. kills mosquitoes on the skin
 - D. adds vitamins to the skin

- 35.** Wearing a nose mask in a crowded place helps to control a viral disease because the mask
- A. kills the virus in the whole room
 - B. traps the droplets breathed out and keeps out those from other people
 - C. cools the air that is breathed in
 - D. supplies oxygen to the wearer
- 36.** Besides illness and death, a serious effect of a pandemic on a country is that
- A. the population of mosquitoes falls
 - B. the rainfall of the country increases
 - C. the soil of the country loses its fertility
 - D. schools and businesses close and the economy suffers
- 37.** The MOST important role of an individual in controlling a viral disease is to
- A. share medicines with neighbours
 - B. hide any illness from the health workers
 - C. observe the safety measures and report any symptoms early
 - D. wait for the government to act alone
- 38.** A useful role for community members during an outbreak is to
- A. drive infected persons out of the community
 - B. hold large gatherings to discuss the disease
 - C. stop all health workers from entering the area
 - D. educate one another and support those in isolation with food and care
- 39.** Which of the following is a role of government in managing a pandemic?
- A. Preventing health workers from giving information
 - B. Closing all health facilities to save money
 - C. Providing vaccines, testing centres and treatment facilities
 - D. Asking every family to make its own medicine
- 40.** A class wishes to educate the community about a viral disease. The MOST suitable material to design and produce is a
- A. list of the names of sick people in the town
 - B. chart of the prices of medicines
 - C. map of the roads in the district
 - D. poster showing the symptoms and the ways of preventing the disease

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 13: Life Cycle Ho

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The series of changes an organism passes through from the egg to the adult is called
 - A. a life cycle
 - B. a food chain
 - C. a life span
 - D. a water cycle
2. The change in form that an insect undergoes as it grows from the egg to the adult is known as
 - A. fertilisation
 - B. photosynthesis
 - C. metamorphosis
 - D. respiration
3. An insect whose life cycle has the four stages egg, larva, pupa and adult shows
 - A. incomplete metamorphosis
 - B. continuous growth
 - C. no metamorphosis
 - D. complete metamorphosis
4. The life cycle of an organism is described as a cycle because
 - A. the organism returns to the egg stage as it grows old
 - B. it takes exactly one year to finish
 - C. the adult produces eggs and the same stages begin again
 - D. the organism grows round in shape
5. Which of the following animals has a life cycle most like that of the housefly?
 - A. The mosquito
 - B. The goat
 - C. The lizard
 - D. The fowl

- 6.** How many stages are there in the life cycle of the housefly?
- A. Two
 - B. Three
 - C. Four
 - D. Five
- 7.** The correct order of the stages in the life cycle of the housefly is
- A. larva, egg, pupa, adult
 - B. egg, larva, pupa, adult
 - C. egg, larva, adult, pupa
 - D. egg, pupa, larva, adult
- 8.** The first stage in the life cycle of the housefly is the
- A. larva
 - B. adult
 - C. pupa
 - D. egg
- 9.** The young housefly that hatches from the egg is called
- A. a nymph
 - B. a larva
 - C. an adult
 - D. a pupa
- 10.** The larva of the housefly is commonly known as
- A. a caterpillar
 - B. a maggot
 - C. a tadpole
 - D. a grub-fly
- 11.** The stage of the housefly that does not feed and appears to be at rest is the
- A. larva
 - B. egg
 - C. pupa
 - D. adult
- 12.** The adult housefly comes out of the
- A. soil only
 - B. pupa
 - C. egg
 - D. larva

- 13.** The stage of the life cycle in which the housefly lays eggs is the
- A. larva stage
 - B. egg stage
 - C. pupa stage
 - D. adult stage
- 14.** A pupil arranged cut-outs of the stages of the housefly in this order: pupa, egg, adult, larva. The arrangement is wrong because
- A. the larva must come before the egg
 - B. the pupa and the adult are the same stage
 - C. the egg comes first and the pupa comes after the larva
 - D. the adult comes first in every life cycle
- 15.** The eggs of the housefly are usually laid
- A. on dry stones in the sun
 - B. inside clean drinking water
 - C. on the leaves of growing crops only
 - D. in rotting rubbish, manure and other decaying matter
- 16.** Which of the following best describes the larva of the housefly?
- A. A tiny white oval object laid in a group
 - B. A winged insect with six legs
 - C. A hard, dark, motionless case
 - D. A white, legless, worm-like creature
- 17.** During the pupa stage, the housefly
- A. feeds heavily on rubbish
 - B. changes inside a hard case into the adult form
 - C. flies about looking for food
 - D. lays many eggs
- 18.** Which stage of the housefly feeds most and grows fastest?
- A. The larva
 - B. The pupa
 - C. The egg
 - D. The adult
- 19.** The complete life cycle of the housefly may take only about one to three weeks. This is important because
- A. each housefly lives for many years
 - B. only a few eggs are laid at a time
 - C. the housefly cannot spread diseases quickly
 - D. very large numbers of houseflies can be produced in a short time

- 20.** The warm, moist conditions of a refuse dump affect the life cycle of the housefly by
- A. making the stages develop faster
 - B. making the adult unable to fly
 - C. stopping the pupa from forming
 - D. killing the eggs at once
- 21.** Houseflies are mostly found around
- A. refuse dumps, faeces and uncovered food
 - B. clean, dry and well-swept rooms
 - C. the open sea
 - D. deep wells of clear water
- 22.** Which of the following is NOT a feeding place of the housefly?
- A. An open pan of cooked rice
 - B. A dead animal by the roadside
 - C. A sealed tin of milk
 - D. A heap of animal manure
- 23.** The mouth parts of the housefly are suited for
- A. tearing flesh from bones
 - B. sucking up liquid and softened food
 - C. biting and chewing hard grains
 - D. piercing the skin and sucking blood
- 24.** A housefly feeding on solid sugar first
- A. swallows it whole
 - B. bites it into small pieces
 - C. warms it with its wings
 - D. pours saliva on it to dissolve it
- 25.** A housefly carries germs mainly on its
- A. eyes only
 - B. eggs only
 - C. legs, body hairs and mouth parts
 - D. wings only
- 26.** Houseflies move between a refuse dump and the food on a table. This habit is dangerous because
- A. the flies lay eggs that people can see
 - B. the flies make the food cold
 - C. the flies eat all the food
 - D. germs picked up from the refuse are left on the food

- 27.** A gutter behind a chop bar is blocked with food waste and water. The best reason why many houseflies are seen there is that the gutter
- A. has clean water for the flies to drink
 - B. provides food and a breeding place for the flies
 - C. keeps away the enemies of the flies
 - D. is warmer than the air around
- 28.** An insect that lives close to human beings and causes harm and discomfort is described as
- A. a predator
 - B. a menace
 - C. a decomposer
 - D. a pollinator
- 29.** Which of the following diseases can be spread by the housefly?
- A. Dysentery
 - B. Measles
 - C. Ringworm
 - D. Malaria
- 30.** Food that has been visited by houseflies may cause
- A. sickle cell disease
 - B. food poisoning
 - C. night blindness
 - D. asthma
- 31.** Apart from spreading disease, houseflies are a problem because they
- A. bite people at night and suck blood
 - B. destroy clothes in the wardrobe
 - C. are a nuisance in the home and environment
 - D. eat the roots of crops
- 32.** A child in a home where food is left uncovered keeps having diarrhoea. The most likely explanation is that
- A. houseflies have bitten the child
 - B. the food was cooked for too long
 - C. the child drinks too much boiled water
 - D. houseflies have carried germs from faeces or refuse onto the food
- 33.** Which of the following is the strongest evidence that houseflies spread disease in a community?
- A. Houseflies make a buzzing sound while flying
 - B. Houseflies fly faster in the afternoon
 - C. Houseflies are seen in every town
 - D. Communities with poor refuse disposal and many flies record more cases of diarrhoea

- 34.** Which disease of the ones listed is spread by the mosquito and NOT by the housefly?
- A. Malaria
 - B. Dysentery
 - C. Typhoid fever
 - D. Cholera
- 35.** The best single way to stop houseflies from breeding in a compound is to
- A. spray perfume in the rooms each morning
 - B. sweep the compound once a month
 - C. collect and dispose of refuse properly
 - D. keep the windows open all day
- 36.** Covering food with a net cover helps to control the housefly menace because it
- A. keeps the flies away from the food
 - B. prevents the eggs from hatching
 - C. stops the food from getting cold
 - D. kills the flies that come near
- 37.** Which of the following practices will NOT help to reduce the number of houseflies in a community?
- A. Throwing kitchen waste behind the house
 - B. Using a pit latrine with a tight cover
 - C. Clearing choked gutters regularly
 - D. Covering refuse containers with lids
- 38.** Faeces left in the open is a serious health problem mainly because
- A. it attracts rain to the area
 - B. it makes the soil too rich for crops
 - C. houseflies breed in it and carry the germs to food
 - D. it dries up quickly in the sun
- 39.** A Basic 7 class wants to teach their community how to reduce the housefly menace. The MOST effective plan would be to
- A. ask each pupil to kill ten flies at home
 - B. hold a clean-up of the refuse dump and gutters and give a talk on covering food and refuse
 - C. buy insecticide and spray the classroom only
 - D. write the information in their exercise books

- 40.** A household covers its food carefully but still throws its kitchen waste in an open heap beside the kitchen. The weakness of this arrangement is that
- A. the flies still breed in the heap and will reach the food whenever it is opened
 - B. covering food is of no use at all
 - C. the heap will stop the flies from laying eggs
 - D. the covered food will go bad faster

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 14: Living Cells

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The smallest unit of a living organism that can carry out all the activities of life is
 - A. the system
 - B. the cell
 - C. the organ
 - D. the tissue
2. The instrument used in the laboratory to see cells clearly is
 - A. the microscope
 - B. the barometer
 - C. the stethoscope
 - D. the thermometer
3. A thin piece of onion was placed on a glass slide and viewed with a microscope. The many small boxes seen were
 - A. drops of water
 - B. crystals of salt
 - C. grains of sugar
 - D. cells
4. Which of the following statements about living things is correct?
 - A. Only animals are made up of cells
 - B. Both plants and animals are made up of cells
 - C. Only plants are made up of cells
 - D. Neither plants nor animals are made up of cells
5. A group of similar cells working together to perform the same work forms
 - A. an organism
 - B. a tissue
 - C. an organelle
 - D. an organ

- 6.** The small structures found inside a cell, each doing a special work, are called
- A. systems
 - B. tissues
 - C. organs
 - D. organelles
- 7.** Which of the following parts is found in an animal cell?
- A. Cell membrane
 - B. Cell wall
 - C. Chloroplast
 - D. Large central vacuole
- 8.** The part of an animal cell that forms its outer boundary is the
- A. cell wall
 - B. nucleus
 - C. cell membrane
 - D. cytoplasm
- 9.** The jelly-like material in which the organelles of a cell float is the
- A. nucleus
 - B. cell sap
 - C. chloroplast
 - D. cytoplasm
- 10.** The largest organelle in a typical animal cell is usually the
- A. ribosome
 - B. mitochondrion
 - C. nucleus
 - D. vacuole
- 11.** Which of the following is NOT found in an animal cell?
- A. Mitochondrion
 - B. Cell wall
 - C. Cytoplasm
 - D. Nucleus
- 12.** The sausage-shaped organelles found in the cytoplasm of an animal cell are the
- A. nuclei
 - B. chloroplasts
 - C. mitochondria
 - D. cell walls

- 13.** An animal cell has an irregular shape MAINLY because
- A. it has no cell wall to hold it in a fixed shape
 - B. it has no cell membrane
 - C. it has no nucleus
 - D. its cytoplasm is too thick to move
- 14.** A cell was taken from the cheek lining of a goat. The cell would contain
- A. chloroplasts, a cell wall and a nucleus
 - B. a cell wall, cytoplasm and a large vacuole
 - C. a nucleus, cytoplasm and a cell membrane
 - D. a cell wall, chloroplasts and a large vacuole
- 15.** The part of the cell that controls all the activities of the cell is the
- A. nucleus
 - B. cytoplasm
 - C. ribosome
 - D. cell membrane
- 16.** The cell membrane is important because it
- A. stores the water of the cell
 - B. gives the cell its green colour
 - C. controls the substances that enter and leave the cell
 - D. makes food for the cell
- 17.** The organelle in which energy is released from food in a cell is the
- A. mitochondrion
 - B. cell membrane
 - C. vacuole
 - D. nucleus
- 18.** Most of the chemical reactions of a cell take place in the
- A. cell membrane
 - B. cytoplasm
 - C. nucleus
 - D. cell wall
- 19.** Proteins are built up in a cell at the
- A. chloroplasts
 - B. cell membrane
 - C. vacuoles
 - D. ribosomes

- 20.** A muscle cell has very many mitochondria. This is because the muscle cell
- A. has no nucleus of its own
 - B. needs to store a large amount of water
 - C. needs a large amount of energy
 - D. has to make its own food
- 21.** If the nucleus is removed from a living cell, the cell will MOST likely
- A. produce more energy than before
 - B. change into a plant cell
 - C. die, because its activities are no longer controlled
 - D. continue to grow faster than before
- 22.** The outermost part of a plant cell is the
- A. cell membrane
 - B. nucleus
 - C. cytoplasm
 - D. cell wall
- 23.** The cell wall of a plant cell is made MAINLY of
- A. cellulose
 - B. starch
 - C. protein
 - D. fat
- 24.** The green organelles found in the cells of a leaf are the
- A. mitochondria
 - B. ribosomes
 - C. nuclei
 - D. chloroplasts
- 25.** The green colouring matter found inside the chloroplast is
- A. haemoglobin
 - B. cell sap
 - C. cytoplasm
 - D. chlorophyll
- 26.** A large space filled with cell sap in the middle of a mature plant cell is the
- A. nucleus
 - B. cell wall
 - C. vacuole
 - D. chloroplast

- 27.** In a mature plant cell the nucleus is usually found
- A. outside the cell wall
 - B. exactly in the centre of the cell
 - C. inside the large vacuole
 - D. pushed to one side, near the cell membrane
- 28.** Which part of a plant cell would NOT be found in a cell taken from the root of the plant?
- A. Vacuole
 - B. Chloroplast
 - C. Nucleus
 - D. Cell wall
- 29.** The chief function of the cell wall is to
- A. release energy from food
 - B. control what enters and leaves the cell
 - C. give the cell a fixed shape and support
 - D. make food for the plant
- 30.** Food is made in the plant cell in the
- A. chloroplasts
 - B. nucleus
 - C. mitochondria
 - D. vacuole
- 31.** The large vacuole of a plant cell is important because it
- A. makes the proteins of the cell
 - B. stores cell sap and keeps the cell firm
 - C. controls the activities of the cell
 - D. releases energy from food
- 32.** A plant that has not been watered for several days becomes limp MAINLY because
- A. the vacuoles of its cells have lost water
 - B. the cell walls have been destroyed
 - C. the chloroplasts have lost their colour
 - D. the nuclei of the cells have died
- 33.** The cell membrane of a plant cell lies
- A. just inside the cell wall
 - B. just outside the cell wall
 - C. inside the large vacuole
 - D. inside the nucleus

- 34.** A plant cell placed in the dark for a long time would be unable to
- A. keep its shape
 - B. make its own food
 - C. release energy from stored food
 - D. take in water
- 35.** Which of the following parts is found in a plant cell but NOT in an animal cell?
- A. Nucleus
 - B. Cell membrane
 - C. Cytoplasm
 - D. Chloroplast
- 36.** Which parts are found in BOTH plant and animal cells?
- A. Chloroplast, vacuole and cell wall
 - B. Nucleus, cytoplasm and cell membrane
 - C. Cell wall, nucleus and chloroplast
 - D. Cell wall, cytoplasm and chloroplast
- 37.** A pupil looked at a cell under a microscope and saw a cell wall, chloroplasts and a large central vacuole. The cell was
- A. a red blood cell
 - B. a plant cell
 - C. a cheek cell
 - D. an animal cell
- 38.** Plant cells usually have a regular box-like shape while animal cells do not. The reason is that the plant cell has
- A. a firm cell wall outside the cell membrane
 - B. no cytoplasm in its cell
 - C. a larger nucleus than the animal cell
 - D. more mitochondria than the animal cell
- 39.** A pupil was asked to model a plant cell and an animal cell with local materials. The BEST way to show the difference between them would be to
- A. use a larger nucleus in the plant cell model
 - B. colour both models green
 - C. leave out the cytoplasm in the animal cell model
 - D. put a firm box round the model of the plant cell only
- 40.** Which observation would BEST convince a class that both plants and animals are made up of cells?
- A. Feeling the surface of a leaf and the skin of the arm
 - B. Looking at a chart of a plant cell only
 - C. Viewing slides of leaf tissue and cheek tissue and seeing similar small

units in both

- D. Reading in a textbook that living things are made of cells

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 15: Materials

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Anything that has mass and occupies space is called
 - A. a force
 - B. a shadow
 - C. matter
 - D. energy
2. The three states in which materials are commonly found are
 - A. wood, metal and glass
 - B. hard, soft and rough
 - C. natural, artificial and waste
 - D. solid, liquid and gas
3. A pupil described a material as hard, rough and grey, and said that it kept its own shape. The material is MOST likely
 - A. palm oil
 - B. a stone
 - C. cooking gas
 - D. kerosene
4. Which of the following properties would a pupil record by LOOKING at a material only?
 - A. Its smell
 - B. Its colour
 - C. Its temperature
 - D. Its mass
5. Which group contains only solids?
 - A. Chalk, sand, iron nail
 - B. Sand, petrol, oxygen
 - C. Chalk, water, iron nail
 - D. Iron nail, air, palm oil

- 6.** Which of the following is a liquid at room temperature?
- A. Oxygen
 - B. Charcoal
 - C. Groundnut oil
 - D. Common salt
- 7.** A material that has no colour, cannot be held in the hand and spreads to fill the whole room is
- A. a liquid
 - B. a gas
 - C. a solid
 - D. a mixture of solids
- 8.** A material that has a fixed shape and a fixed volume is
- A. a solid
 - B. a gas
 - C. a vapour
 - D. a liquid
- 9.** A liquid has
- A. both a fixed shape and a fixed volume
 - B. a fixed shape but no fixed volume
 - C. neither a fixed volume nor a fixed mass
 - D. a fixed volume but no fixed shape
- 10.** Water poured from a bucket into a bottle takes the shape of the bottle. This shows that a liquid
- A. has no mass of its own
 - B. has no shape of its own
 - C. has no volume of its own
 - D. changes into a gas in the bottle
- 11.** In which state of matter are the particles packed closely together in a regular pattern?
- A. Vapour
 - B. Gas
 - C. Liquid
 - D. Solid
- 12.** The particles of a gas are
- A. far apart and move about freely
 - B. joined in long chains that cannot move
 - C. close together but able to slide over one another
 - D. close together and fixed in position

- 13.** A cylinder of cooking gas feels heavy because
- A. a gas has no mass but the air presses on it
 - B. a gas has mass even though it cannot be seen
 - C. the gas has a fixed shape inside the cylinder
 - D. the gas has turned into a solid inside
- 14.** Which property is shared by solids, liquids and gases?
- A. They can all be compressed easily
 - B. They all have mass and occupy space
 - C. They all have a fixed shape
 - D. They can all be poured from one container into another
- 15.** The liquid that is most important for the survival of humans is
- A. kerosene
 - B. petrol
 - C. water
 - D. palm oil
- 16.** Which of the following is a use of water in the human body?
- A. It hardens the bones of the body
 - B. It changes food into oxygen
 - C. It helps in the digestion and transport of food
 - D. It makes the body produce light
- 17.** Blood is an important liquid in the body because it
- A. carries food and oxygen to all parts of the body
 - B. stores all the fat of the body
 - C. removes bones that are damaged
 - D. keeps the bones stiff and strong
- 18.** A farmer at Techiman uses a liquid obtained from the oil palm fruit for cooking. The liquid is
- A. palm oil
 - B. vinegar
 - C. kerosene
 - D. petrol
- 19.** Which of the following shows the need to preserve liquids in a community?
- A. Pouring left-over drinking water on the ground
 - B. Leaving the tap running while brushing the teeth
 - C. Covering the water in a storage drum and using it carefully
 - D. Washing the compound with pipe-borne water every morning

- 20.** A tanker of petrol is important to a town MAINLY because petrol is used
- A. for drinking during the dry season
 - B. as a fuel in vehicles and machines
 - C. for watering crops on a farm
 - D. for washing cooking utensils
- 21.** Which practice would BEST protect the liquid stored in a water tank at a school?
- A. Keeping the tank covered and cleaning it regularly
 - B. Siting the tank beside the refuse dump
 - C. Allowing pupils to dip their cups into the tank
 - D. Leaving the lid open so that air can enter
- 22.** Which solid material is used MAINLY for roofing buildings in many Ghanaian towns?
- A. Glass
 - B. Cotton wool
 - C. Rubber
 - D. Aluminium sheet
- 23.** A solid obtained from plants and used for making tables and chairs is
- A. clay
 - B. cement
 - C. iron
 - D. wood
- 24.** Common salt is an important solid in the home because it
- A. adds taste to food and supplies minerals the body needs
 - B. makes food cook without fire
 - C. keeps the kitchen cool
 - D. turns food into a liquid
- 25.** Which of the following solids is used by a mason to build a wall?
- A. Palm fronds
 - B. Rubber tyres
 - C. Cement blocks
 - D. Cotton cloth
- 26.** Iron is widely used for making farm tools such as cutlasses MAINLY because it is
- A. shiny and colourful
 - B. cheap and smooth
 - C. hard and strong
 - D. light and soft

- 27.** A Basic 7 class was asked to explain why useful solids in the environment should be preserved. The BEST reason is that
- A. they are found only in the forest zone
 - B. they are limited in supply and are needed by humans and other living things
 - C. they are too heavy to be carried away
 - D. they lose their colour when they are used
- 28.** A substance that cannot be broken down into simpler substances by a chemical method is
- A. an element
 - B. a mixture
 - C. an alloy
 - D. a solution
- 29.** The chemical symbol of sodium is
- A. So
 - B. Sn
 - C. S
 - D. Na
- 30.** The element whose chemical symbol is K is
- A. potassium
 - B. carbon
 - C. calcium
 - D. krypton
- 31.** Which of the following gives the correct symbol of calcium?
- A. Cl
 - B. C
 - C. Cm
 - D. Ca
- 32.** The first element in the periodic table is
- A. oxygen
 - B. helium
 - C. hydrogen
 - D. lithium
- 33.** The elements in the periodic table are arranged in order of increasing
- A. atomic number
 - B. the date on which the element was discovered
 - C. mass of one litre of the element
 - D. the first letter of the name of the element

- 34.** An element has atomic number 8. The element is
- A. sulphur
 - B. carbon
 - C. oxygen
 - D. neon
- 35.** Which of the following is a property of MOST metals?
- A. They break easily into powder
 - B. They are poor conductors of heat
 - C. They are always liquids at room temperature
 - D. They conduct heat and electricity
- 36.** Which of the following elements is a non-metal?
- A. Aluminium
 - B. Sulphur
 - C. Potassium
 - D. Magnesium
- 37.** Helium, neon and argon are grouped together in the periodic table because they
- A. burn easily in air
 - B. are all liquids at room temperature
 - C. are all metals
 - D. are unreactive gases
- 38.** Elements that are found in the same group of the periodic table
- A. have exactly the same mass
 - B. have similar chemical properties
 - C. are all metals
 - D. have the same atomic number
- 39.** Sodium, magnesium and aluminium are all found in Period 3 of the periodic table. This means that they
- A. belong to the same group
 - B. have the same number of protons
 - C. are in the same horizontal row of the table
 - D. are all noble gases
- 40.** A pupil found that a shiny element conducted electricity, could be beaten into sheets and did not break when it was bent. The element is BEST described as
- A. a mixture
 - B. a non-metal
 - C. a noble gas
 - D. a metal

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a labelled sketch or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 16: Science and

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A career in science is BEST described as
 - A. work that needs no training at all
 - B. work done only by teachers
 - C. any work done inside a building
 - D. work in which a person uses scientific knowledge and skills to earn a living
2. A scientist who studies living things such as plants and animals is a
 - A. meteorologist
 - B. geologist
 - C. biologist
 - D. physicist
3. The scientist who studies substances, how they are made up and how they react is a
 - A. biologist
 - B. astronomer
 - C. surveyor
 - D. chemist
4. A person who designs and supervises the building of roads, bridges and dams is a
 - A. veterinary officer
 - B. civil engineer
 - C. nurse
 - D. pharmacist
5. A scientist who treats sick animals on a farm is a
 - A. veterinary doctor
 - B. agronomist
 - C. pharmacist
 - D. radiographer

- 6.** The health worker who prepares and dispenses medicines in a hospital is the
- A. physiotherapist
 - B. pharmacist
 - C. midwife
 - D. laboratory technician
- 7.** A person who studies the weather and prepares the daily forecast is a
- A. food scientist
 - B. meteorologist
 - C. zoologist
 - D. geologist
- 8.** A scientist who works to improve the processing and preservation of foods such as cassava and maize is a
- A. civil engineer
 - B. astronomer
 - C. surveyor
 - D. food scientist
- 9.** Which of the following careers is MOST directly concerned with the study of rocks and minerals?
- A. Food science
 - B. Computer programming
 - C. Nursing
 - D. Geology
- 10.** Professor Francis Allotey, a Ghanaian scientist of international repute, worked chiefly in the field of
- A. surgery and nursing
 - B. music and drama
 - C. mathematics and physics
 - D. law and government
- 11.** Professor Ibok Oduro of KNUST is known for her work in
- A. food science and the use of local food crops
 - B. the mining of gold
 - C. the design of aeroplane engines
 - D. the printing of newspapers
- 12.** Professor Ewurama Addy, a Ghanaian woman scientist, is known for her work in
- A. forecasting the weather
 - B. repairing motor vehicles
 - C. building bridges
 - D. biochemistry and the biological sciences

- 13.** Professor Jophus Anamuah-Mensah is best known in Ghana as a
- A. science educator
 - B. military officer
 - C. professional footballer
 - D. textile trader
- 14.** Professors Theophilus Ossei-Anto and Christian Anthony-Krueger have both contributed to
- A. the building of the Akosombo Dam
 - B. the printing of currency notes
 - C. the teaching and learning of science in Ghana
 - D. the export of cocoa beans
- 15.** Studying the lives of Ghanaian scientists is useful to pupils chiefly because it
- A. helps them to pass examinations without studying
 - B. shows them that Ghanaians can succeed in science and gives them models to follow
 - C. shows that only professors can study science
 - D. proves that science began in Ghana
- 16.** A science educator is a person who
- A. sells science textbooks
 - B. repairs laboratory apparatus
 - C. studies and improves the way science is taught and learnt
 - D. builds laboratories for a living
- 17.** Which institution in Ghana is BEST known for training engineers and scientists?
- A. The Ghana Standards Authority library
 - B. The National Theatre
 - C. The Ghana Police Training School
 - D. Kwame Nkrumah University of Science and Technology
- 18.** Alexander Fleming is remembered for the discovery of
- A. the electric bulb
 - B. the law of gravity
 - C. penicillin, the first antibiotic
 - D. the structure of the atom
- 19.** Penicillin has improved human life chiefly because it
- A. cures bacterial infections that once killed many people
 - B. supplies vitamins to the body
 - C. prevents all accidents on the roads
 - D. removes waste from the environment

- 20.** Albert Einstein is famous for his work in
- A. the classification of plants
 - B. physics, especially the theory of relativity
 - C. the discovery of vaccines
 - D. the invention of the telephone
- 21.** Charles Darwin is remembered for his explanation of
- A. how electricity flows in a wire
 - B. how living things change over time by natural selection
 - C. how the heart pumps blood
 - D. how rain is formed
- 22.** Stephen Hawking was a scientist noted for his study of
- A. the design of motor cars
 - B. the growth of crops
 - C. the universe, including black holes
 - D. the treatment of malaria
- 23.** Louis Pasteur's work on germs led directly to
- A. the discovery of gold in rocks
 - B. the pasteurisation of milk and the use of vaccines
 - C. the building of the first computer
 - D. the invention of the aeroplane
- 24.** The MAIN lesson pupils can learn from the lives of great scientists is that
- A. a scientist must be born outside Africa
 - B. discoveries are made only by accident
 - C. science needs no experiments at all
 - D. careful observation, hard work and curiosity lead to discovery
- 25.** A pupil preparing a presentation on a named scientist should include
- A. the scientist's field of work, the discovery made and how it has helped people
 - B. only the scientist's date of birth
 - C. a list of the scientist's relatives
 - D. the scientist's favourite food
- 26.** Technology is BEST described as
- A. the study of living things only
 - B. the use of scientific knowledge to make useful things and solve problems
 - C. the collection of old machines
 - D. the writing of science examinations

- 27.** The use of mobile phones for money transfer in Ghana is an example of the impact of science and technology on
- A. the movement of the planets
 - B. the growth of trees in the forest
 - C. trade and daily life in the community
 - D. the rainfall of the country
- 28.** A refrigerator improves life in the home chiefly because it
- A. cooks food without fire
 - B. removes salt from food
 - C. makes food richer in protein
 - D. keeps food cold so that it does not spoil quickly
- 29.** The use of solar panels in schools without a reliable supply of electricity shows that science and technology can
- A. remove the need for teachers
 - B. make the Sun shine more brightly
 - C. provide practical solutions to local problems
 - D. increase the number of power cuts
- 30.** The use of tractors, improved seeds and irrigation on farms has MAINLY led to
- A. higher yields and more food for the population
 - B. a fall in the quality of all crops
 - C. the end of rainfall in farming areas
 - D. the loss of all farming jobs
- 31.** Which of the following shows the impact of science in the school?
- A. The sweeping of the compound each morning
 - B. The wearing of school uniform
 - C. The use of computers and projectors in teaching
 - D. The singing of the national anthem
- 32.** A NEGATIVE effect of some modern technology on the community is
- A. easier travel from town to town
 - B. pollution of the air and water by factories and vehicles
 - C. faster communication between people
 - D. better treatment of diseases
- 33.** Innovation in science means
- A. memorising scientific facts
 - B. copying a classmate's work
 - C. repeating exactly what others have done
 - D. putting a new or improved idea into practical use

- 34.** Ghana needs to train more agricultural scientists MAINLY in order to
- A. stop the export of cocoa
 - B. reduce the number of farms in the country
 - C. increase food production and reduce the importation of food
 - D. increase the price of fertiliser
- 35.** Training more medical and laboratory scientists in Ghana would MOST directly
- A. improve the diagnosis and treatment of diseases in the country
 - B. lower the quality of local foods
 - C. reduce the number of schools
 - D. increase the rainfall of the country
- 36.** A reason for Ghana to focus on careers in renewable energy is that they would help the country to
- A. use up its crude oil more quickly
 - B. produce cleaner electricity and depend less on imported fuel
 - C. stop the use of all machines
 - D. increase the smoke in the cities
- 37.** A pupil who wishes to become an engineer should pay particular attention to
- A. science and mathematics
 - B. handwriting only
 - C. games and sports only
 - D. memorising long poems
- 38.** The BEST way for a Basic 7 pupil to prepare for a career in science is to
- A. take part in practical work, read widely and ask questions
 - B. wait until senior high school before studying science
 - C. avoid difficult topics
 - D. copy answers from friends
- 39.** A school science club wishing to learn about careers in science would benefit MOST from
- A. reading only the names of scientists
 - B. collecting money from pupils
 - C. holding a football competition each week
 - D. inviting scientists and visiting laboratories, farms and factories
- 40.** The MAIN purpose of building a portfolio on the impact of science and technology is to
- A. gather and organise evidence of what has been found out, for presentation
 - B. keep the findings away from other pupils
 - C. replace the class textbook
 - D. avoid doing any research

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 17: Solar Sys

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. The solar system is made up of
 - A. the Earth and the Moon only
 - B. all the stars in the sky
 - C. the clouds and the air above the Earth
 - D. the Sun and all the bodies that move round it
2. At the centre of the solar system is
 - A. the Sun
 - B. Mercury
 - C. the Earth
 - D. the Moon
3. The Sun is best described as
 - A. a moon
 - B. a star
 - C. a cloud of water
 - D. a planet
4. The Sun is important to the solar system because it
 - A. moves round the Earth each day
 - B. supplies the planets with water
 - C. reflects light from the planets
 - D. supplies light and heat and holds the planets in their paths
5. A planet differs from a star in that a planet
 - A. gives out its own light and heat
 - B. is always nearer to the Earth than the Moon
 - C. is always larger than a star
 - D. does not give out its own light but shines by reflecting sunlight
6. The force that keeps the planets moving round the Sun instead of flying away is
 - A. the heat of the Sun
 - B. the magnetism of the Earth
 - C. the gravitational pull of the Sun
 - D. the push of the wind in space

- 7.** Which of the following is NOT a member of the solar system?
- A. The Moon
 - B. The Milky Way
 - C. Mars
 - D. The Sun
- 8.** The planets nearest to the Sun are called
- A. the outer planets
 - B. the stars
 - C. the satellites
 - D. the inner planets
- 9.** How many inner planets are there in the solar system?
- A. Six
 - B. Four
 - C. Two
 - D. Eight
- 10.** The inner planets, in order from the Sun, are
- A. Venus, Mercury, Mars, Earth
 - B. Mercury, Venus, Earth, Mars
 - C. Mercury, Earth, Venus, Mars
 - D. Earth, Mars, Mercury, Venus
- 11.** The inner planets are also called the rocky planets because they
- A. are covered with ice
 - B. are made mainly of gas
 - C. have solid, rocky surfaces
 - D. are made of burning rock
- 12.** The planet nearest to the Sun is
- A. Earth
 - B. Mars
 - C. Venus
 - D. Mercury
- 13.** The fourth planet from the Sun is
- A. Earth
 - B. Venus
 - C. Mercury
 - D. Mars

- 14.** Which inner planet is known to support life?
- A. Earth
 - B. Mars
 - C. Venus
 - D. Mercury
- 15.** Mars is often called the red planet because
- A. it is very close to the Sun
 - B. it is the hottest planet
 - C. the rocks and dust on its surface are reddish in colour
 - D. it gives out red light of its own
- 16.** Which of the following is a property shared by all the inner planets?
- A. They all have many moons
 - B. They all have rings around them
 - C. They are all small and have solid surfaces
 - D. They are all covered with liquid water
- 17.** A very large group of stars, dust and gas held together by gravity is called
- A. a planet
 - B. a solar system
 - C. a satellite
 - D. a galaxy
- 18.** The galaxy in which our solar system is found is called
- A. the Milky Way
 - B. the Great Bear
 - C. the Orion Belt
 - D. the Southern Cross
- 19.** On a clear, dark night away from town lights, the Milky Way appears as
- A. a faint band of light stretching across the sky
 - B. a single very bright star
 - C. a line of moving lights
 - D. a round red object
- 20.** Which of the following statements about the Milky Way is correct?
- A. The Milky Way is the path followed by the Earth
 - B. The Milky Way is found inside the solar system
 - C. The solar system is only a very small part of the Milky Way
 - D. The Milky Way is another name for the Sun

- 21.** Arrange the following from the smallest to the largest: galaxy, planet, solar system.
- A. Planet, galaxy, solar system
 - B. Planet, solar system, galaxy
 - C. Galaxy, solar system, planet
 - D. Solar system, planet, galaxy
- 22.** Street lights make it difficult to see the Milky Way from a town in Ghana because
- A. tall buildings block the light of the stars
 - B. the Milky Way moves away from towns
 - C. the bright lights of the town drown the faint light from the stars
 - D. the Milky Way can be seen only in the rainy season
- 23.** The shape of the path along which a planet moves round the Sun is
- A. elliptical (oval)
 - B. a perfect circle
 - C. a square
 - D. a straight line
- 24.** The path along which a planet travels round the Sun is called its
- A. axis
 - B. orbit
 - C. equator
 - D. satellite
- 25.** The turning of a planet on its own axis is called
- A. rotation
 - B. radiation
 - C. revolution
 - D. reflection
- 26.** The movement of a planet round the Sun is called
- A. orbit tilting
 - B. rotation
 - C. vibration
 - D. revolution
- 27.** The rotation of the Earth on its axis causes
- A. the seasons of the year
 - B. the tides of the sea
 - C. day and night
 - D. the phases of the Moon

- 28.** A planet that is far from the Sun takes a longer time to go round the Sun than one that is near. This is mainly because the far planet
- A. has a much longer path to travel
 - B. stops from time to time on the way
 - C. is pulled backwards by the stars
 - D. is heavier than the near planet
- 29.** One complete revolution of the Earth round the Sun takes about
- A. 24 hours
 - B. 365 days
 - C. 88 days
 - D. 30 days
- 30.** Which of the following is a property of Mercury?
- A. It is the smallest of the inner planets and has almost no atmosphere
 - B. It has a thick atmosphere of oxygen
 - C. It is covered with large oceans of water
 - D. It is the largest planet in the solar system
- 31.** Mercury has a very hot day side and a very cold night side mainly because it
- A. is far from the Sun
 - B. turns very fast on its axis
 - C. is covered with white ice
 - D. has almost no atmosphere to hold in the heat
- 32.** Mercury takes about 88 Earth days to go round the Sun once. This means that Mercury
- A. does not move round the Sun at all
 - B. has a much shorter year than the Earth
 - C. has a much longer year than the Earth
 - D. has a year of the same length as that of the Earth
- 33.** Venus is the hottest planet in the solar system because
- A. it is the planet nearest to the Sun
 - B. its thick atmosphere traps the heat from the Sun
 - C. it gives out heat of its own
 - D. it is made of burning coal
- 34.** Venus is often called the morning star or the evening star because it
- A. appears only once each year
 - B. is the brightest star in the sky
 - C. can be seen shining brightly near sunrise or sunset
 - D. gives out its own light at dawn

- 35.** Which statement about the sizes of Venus and the Earth is correct?
- A. Venus is nearly the same size as the Earth
 - B. Venus is the largest planet in the solar system
 - C. Venus is many times larger than the Earth
 - D. Venus is smaller than the Moon
- 36.** Mercury and Venus are alike in that both
- A. have surfaces covered with water
 - B. have many moons
 - C. are larger than the Earth
 - D. move round the Sun in paths inside that of the Earth
- 37.** A model of the solar system is useful mainly because it
- A. helps us to see the arrangement and movement of bodies that are too far away to visit
 - B. can be used to measure the heat of the Sun
 - C. shows the true sizes and distances of the planets
 - D. makes the planets move faster
- 38.** In building a model of the solar system from balls of different sizes, the largest ball should stand for
- A. Venus
 - B. the Earth
 - C. the Sun
 - D. Mercury
- 39.** A pupil at Koforidua made a model in which the four inner planets were placed in a straight line beside the Sun. The main fault in this model is that
- A. the planets should be bigger than the Sun
 - B. the Sun should be placed at the edge of the model
 - C. the planets move round the Sun along separate paths and are not in a line
 - D. the planets should all be the same size
- 40.** Which material would be the MOST suitable for showing the path of a planet in a model of the solar system?
- A. A square piece of cardboard
 - B. A loop of wire bent into an oval shape
 - C. A straight stick
 - D. A short piece of rope tied in a knot

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 18: Understanding the

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. A landform is BEST described as
 - A. a map drawn by a surveyor
 - B. a kind of soil used for farming
 - C. a natural feature of the surface of the Earth, such as a mountain or a plain
 - D. a building put up by people
2. A wide stretch of flat, low land is called a
 - A. plateau
 - B. valley
 - C. mountain
 - D. plain
3. A piece of high land with a broad, flat top is a
 - A. plateau
 - B. valley
 - C. delta
 - D. plain
4. The low land lying between two hills or mountains is a
 - A. plateau
 - B. plain
 - C. peak
 - D. valley
5. The highest mountain in Ghana is
 - A. Mount Afadja (Afadjato)
 - B. Mount Kenya
 - C. Mount Kilimanjaro
 - D. Mount Cameroon

- 6.** The Accra Plains and the Afram Plains are examples of the landform called the
- A. plateau
 - B. plain
 - C. valley
 - D. mountain
- 7.** Which group of animals would one expect to find MOST commonly on a grassy plain?
- A. bats, baboons and rock hyraxes
 - B. grasscutters, antelopes and cattle
 - C. crabs, oysters and starfish
 - D. tilapia, catfish and mudfish
- 8.** Mangrove plants and crabs are found MAINLY in
- A. the roof of a cave
 - B. the middle of a desert
 - C. dry rocky hill tops
 - D. muddy ground along the coast and river mouths
- 9.** The special features that help a living thing to survive where it lives are called
- A. habitats
 - B. associations
 - C. adaptations
 - D. competitions
- 10.** The thick fur of an animal living on a cold mountain helps it to
- A. keep its body warm
 - B. store water in its body
 - C. find its food in the dark
 - D. run faster on the rocks
- 11.** Mountain goats and rock hyraxes can move about on steep rocky slopes because they have
- A. webbed feet for swimming
 - B. strong legs and hooves or pads that grip the rock
 - C. wide flat feet for walking on mud
 - D. long tails for hanging from trees
- 12.** The long legs of the egret that feeds in a marshy plain help it to
- A. dig deep holes in dry soil
 - B. climb tall trees quickly
 - C. wade through shallow water without wetting its body
 - D. fly faster than other birds

- 13.** An antelope living on an open plain escapes from its enemies MAINLY by
- A. swimming across the plain
 - B. burrowing into the rock
 - C. changing into another animal
 - D. running very fast on its long slender legs
- 14.** A fish living in a fast-flowing stream in a valley is helped to stay in place by its
- A. streamlined body and strong fins
 - B. hard shell and eight legs
 - C. hollow bones and feathers
 - D. thick fur and long ears
- 15.** The sandy colour of many animals of the dry northern plains is useful because it
- A. makes them attractive to hunters
 - B. helps them to dig faster
 - C. helps them to blend with their surroundings and escape notice
 - D. keeps insects away from them
- 16.** Burrowing into the ground during the hottest part of the day helps a small animal of the plains to
- A. grow a longer tail
 - B. find more sunlight
 - C. increase the loss of water from its body
 - D. avoid the heat and reduce the loss of water from its body
- 17.** Plants growing in a dry, open plain often have long roots in order to
- A. catch insects for food
 - B. hold leaves above the clouds
 - C. produce more flowers in the dry season
 - D. reach water deep in the soil
- 18.** The small, needle-like leaves of some plants in dry places help them to
- A. lose more water by transpiration
 - B. lose less water by transpiration
 - C. trap more rain water
 - D. store more seeds
- 19.** The thick fleshy stem of a cactus-like plant is useful for
- A. killing insects
 - B. storing water
 - C. attracting rainfall
 - D. storing oxygen

- 20.** Mangrove plants growing in soft muddy soil are supported by
- A. a single straight tap root only
 - B. woody stems that lie flat in the mud
 - C. floating leaves
 - D. stilt (prop) roots that spread out above the mud
- 21.** Trees of the thick forest in a wet valley are usually
- A. short, with needle-like leaves
 - B. tall, with broad leaves that catch the sunlight
 - C. leafless throughout the year
 - D. grown only in sandy soil
- 22.** Grasses survive the yearly bush fires of the savanna plains MAINLY because they
- A. grow taller than the flames
 - B. are too wet to burn at all
 - C. have underground parts that sprout again after the fire
 - D. are protected by their flowers
- 23.** Some plants on rocky hillsides grow close to the ground. This helps them to
- A. lose more water quickly
 - B. resist being blown away by strong winds
 - C. receive less sunlight
 - D. escape from grazing animals completely
- 24.** A plant that grows on a plateau with shallow, stony soil is likely to have
- A. roots that spread out widely near the surface
 - B. no roots at all
 - C. roots that float in water
 - D. one very deep root only
- 25.** An association in which BOTH living things benefit is called
- A. predation
 - B. parasitism
 - C. commensalism
 - D. mutualism
- 26.** In parasitism, the parasite
- A. is harmed while the host benefits
 - B. and the host are both harmed
 - C. benefits while the host is harmed
 - D. and the host both benefit equally

- 27.** An association in which one living thing benefits while the other is neither helped nor harmed is
- A. mutualism
 - B. parasitism
 - C. competition
 - D. commensalism
- 28.** A bee visiting a flower for nectar and carrying pollen to the next flower is an example of
- A. predation
 - B. mutualism
 - C. parasitism
 - D. commensalism
- 29.** A tapeworm living in the intestine of a goat shows
- A. mutualism
 - B. competition
 - C. commensalism
 - D. parasitism
- 30.** A cattle egret that feeds on the insects disturbed by a grazing cow, without affecting the cow, shows
- A. mutualism
 - B. commensalism
 - C. parasitism
 - D. predation
- 31.** Which of the following is a parasite of a flowering plant?
- A. an earthworm in the soil below the tree
 - B. a bee resting on a flower
 - C. mistletoe growing on the branch of a tree
 - D. a bird building a nest in the branches
- 32.** Root nodule bacteria living in the roots of a groundnut plant supply the plant with nitrogen and receive food from it. This is
- A. commensalism
 - B. a food chain
 - C. mutualism
 - D. parasitism
- 33.** A lion hunting and killing an antelope is BEST described as
- A. predation
 - B. mutualism
 - C. commensalism
 - D. parasitism

- 34.** The place where a plant or an animal naturally lives is its
- A. habitat
 - B. population
 - C. adaptation
 - D. association
- 35.** Two goats feeding on the same small patch of grass are said to be in
- A. parasitism
 - B. competition
 - C. mutualism
 - D. commensalism
- 36.** A chameleon changing its colour to match the branch it rests on is an example of
- A. camouflage
 - B. pollination
 - C. hibernation
 - D. migration
- 37.** Birds flying from one region to another when the dry season comes are showing
- A. migration
 - B. camouflage
 - C. germination
 - D. parasitism
- 38.** A pupil studying the animals of a nearby valley should FIRST
- A. copy the names of animals from a friend
 - B. write the report before the visit
 - C. visit the valley and observe and record the animals seen there
 - D. collect money from the class
- 39.** Which of the following is the BEST record to keep in an album of the plants and animals of a landform?
- A. a labelled picture or drawing of each one, with the place and date it was seen
 - B. the price of each animal in the market
 - C. a list of names only, without any detail
 - D. the names of the pupils in the class only
- 40.** If the trees of a valley are all cut down, the animals living there will MOST likely
- A. change into plants
 - B. grow larger and multiply quickly
 - C. lose their shelter and food and move away or die
 - D. remain exactly as they were

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 19: Waste Management

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Waste is BEST described as
 - A. any material that is sold in the market
 - B. any material that is buried in the soil
 - C. any material that is no longer wanted by the person who produced it
 - D. only the rubbish swept from a room
2. Peels, leftover food and dry leaves are examples of
 - A. metallic waste
 - B. electronic waste
 - C. organic (biodegradable) waste
 - D. chemical waste
3. Water sachets, plastic bottles and rubber bags are examples of
 - A. organic waste
 - B. metallic waste
 - C. glass waste
 - D. plastic waste
4. Empty milk tins and old roofing sheets are examples of
 - A. liquid waste
 - B. paper waste
 - C. plastic waste
 - D. metallic waste
5. Used needles, soiled bandages and expired drugs from a clinic are classified as
 - A. paper waste
 - B. domestic waste
 - C. medical waste
 - D. agricultural waste
6. Dirty water from a bathroom and a kitchen is an example of
 - A. gaseous waste
 - B. toxic metallic waste
 - C. solid waste
 - D. liquid waste

- 7.** Smoke from a lorry and fumes from a factory chimney are examples of
- A. gaseous waste
 - B. liquid waste
 - C. solid waste
 - D. organic waste
- 8.** Spoilt mobile phones, old radios and broken television sets are known as
- A. medical waste
 - B. electronic waste
 - C. agricultural waste
 - D. paper waste
- 9.** Biodegradable waste is waste that
- A. cannot be broken down at all
 - B. can be burnt in an open fire
 - C. can be broken down by micro-organisms into simple substances
 - D. can be sold to a recycling factory
- 10.** Which of the following wastes is NON-biodegradable?
- A. A sheet of old newspaper
 - B. A plastic water sachet
 - C. A banana peel
 - D. Maize husks
- 11.** The organisms chiefly responsible for breaking down biodegradable waste are
- A. birds and bats
 - B. bacteria and fungi
 - C. fish and frogs
 - D. grasshoppers and bees
- 12.** A plastic bottle buried in the soil is still whole after twenty years because plastic
- A. absorbs water from the soil
 - B. is a living material
 - C. is heavier than soil
 - D. is not broken down by micro-organisms
- 13.** Waste that rots and gives off a bad smell is MOST likely to be
- A. food remains and other organic waste
 - B. broken glass
 - C. aluminium cans
 - D. pieces of rubber tyre

- 14.** Heaps of rotting refuse near homes are dangerous mainly because they
- A. attract rain to the area
 - B. lower the temperature of the air
 - C. make the soil too rich for crops
 - D. breed houseflies, mosquitoes and rats that spread disease
- 15.** Choked gutters filled with plastic waste in a town lead directly to
- A. an increase in soil fertility
 - B. flooding when it rains
 - C. a fall in the price of plastics
 - D. cleaner drinking water
- 16.** Burning rubber and plastic waste in the open is discouraged because it
- A. cools the surrounding air
 - B. destroys the waste completely with no effect
 - C. releases poisonous gases and smoke into the air
 - D. makes the plastics biodegradable
- 17.** The three Rs of good waste management are
- A. remove, rinse and rest
 - B. reduce, reuse and recycle
 - C. read, record and report
 - D. repair, rebuild and resell
- 18.** Carrying a basket to the market instead of collecting several plastic bags is an example of
- A. recycling waste
 - B. burying waste
 - C. reducing waste
 - D. burning waste
- 19.** Using an empty tin as a container for seedlings is an example of
- A. recycling waste
 - B. reducing waste
 - C. dumping waste
 - D. reusing waste
- 20.** Collecting plastic bottles and sending them to a factory where they are melted and made into new articles is
- A. recycling
 - B. incineration
 - C. composting
 - D. reusing

- 21.** Separating waste into different containers at the point where it is produced is called
- A. waste recycling
 - B. waste burning
 - C. waste collection
 - D. waste segregation (sorting)
- 22.** The MAIN advantage of sorting waste at home is that it
- A. makes it easier to compost, reuse or recycle each kind of waste
 - B. removes the need for dustbins
 - C. makes the waste disappear
 - D. increases the total amount of waste
- 23.** Household refuse kept in a covered container is better than refuse left in an open pan because the cover
- A. keeps out flies and prevents bad smells from spreading
 - B. makes the refuse rot faster
 - C. turns the refuse into compost at once
 - D. adds weight to the refuse
- 24.** Treating waste in a properly designed incinerator is better than open burning because the incinerator
- A. uses no fuel at all
 - B. burns the waste at a high temperature and controls the gases released
 - C. turns all the waste into drinking water
 - D. produces no ash
- 25.** Which of the following is the BEST method of disposing of broken glass at home?
- A. Wrapping it securely and placing it in a bin for hard waste
 - B. Putting it loose into a sack of food waste
 - C. Burying it in the vegetable bed
 - D. Throwing it into the gutter
- 26.** Composting is the process by which
- A. waste water is boiled to make it pure
 - B. plastic waste is melted into new bottles
 - C. organic waste is allowed to decay to form manure
 - D. metal waste is beaten into new tools
- 27.** The materials MOST suitable for making compost are
- A. old batteries and spoilt phones
 - B. grass cuttings, peels, dry leaves and animal droppings
 - C. tins and broken glass
 - D. water sachets and plastic bottles

- 28.** Water is sprinkled on a compost heap in order to
- A. wash away the bad smell
 - B. cool the heap and stop decay
 - C. keep it moist so that the micro-organisms can act
 - D. make the heap heavier
- 29.** A compost heap is turned over from time to time mainly to
- A. allow air to reach all parts of the heap
 - B. remove the manure formed
 - C. make the heap look tidy
 - D. stop the micro-organisms from working
- 30.** The warmth felt inside a compost heap is evidence that
- A. the Sun has heated the whole heap
 - B. micro-organisms are active and respiration is taking place
 - C. water is freezing inside the heap
 - D. the waste is burning with a flame
- 31.** Compost added to garden soil is useful because it
- A. kills all the organisms in the soil
 - B. turns the soil into sand
 - C. supplies plant nutrients and improves the structure of the soil
 - D. makes the soil hold no water
- 32.** The use of a compost pit to treat kitchen waste shows that waste can be
- A. changed into plastic
 - B. stored for ever without change
 - C. made to disappear completely
 - D. turned into a useful resource
- 33.** Dumping refuse into a stream used by a village is dangerous because it
- A. pollutes the water and can cause diseases such as cholera
 - B. increases the number of fish
 - C. purifies the water naturally
 - D. makes the stream flow faster
- 34.** A common disease spread by houseflies that breed on uncovered refuse is
- A. sickle-cell disease
 - B. typhoid fever
 - C. night blindness
 - D. ringworm

- 35.** Mosquitoes that spread malaria breed MOST easily in
- A. water collected in discarded tins and old tyres
 - B. dry heaps of sand
 - C. burning rubbish
 - D. swept and clean compounds
- 36.** Before carrying out a clean-up project in the community, pupils should FIRST
- A. research and plan the work and obtain permission from the authorities
 - B. begin by burning all the waste they find
 - C. clean only their own homes
 - D. collect money from passers-by
- 37.** The MOST suitable protective items for pupils taking part in a clean-up exercise are
- A. sunglasses and wrist watches
 - B. hand gloves, nose masks and boots
 - C. school bags and books
 - D. sandals and short sleeves
- 38.** At the end of a waste management project, the pupils should
- A. keep the findings a secret
 - B. forget about the project at once
 - C. repeat exactly the same work the next day
 - D. write and present a report on what they did and found
- 39.** The BEST way for a school to keep its compound permanently clean is to
- A. burn all waste at the back of the classrooms
 - B. ask pupils to carry their waste home every day
 - C. provide labelled bins and agree on a routine for emptying and sorting them
 - D. sweep only when visitors are expected
- 40.** A community that practises good waste management will MOST likely experience
- A. more flooding in the rainy season
 - B. an increase in malaria cases
 - C. a greater number of refuse heaps
 - D. fewer diseases and a cleaner environment

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

Paper 20: Water Cycle

Time Allowed: 2 hours (Suggested: Section A - 40 min; Section B - 80 min) | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark. Choose the BEST answer and write its letter (A, B, C or D) against the question number on your answer sheet.

1. Which of the following is a natural source of water?
 - A. A water tanker
 - B. A sachet of water
 - C. A tap
 - D. A river
2. The largest natural store of water on the earth is
 - A. the sea
 - B. the stream
 - C. the well
 - D. the lagoon
3. Water that collects under the ground and is reached by digging a well is called
 - A. rain water
 - B. sea water
 - C. ground water
 - D. pipe water
4. Lake Bosomtwe, the Volta Lake and the Pra River are all examples of
 - A. underground streams
 - B. sources of sea water
 - C. man-made reservoirs only
 - D. surface water bodies
5. Which of the following best explains why the water in the sea cannot be drunk straight away?
 - A. It contains a large amount of dissolved salt
 - B. It never evaporates
 - C. It is always too cold
 - D. It contains no dissolved substances

- 6.** The change of water from the liquid state into water vapour is called
- A. precipitation
 - B. evaporation
 - C. condensation
 - D. melting
- 7.** The main source of the energy that drives the water cycle is
- A. the moon
 - B. the wind
 - C. the soil
 - D. the sun
- 8.** Clothes spread on a line in the open dry quickly because the water in them
- A. condenses on the cloth
 - B. freezes in the air
 - C. evaporates into the air
 - D. sinks into the ground
- 9.** The loss of water vapour from the leaves of plants to the atmosphere is called
- A. respiration
 - B. transpiration
 - C. condensation
 - D. germination
- 10.** Which of the following conditions will make water in an open pan evaporate fastest?
- A. A shaded, humid evening
 - B. A cool, still night
 - C. A hot, windy afternoon
 - D. A cold, rainy morning
- 11.** Two equal volumes of water were poured into a wide basin and a narrow bottle and left in the sun. The water in the basin disappeared faster because
- A. it had a larger surface exposed to the air
 - B. it was heavier than the water in the bottle
 - C. it was colder than the water in the bottle
 - D. it contained more dissolved air
- 12.** During evaporation from the sea, the salt in the sea water
- A. is left behind in the sea
 - B. falls back as salty rain
 - C. rises with the water vapour
 - D. changes into water vapour

- 13.** A farmer at Techiman covered the soil around his young plants with dry grass. This practice mainly reduces
- A. condensation in the air
 - B. transpiration from the leaves
 - C. precipitation over the farm
 - D. evaporation of water from the soil
- 14.** The change of water vapour into tiny drops of liquid water is called
- A. evaporation
 - B. transpiration
 - C. precipitation
 - D. condensation
- 15.** Water vapour rising into the atmosphere condenses mainly because the air above
- A. becomes warmer with height
 - B. moves faster with height
 - C. becomes cooler with height
 - D. contains no dust
- 16.** Clouds are made up of
- A. pure oxygen gas
 - B. tiny drops of water and ice crystals
 - C. smoke from factories
 - D. dry dust from the desert
- 17.** Drops of water form on the outside of a cold bottle of mineral water left on a table because
- A. the glass melts a little
 - B. water leaks out through the glass
 - C. the water inside evaporates through the glass
 - D. water vapour in the air condenses on the cold surface
- 18.** Dew that is seen on grass early in the morning is formed by
- A. condensation of water vapour on the cool grass
 - B. light rain that falls at dawn
 - C. evaporation of water from the soil
 - D. transpiration by the grass at night
- 19.** In the water cycle, condensation takes place
- A. on the surface of the sea
 - B. high in the atmosphere where the air is cool
 - C. inside the roots of plants
 - D. deep inside the soil

- 20.** Which process is the direct opposite of evaporation?
- A. Condensation
 - B. Precipitation
 - C. Infiltration
 - D. Transpiration
- 21.** Water that falls from clouds to the earth as rain, hail or snow is known as
- A. transpiration
 - B. precipitation
 - C. evaporation
 - D. condensation
- 22.** Rain falls from a cloud when the water drops in the cloud
- A. join together and become too heavy to stay in the air
 - B. become warmer than the air around them
 - C. change back into water vapour
 - D. are pushed up by the wind
- 23.** The commonest form of precipitation in Ghana is
- A. hail
 - B. rain
 - C. sleet
 - D. snow
- 24.** Rain water that flows over the surface of the land into streams and rivers is called
- A. condensation
 - B. transpiration
 - C. run-off
 - D. evaporation
- 25.** The movement of rain water downwards into the soil is called
- A. evaporation
 - B. transpiration
 - C. infiltration
 - D. precipitation
- 26.** After a heavy downpour at Kumasi, most of the rain water that fell on the tarred roads
- A. soaked into the tar
 - B. was taken up by the roads as ground water
 - C. evaporated immediately
 - D. ran off quickly into gutters and streams

- 27.** Flooding is common in towns where many open spaces have been covered with concrete, mainly because
- A. less water soaks into the ground, so run-off increases
 - B. plants take up less water from the air
 - C. more water evaporates from the concrete
 - D. concrete causes heavier rainfall
- 28.** The correct order of the stages of the water cycle is
- A. collection, precipitation, condensation, evaporation
 - B. evaporation, condensation, precipitation, collection
 - C. condensation, evaporation, collection, precipitation
 - D. precipitation, evaporation, condensation, collection
- 29.** The water cycle is described as a repeated pattern of change in nature because
- A. it happens only in the rainy season
 - B. it takes place once in every year
 - C. new water is made each time it rains
 - D. the same stages follow one another again and again without ending
- 30.** The total amount of water on the earth
- A. decreases each time the sun shines
 - B. remains about the same, as the water is used over and over again
 - C. increases each time it rains
 - D. is made fresh by plants each year
- 31.** Which of the following returns water from the land back to the sea, completing the cycle?
- A. Rivers and streams
 - B. Wind
 - C. Dew
 - D. Clouds
- 32.** A pupil drew a flow chart of the water cycle but ended it at precipitation. The chart is incomplete because it fails to show that
- A. clouds are made of water drops
 - B. rain water is always clean
 - C. the water returns to the sources from which it evaporates again
 - D. plants also lose water
- 33.** The water that a pupil drinks today may once have been part of a river that flowed many years ago. This is possible because
- A. fresh water is created by rainfall
 - B. water does not change state
 - C. water is used again and again in the water cycle

D. rivers never dry up

34. The water cycle is important to farmers mainly because it

- A. removes weeds from the farm
- B. provides rain for the growth of crops
- C. makes the soil harder
- D. provides salt for the soil

35. As water vapour condenses in the atmosphere, energy is released. This released energy helps to

- A. form salt in the air
- B. increase run-off on land
- C. warm the surrounding environment
- D. cool the sea water

36. The water cycle raises the water table of an area by

- A. allowing rain water to soak into the ground and collect below the surface
- B. keeping the soil surface dry
- C. carrying water away in rivers
- D. increasing evaporation from the sea

37. Water moving through the cycle is described as a carrier of nutrients because it

- A. stores nutrients only in clouds
- B. destroys the nutrients it meets
- C. turns mineral salts into gas
- D. dissolves mineral salts and carries them to soils, rivers and living things

38. Which of the following best explains how the water cycle helps to provide clean water for a community?

- A. Water is boiled in the clouds before it falls
- B. Water that evaporates leaves its dissolved salts and dirt behind and returns as fairly pure rain
- C. Clouds filter out the water and keep the clean part
- D. Rain water collects salt from the air as it falls

39. A town has cut down the trees on the hills around it and the rains have become less reliable. The most likely reason is that

- A. the trees used to block the clouds
- B. the soil now holds too much water
- C. trees prevent water from evaporating from rivers
- D. with fewer trees there is less transpiration, so less water vapour enters the air

- 40.** A community wants to reduce the flooding that follows every heavy rain while also raising its water table. The BEST action to take is to
- A. pump the flood water into the nearest stream each time
 - B. build higher walls round the houses
 - C. plant trees and grasses and keep open ground so that rain water soaks in
 - D. pave all the open ground with concrete

SECTION B: THEORY

Answer ALL questions. Write your answers in your answer booklet. The marks for each question are shown in brackets.

Question 1

- (a) Write your name and class clearly in the spaces provided on your answer sheet.

Question 2

- (a) This paper has TWO sections: Section A and Section B.

Question 3

- (a) Answer ALL the questions in both sections.

Question 4

- (a) Section A has 40 multiple-choice questions. Each question carries 1 mark. Choose the best answer (A, B, C or D) and record it on the answer sheet provided.

Question 5

- (a) Section B has 6 theory questions totalling 60 marks. Write your answers clearly and neatly in your answer booklet. Study the figures carefully before you answer the questions on them.

Question 6

- (a) Use a pencil and a ruler where a diagram, a flow chart or a table is required. Do not use correction fluid.

Question 7

- (a) Hand in your answer sheet and your answer booklet at the end of the examination.

ANSWER KEY

and Marking Schemes

This section contains the correct answers for all multiple-choice questions and detailed marking schemes for all theory questions.

Agricultural Tool

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	15.1 Simple Agricultural Tools and Their Uses	Remember
2	D	15.1 Simple Agricultural Tools and Their Uses	Remember
3	B	15.1 Simple Agricultural Tools and Their Uses	Remember
4	A	15.1 Simple Agricultural Tools and Their Uses	Understand
5	B	15.1 Simple Agricultural Tools and Their Uses	Remember
6	D	15.1 Simple Agricultural Tools and Their Uses	Understand
7	A	15.1 Simple Agricultural Tools and Their Uses	Remember
8	C	15.1 Simple Agricultural Tools and Their Uses	Analyse
9	B	15.1 Simple Agricultural Tools and Their Uses	Remember
10	B	15.1 Simple Agricultural Tools and Their Uses	Remember
11	D	15.2 The Meaning and Importance of Handlin...	Remember
12	D	15.2 The Meaning and Importance of Handlin...	Remember
13	C	15.2 The Meaning and Importance of Handlin...	Analyse
14	D	15.2 The Meaning and Importance of Handlin...	Understand
15	C	15.2 The Meaning and Importance of Handlin...	Understand
16	D	15.2 The Meaning and Importance of Handlin...	Understand
17	A	15.2 The Meaning and Importance of Handlin...	Analyse
18	C	15.3 Basic Rules for Handling Tools Safely	Understand
19	A	15.3 Basic Rules for Handling Tools Safely	Understand
20	A	15.3 Basic Rules for Handling Tools Safely	Apply
21	D	15.3 Basic Rules for Handling Tools Safely	Understand
22	C	15.3 Basic Rules for Handling Tools Safely	Apply
23	D	15.3 Basic Rules for Handling Tools Safely	Remember
24	A	15.3 Basic Rules for Handling Tools Safely	Apply
25	B	15.3 Basic Rules for Handling Tools Safely	Apply
26	D	15.4 Maintenance Practices for Particular ...	Understand
27	B	15.4 Maintenance Practices for Particular ...	Remember
28	D	15.4 Maintenance Practices for Particular ...	Understand
29	C	15.4 Maintenance Practices for Particular ...	Understand
30	B	15.4 Maintenance Practices for Particular ...	Apply
31	C	15.4 Maintenance Practices for Particular ...	Remember
32	A	15.4 Maintenance Practices for Particular ...	Analyse
33	C	15.4 Maintenance Practices for Particular ...	Apply
34	A	15.4 Maintenance Practices for Particular ...	Understand
35	B	15.5 Applying Tool Handling and Maintenanc...	Apply
36	B	15.5 Applying Tool Handling and Maintenanc...	Understand
37	C	15.5 Applying Tool Handling and Maintenanc...	Remember

Q	Ans	Curriculum Area	Cognitive Level
38	A	15.5 Applying Tool Handling and Maintenanc...	Analyse
39	B	15.5 Applying Tool Handling and Maintenanc...	Evaluate
40	A	15.5 Applying Tool Handling and Maintenanc...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 7 (17%) | Evaluate: 2 (5%) | Remember: 13 (32%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [4 marks]

- A - **cutlass (machete)** (1 mark).
- B - **hoe** (1 mark).
- C - **rake** (1 mark).
- D - **watering can** (1 mark).
- *Marking note: Accept 'matchet' for A.*

(b) [4 marks]

- A - **clearing weeds and bushes (accept harvesting)** (1 mark).
- B - **weeding, making ridges or mounds** (1 mark).
- C - **gathering leaves and trash, levelling a seedbed** (1 mark).
- D - **watering seedlings and nursery beds** (1 mark).
- *Marking note: 'Farming' is too general and earns no mark.*

(c) [2 marks]

- Any TWO of: **feeding or drinking trough**; **broom, brush or scraper for cleaning pens**; **shovel or spade**; **bucket**; **wheelbarrow**; **rope or halter** (1 mark each).
- *Marking note: A tool already shown in Fig. 1 earns no mark.*

(d) [2 marks]

- **They are operated by hand and need no engine or fuel** (1 mark).
- **They are cheap, easy to use and easy to repair and maintain** (1 mark).
- *Marking note: 'They are small' alone earns no mark.*

Question 2 [Total: 10 marks]

(a) [3 marks]

- P - **the blade is rusty (not cleaned and oiled)** (1 mark).
- Q - **the handle is split or cracked** (1 mark).
- R - **the blade is left lying in water** (1 mark).
- *Marking note: 'The cutlass is old' earns no mark.*

(b) [3 marks]

- P - **wash off the soil, dry the blade and oil it after use** (1 mark).
- Q - **keep the handle dry and oiled; refit or replace it when it splits** (1 mark).
- R - **store the tool in a dry store, hung on a rack, not on wet ground** (1 mark).
- *Marking note: The practice must match the fault it is written against.*

(c) [2 marks]

- Any TWO of: ****the tools are hung on racks or nails off the floor****; ****the tools are clean and free of rust****; ****oil is kept ready for oiling the tools****; ****the store is dry and under cover**** (1 mark each).
- *Marking note: 'The store is nice' earns no mark.*

(d) [2 marks]

- Any TWO of: ****the tool is spoilt early and money must be spent replacing it****; ****work on the farm becomes slow and tiring with a blunt, rusty tool****; ****it is dangerous and can cause injury**** (1 mark each).
- *Marking note: Two answers that both say 'it will spoil' count once.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- (i) ****The way a tool is held, carried and used during work**** (1 mark).
- (ii) ****The care given to a tool to keep it in good working condition**** (1 mark).
- *Marking note: Examples alone, with no explanation, earn no mark.*

(b) [4 marks]

- Any FOUR of: ****check the tool and the handle before use****; ****carry a cutlass with the blade down and away from the body****; ****give a cutlass to another person handle first****; ****work a safe distance apart****; ****use a tool only for its proper work****; ****wear boots and hand gloves****; ****keep idle tools together with the edges down or covered**** (1 mark each).
- *Marking note: 'Be careful with the cutlass' is too vague for a mark.*

(c) [2 marks]

- Any TWO of: ****the edge is chipped, bent or blunted****; ****the blade may break****; ****the blade may fly off the handle and injure someone**** (1 mark each).
- *Marking note: The rule broken is that a tool must be used only for its proper work.*

(d) [2 marks]

- Any TWO of: ****they prevent injury to the user and to others****; ****they protect the tools from damage****; ****work is in fact finished faster and better with safe, sound tools**** (1 mark each).
- *Marking note: 'Because the teacher said so' earns no mark.*

Question 4 [Total: 10 marks]

(a) [4 marks]

- Cutlass - ****sharpen with a file or whetstone; clean and oil the blade**** (1 mark).
- Hoe - ****clean, dry and oil the blade; keep the wooden handle dry and oiled**** (1 mark).
- Knapsack sprayer - ****wash out the tank and nozzle after use and oil the moving parts**** (1 mark).
- Wheelbarrow - ****oil the wheel and axle and tighten the bolts**** (1 mark).
- ***Marking note: A practice written against the wrong tool earns no mark.***

(b) [3 marks]

- ****Scrape and wash off the soil, then dry the blade**** (1 mark).
- ****Rub the blade with oil and check or wipe the handle**** (1 mark).
- ****Hang it in a dry store, off the floor**** (1 mark).
- ***Marking note: The three steps must be in a sensible order.***

(c) [3 marks]

- ****Rust eats away the metal, so the tool becomes blunt, weak and finally breaks**** (1 mark).
- ****Condition 1: moisture, from water, rain, dew or damp soil**** (1 mark).
- ****Condition 2: air (oxygen) in contact with the bare metal, especially under soil left on it**** (1 mark).
- ***Marking note: 'Dirty tools' alone is not a condition; the moisture must be named.***

Question 5 [Total: 10 marks]**(a)** [3 marks]

- Any THREE of: ****the tools used and the work each does****; ****how the farmer holds, carries and uses them, and the protective clothing worn****; ****what is done to the tools after work (cleaning, sharpening, oiling, repair)****; ****where and how the tools are stored****; ****the condition of the tools**** (1 mark each).
- ***Marking note: 'Everything about the farm' earns no mark.***

(b) [3 marks]

- Any THREE of: ****greet the farmer and ask permission****; ****stay together and follow instructions****; ****do not touch tools or animals without being told****; ****ask questions politely and do not criticise the farmer****; ****do not trample the crops or frighten the birds****; ****thank the farmer before leaving**** (1 mark each).
- ***Marking note: Accept any clear rule of courtesy or safety.***

(c) [4 marks]

- (i) Any TWO of: ****oil the wheelbarrow wheel and tighten the bolts****; ****refit or replace the loose hoe handles****; ****wash out the sprayer after every use**** (1 mark each).
- (ii) ****A correct reason beside each of the two recommendations given**** (1 mark)

each).

- *Marking note: The reason must match the recommendation it is written against.*

Question 6 [Total: 8 marks]

(a) [4 marks]

- **Four clear rules, each stating an action** (1 mark each), for example: **sign for and return each tool to its rack**; **clean and dry tools before returning them**; **oil the metal parts**; **report damaged tools at once**; **use each tool only for its proper work**; **keep the store dry and locked**.

- *Marking note: A rule with no action, such as 'tools are important', earns no mark.*

(b) [2 marks]

- **Two tools named with a correct use for each** (1 mark each), for example **hoe - to dig and loosen the soil**; **rake - to level the bed**.

- *Marking note: A tool named without its use earns no mark.*

(c) [2 marks]

- **Well-kept tools last longer, so money is not spent replacing them each season** (1 mark).

- **Work is done faster and with fewer accidents, so less is spent on hired labour and on treatment** (1 mark).

- *Marking note: A general statement about saving money, with no link to the tools, earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 5: Agricultural Tools |
 Class: Basic 7 (JHS 1)

Content Standard: **B7/JHS1.4.5.1** - Demonstrate knowledge and skills in handling and maintenance of basic and simple agricultural tools.

> S

Animal Production

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	6.1 Domestic Animals and their Breeds	Remember
2	B	6.1 Domestic Animals and their Breeds	Understand
3	C	6.1 Domestic Animals and their Breeds	Remember
4	B	6.1 Domestic Animals and their Breeds	Remember
5	A	6.1 Domestic Animals and their Breeds	Remember
6	C	6.1 Domestic Animals and their Breeds	Apply
7	D	6.2 Ruminants	Remember
8	B	6.2 Ruminants	Remember
9	D	6.2 Ruminants	Remember
10	C	6.2 Ruminants	Understand
11	A	6.2 Ruminants	Understand
12	B	6.2 Ruminants	Analyse
13	C	6.2 Ruminants	Remember
14	B	6.3 Monogastric Animals and Poultry	Remember
15	C	6.3 Monogastric Animals and Poultry	Remember
16	D	6.3 Monogastric Animals and Poultry	Remember
17	A	6.3 Monogastric Animals and Poultry	Understand
18	A	6.3 Monogastric Animals and Poultry	Remember
19	B	6.3 Monogastric Animals and Poultry	Understand
20	D	6.3 Monogastric Animals and Poultry	Analyse
21	D	6.4 Comparing Ruminants, Monogastrics and ...	Understand
22	B	6.4 Comparing Ruminants, Monogastrics and ...	Understand
23	C	6.4 Comparing Ruminants, Monogastrics and ...	Evaluate
24	D	6.4 Comparing Ruminants, Monogastrics and ...	Apply
25	A	6.4 Comparing Ruminants, Monogastrics and ...	Analyse
26	B	6.4 Comparing Ruminants, Monogastrics and ...	Understand
27	D	6.5 Domestic Uses of Animals	Remember
28	A	6.5 Domestic Uses of Animals	Understand
29	A	6.5 Domestic Uses of Animals	Remember
30	D	6.5 Domestic Uses of Animals	Understand
31	B	6.5 Domestic Uses of Animals	Apply
32	C	6.5 Domestic Uses of Animals	Understand
33	A	6.5 Domestic Uses of Animals	Remember
34	D	6.6 Commercial Uses and By-Products	Remember
35	A	6.6 Commercial Uses and By-Products	Remember
36	D	6.6 Commercial Uses and By-Products	Understand
37	C	6.6 Commercial Uses and By-Products	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	A	6.6 Commercial Uses and By-Products	Understand
39	C	6.6 Commercial Uses and By-Products	Apply
40	B	6.6 Commercial Uses and By-Products	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 3 (7%) | Apply: 4 (10%) | Evaluate: 2 (5%) | Remember: 17 (42%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [4 marks]

- A - **goat** (1 mark).
- B - **cow / cattle** (1 mark).
- C - **pig** (1 mark).
- D - **fowl (chicken)** (1 mark).
- *Marking note: Accept 'she-goat', 'nanny goat' for A and 'hen' for D.*

(b) [3 marks]

- A - **ruminant** (1 mark).
- C - **monogastric animal** (1 mark).
- D - **poultry** (accept: poultry, a monogastric bird) (1 mark).
- *Marking note: 'Monogastric' written for D is accepted, but 'ruminant' is not.*

(c) [3 marks]

- (i) Any ONE of: **B has four legs and D has two**; **B is covered with hair while D has feathers**; **D has a beak, comb and wings**; **B has horns and hooves** (1 mark).
- (ii) Any TWO of: **both are ruminants (chew the cud)**; **both are four-legged mammals covered with hair**; **both have hooves and horns**; **both feed mainly on grass**; **both give birth to live young and suckle them** (1 mark each).
- *Marking note: In (ii) a difference written in place of a similarity earns no mark.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- P - **fowl** (accept: duck, guinea fowl, turkey) (1 mark).
- Q - **cow** (accept: goat, sheep) (1 mark).
- R - **goat** (accept: sheep, cow) (1 mark).
- S - **cattle** (accept: goats, sheep, pigs, poultry) (1 mark).
- *Marking note: A wild animal named anywhere in this part earns no mark.*

(b) [3 marks]

- (i) **A useful material obtained from an animal apart from the main product (meat, milk or eggs), such as hide, skin, horn, bone or dung** (1 mark).
- (ii) **R and S** (1 mark) **because the skin is obtained after the animal is slaughtered for meat and the dung is the animal's waste, neither being the main product** (1 mark).

- *Marking note: Only one mark for naming R and S without a reason.*

(c) [3 marks]

- **Heap it and allow it to rot (compost it), turning it from time to time, before spreading** (1 mark).

- Any TWO reasons: **fresh dung scorches the roots of the crop**; **rotting destroys weed seeds and disease germs**; **it removes the smell and the flies**; **the nutrients are then in a form the plant can take up** (1 mark each, maximum 2 marks).

- *Marking note: 'Dry it in the sun' alone is not accepted as the treatment.*

(d) [2 marks]

- Any TWO, each with a use: **hide or skin** - *leather for bags, sandals and belts*; **horns or hooves** - *combs, buttons, ornaments*; **bones** - *bone meal or glue*; **feathers** - *stuffing pillows, decoration* (1 mark each).

- *Marking note: A by-product named without a use earns no mark.*

Question 3 [Total: 10 marks]

(a) [4 marks]

- Body covering - **hair / feathers** (1 mark).

- Number of legs - **four / two** (1 mark).

- Type of stomach - **four-chambered (ruminant) stomach / simple stomach with a gizzard** (1 mark).

- Main feed - **grass and crop residues / prepared mash (with grit)** (1 mark).

- *Marking note: Both halves of a row must be correct before the mark for that row is given.*

(b) [3 marks]

- **The fowl has no teeth and swallows its food whole** (1 mark).

- **The grit in the gizzard grinds the food into small pieces** (1 mark).

- **Cattle chew with their teeth and chew the cud, so they do not need grit** (1 mark).

- *Marking note: Accept 'the gizzard uses the stones like teeth' for the second mark.*

(c) [3 marks]

- **The pigs** (1 mark).

- **A pig has a simple stomach and cannot digest fibre, so it needs bran and other concentrates, which are costly** (1 mark).

- **Cattle can still use crop residues and browse because the rumen breaks down fibre (or: fowls eat only small quantities each)** (1 mark).

- *Marking note: A candidate who names the cattle may earn at most 1 mark, for a reasoned point about the scarcity of grass.*

Question 4

(a) [4 marks]

- (i) Ama - **GH¢1,300** (1 mark); Kojo - **GH¢2,700** (1 mark).
- (ii) Ama - **GH¢800** (1 mark); Kojo - **GH¢700** (1 mark).
- *Marking note: Follow through: a correct subtraction from a wrong total in (i) earns the mark in (ii).*

(b) [3 marks]

- Any TWO of: **goats are ruminants and feed on grass, browse and crop residues, which cost nothing**; **goats can be grazed or tethered on free pasture**; **broilers are monogastric and need bought, prepared feed daily** (1 mark each).
- **Broilers grow quickly, so Kojo sold several batches of many birds within the year** (1 mark).
- *Marking note: 'Goats eat less' alone, without the reason, earns no mark.*

(c) [3 marks]

- A clear recommendation (1 mark).
- Any TWO supporting reasons drawn from the table, for example: **the goats gave the higher profit (GH¢800 against GH¢700)**; **the goats needed far less cash for feed (GH¢400 against GH¢2,100)**; **the broilers returned money several times within the year**; **the broilers had larger total sales** (1 mark each).
- *Marking note: Either enterprise is acceptable; the marks are for the reasons, which must use the figures.*

Question 5 [Total: 10 marks]**(a)** [4 marks]

- **Divide into groups and share out the area to be visited** (1 mark).
- **Prepare a record sheet and obtain permission before visiting** (1 mark).
- **Observe and count the animals and ask the keepers about their uses, recording at once** (1 mark).
- **Bring the records together, add up the numbers and present them in a table or chart** (1 mark).
- *Marking note: Steps given in a workable order; a single step split into two earns one mark only.*

(b) [3 marks]

- Any THREE of: **name of animal**; **group (ruminant, monogastric, poultry)**; **number counted**; **home or farm visited**; **uses of the animal**; **by-products obtained** (1 mark each).
- *Marking note: 'Information' or 'remarks' is too vague and earns no mark.*

(c) [3 marks]

- Any THREE of: **ask permission and greet the household politely**; **do not enter pens or chase, touch or frighten the animals**; **keep away from dangerous animals such as a bull or a nursing mother**; **wash hands with soap after the visit**; **wear shoes and avoid dung**; **stay in groups and report to the teacher** (1 mark each).

- *Marking note: 'Be careful' or 'behave well' earns no mark without the action.*

Question 6 [Total: 8 marks]**(a) [3 marks]**

- Any THREE of: **the space and housing available**; **the feed she can get cheaply**; **the money she has**; **the market for the product**; **the time and labour needed**; **the neighbours and the rules of the town** (1 mark each).

- *Marking note: A reason that is not a real consideration, such as 'the colour of the animal', earns no mark.*

(b) [3 marks]

- Any THREE of: **it protects the animals from rain, sun, thieves and stray animals**; **it prevents diseases and parasites, so the animals grow well**; **it keeps flies, smells and germs away from the household and the neighbours**; **the droppings can be collected and used as manure**; **the animals do not stray and destroy the neighbours' property** (1 mark each).

- *Marking note: Each reason must be different; 'it keeps them safe' twice earns 1 mark.*

(c) [2 marks]

- Any TWO of: **they supply protein food (meat, eggs, milk) and improve nutrition**; **they create employment and income**; **their hides, skins and horns supply raw materials for industry and export**; **their dung provides manure for crops**; **they reduce the money spent on imported meat** (1 mark each).

- *Marking note: A domestic benefit to one family only, such as 'Afia will eat the eggs', does not answer this part.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 2: Cycles | **Sub-strand:** 4: Animal Production | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.2.4.1**

Climate Change and Green

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	19.1 Sustainable and Non-Sustainable Energy...	Understand
2	B	19.1 Sustainable and Non-Sustainable Energy...	Remember
3	B	19.1 Sustainable and Non-Sustainable Energy...	Remember
4	C	19.1 Sustainable and Non-Sustainable Energy...	Remember
5	D	19.1 Sustainable and Non-Sustainable Energy...	Understand
6	C	19.1 Sustainable and Non-Sustainable Energy...	Understand
7	B	19.1 Sustainable and Non-Sustainable Energy...	Understand
8	A	19.1 Sustainable and Non-Sustainable Energy...	Remember
9	A	19.2 Scientific Ideas Used to Protect the ...	Remember
10	C	19.2 Scientific Ideas Used to Protect the ...	Understand
11	D	19.2 Scientific Ideas Used to Protect the ...	Understand
12	A	19.2 Scientific Ideas Used to Protect the ...	Evaluate
13	D	19.2 Scientific Ideas Used to Protect the ...	Understand
14	A	19.2 Scientific Ideas Used to Protect the ...	Understand
15	B	19.2 Scientific Ideas Used to Protect the ...	Understand
16	A	19.2 Scientific Ideas Used to Protect the ...	Remember
17	D	19.3 The Greenhouse Effect and Global Warming	Remember
18	D	19.3 The Greenhouse Effect and Global Warming	Remember
19	B	19.3 The Greenhouse Effect and Global Warming	Remember
20	A	19.3 The Greenhouse Effect and Global Warming	Understand
21	A	19.3 The Greenhouse Effect and Global Warming	Understand
22	D	19.3 The Greenhouse Effect and Global Warming	Remember
23	B	19.3 The Greenhouse Effect and Global Warming	Understand
24	D	19.3 The Greenhouse Effect and Global Warming	Analyse
25	B	19.3 The Greenhouse Effect and Global Warming	Remember
26	B	19.4 Minimising Greenhouse Effects and Cop...	Understand
27	C	19.4 Minimising Greenhouse Effects and Cop...	Analyse
28	C	19.4 Minimising Greenhouse Effects and Cop...	Analyse
29	B	19.4 Minimising Greenhouse Effects and Cop...	Evaluate
30	C	19.4 Minimising Greenhouse Effects and Cop...	Evaluate
31	C	19.4 Minimising Greenhouse Effects and Cop...	Analyse
32	A	19.4 Minimising Greenhouse Effects and Cop...	Evaluate
33	D	19.4 Minimising Greenhouse Effects and Cop...	Understand
34	B	19.5 Local Projects that Sustain Energy, a...	Understand
35	A	19.5 Local Projects that Sustain Energy, a...	Remember
36	A	19.5 Local Projects that Sustain Energy, a...	Remember
37	D	19.5 Local Projects that Sustain Energy, a...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	C	19.5 Local Projects that Sustain Energy, a...	Apply
39	C	19.5 Local Projects that Sustain Energy, a...	Evaluate
40	D	19.5 Local Projects that Sustain Energy, a...	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 2 (5%) | Evaluate: 5 (12%) | Remember: 13 (32%) | Understand: 16 (40%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [4 marks]

- A - **the Sun (solar energy)** (1 mark).
- B - **the wind (moving air)** (1 mark).
- C - **charcoal or firewood (wood fuel)** (1 mark).
- D - **biogas from animal dung or other organic waste** (1 mark).
- **Marking note: 'Fire' is not accepted for C; the fuel must be named.**

(b) [4 marks]

- A - **renewable** (1 mark) **because sunlight arrives every day and is not used up** (1 mark).
- C - **not renewable in practice** (1 mark) **because the trees are cut down faster than they can grow again** (1 mark).
- **Marking note: Accept 'renewable only if the trees are replanted' for C, with the reason, for full marks.**

(c) [2 marks]

- **D (the biogas plant)** (1 mark).
- **Reason: the cattle dung needed is already available, and the gas gives cooking fuel while the slurry gives manure** (1 mark).
- **Marking note: A letter with no reason earns 1 mark only.**

(d) [2 marks]

- Any TWO of: **trees are cut down, destroying the forest and exposing the soil**; **smoke pollutes the air and harms the lungs**; **carbon dioxide released adds to the greenhouse effect** (1 mark each).
- **Marking note: 'It is dirty' is too vague for a mark.**

Question 2 [Total: 11 marks]

(a) [4 marks]

- W - **radiation (heat and light) coming in from the Sun** (1 mark).
- X - **the layer of greenhouse gases in the atmosphere** (1 mark).
- Y - **heat radiated upwards from the warmed Earth's surface** (1 mark).
- Z - **heat trapped and sent back down to the surface** (1 mark).
- **Marking note: 'The sky' is not accepted for X.**

(b) [2 marks]

- Any TWO of: **carbon dioxide**; **methane**; **water vapour**; **nitrous oxide**; **chlorofluorocarbons (CFCs)** (1 mark each).
- **Marking note:** Oxygen and nitrogen are not greenhouse gases.*

(c) [2 marks]

- **The gases let the Sun's radiation in but hold back the heat leaving the Earth** (1 mark).
- **As more gas is added, more heat is trapped and the average temperature of the Earth rises** (1 mark).
- **Marking note:** A candidate who writes only 'the Earth becomes hot' earns 1 mark.*

(d) [3 marks]

- Any THREE of: **burning fuels in vehicles, generators and factories**; **burning firewood and charcoal for cooking**; **cutting down and burning forests**; **open burning of rubbish**; **large herds of cattle and flooded rice fields releasing methane** (1 mark each).
- **Marking note:** Two answers that are both 'burning things' count once.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- **A long-term change in the usual pattern of weather (temperature, rainfall, wind) of a place** (1 mark).
- **Measured over many years, not from day to day** (1 mark).
- **Marking note:** A definition of weather alone earns no mark.*

(b) [4 marks]

- **Warming has changed the rainfall pattern, so the rains are late, short or stormy** (1 mark) **and the farmer cannot predict the planting season** (1 mark).
- **Melting ice and the expansion of warm sea water raise the level of the sea** (1 mark) **and stronger waves wash away the shore and flood the village** (1 mark).
- **Marking note:** Credit any sound explanation linked to warming.*

(c) [4 marks]

- Any FOUR of: **plant trees and protect the forest and mangroves**; **harvest water and irrigate in the dry spells**; **plant early-maturing or drought-tolerant varieties and use the weather forecast**; **build sea defence works and avoid building close to the shore**; **use solar lamps and improved stoves**; **stop the open burning of rubbish** (1 mark each).
- **Marking note:** 'Pray for rain' earns no mark.*

Question 4

(a) [4 marks]

- Any FOUR of: **switch off lights and fans in empty rooms and at closing**; **use LED bulbs in place of filament bulbs**; **use daylight by opening windows and curtains**; **service appliances and keep refrigerator doors shut**; **unplug chargers and idle appliances**; **put up reminder notices** (1 mark each).
- *Marking note: 'Stop using electricity' is not a practical measure.**

(b) [3 marks]

- Any THREE of: **less fuel burnt in thermal stations, so less carbon dioxide and smoke**; **less pressure to build new dams and stations that destroy land and habitats**; **water in the reservoirs is conserved**; **less waste and worn-out equipment to dispose of** (1 mark each).
- *Marking note: 'It saves money' is excluded by the question.**

(c) [2 marks]

- **A clear slogan of not more than eight words calling for energy to be saved** (1 mark).
- **The message explained: everyone must switch off what is not in use, and saving energy protects the environment** (1 mark).
- *Marking note: A slogan longer than eight words loses the first mark.**

Question 5 [Total: 10 marks]**(a)** [3 marks]

- **Sunlight passes through the clear cover and is absorbed by the dark inside surface, heating the air** (1 mark).
- **The hot air takes up moisture from the fruit on the trays** (1 mark).
- **The moist air escapes through vents while fresh air is drawn in, and the fruit is kept free of rain, dust and flies** (1 mark).
- *Marking note: A mere statement that 'the Sun dries it' earns 1 mark.**

(b) [3 marks]

- Any THREE materials, each with its correct use, for example: **wood or bamboo - the frame**; **clear polythene or glass - the cover**; **black paint or a black metal sheet - the absorbing surface**; **wire mesh or woven cane - the trays** (1 mark each).
- *Marking note: A material named without its use earns no mark.**

(c) [2 marks]

- **Dry equal masses of the same fruit, one in the dryer and one in the open, at the same time** (1 mark).
- **Weigh both daily, or record the drying time, and compare the loss of water and the cleanliness** (1 mark).

- *Marking note: A test with no comparison earns 1 mark at most.*

(d) [2 marks]

- Any TWO of: **less fruit wasted, so more food and income all year**; **no fuel or electricity burnt, so trees are saved and less carbon dioxide released**; **cleaner, safer dried food**; **new jobs in making dryers and selling dried fruit** (1 mark each).

- *Marking note: 'It is a good project' earns no mark.*

Question 6 [Total: 8 marks]

(a) [2 marks]

- **Work that earns a living while protecting or restoring the environment** (1 mark).

- **One sound example, such as raising tree seedlings, installing solar panels or recycling waste** (1 mark).

- *Marking note: 'Any job done outside' earns no mark.*

(b) [4 marks]

- Any FOUR of: **they absorb carbon dioxide and store carbon**; **they release oxygen**; **their roots hold the soil and reduce erosion and flooding**; **they cool the air and help to bring rain**; **they act as windbreaks and hold back the desert**; **they provide habitats, fruit and timber** (1 mark each).

- *Marking note: 'Trees are beautiful' earns no mark.*

(c) [2 marks]

- Any TWO of: **plant at the start of the rains and choose suitable species**; **water the seedlings during dry spells**; **fence or guard them against animals and bush fires**; **weed round them and replace those that die**; **make a named person or class responsible for each tree** (1 mark each).

- *Marking note: 'Plant more trees' does not answer the question.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans and the Environment | **Sub-strand:** 4: Climate Change and Green Economy | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.5.4.1** - Demonstrate understand

Conversion and Conservation

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	13.1 The Law of Conservation of Energy	Remember
2	A	13.1 The Law of Conservation of Energy	Remember
3	B	13.1 The Law of Conservation of Energy	Remember
4	B	13.1 The Law of Conservation of Energy	Understand
5	D	13.1 The Law of Conservation of Energy	Apply
6	A	13.1 The Law of Conservation of Energy	Understand
7	A	13.1 The Law of Conservation of Energy	Analyse
8	B	13.1 The Law of Conservation of Energy	Understand
9	C	13.2 Energy Conversions in Everyday Devices	Understand
10	A	13.2 Energy Conversions in Everyday Devices	Understand
11	A	13.2 Energy Conversions in Everyday Devices	Remember
12	D	13.2 Energy Conversions in Everyday Devices	Understand
13	A	13.2 Energy Conversions in Everyday Devices	Remember
14	B	13.2 Energy Conversions in Everyday Devices	Understand
15	D	13.2 Energy Conversions in Everyday Devices	Understand
16	C	13.2 Energy Conversions in Everyday Devices	Understand
17	A	13.2 Energy Conversions in Everyday Devices	Remember
18	C	13.3 Energy Chains and Simple Calculations	Understand
19	D	13.3 Energy Chains and Simple Calculations	Understand
20	C	13.3 Energy Chains and Simple Calculations	Remember
21	B	13.3 Energy Chains and Simple Calculations	Apply
22	D	13.3 Energy Chains and Simple Calculations	Apply
23	D	13.3 Energy Chains and Simple Calculations	Remember
24	C	13.3 Energy Chains and Simple Calculations	Analyse
25	C	13.3 Energy Chains and Simple Calculations	Understand
26	D	13.4 Useful Energy and Wasted Energy	Analyse
27	C	13.4 Useful Energy and Wasted Energy	Understand
28	B	13.4 Useful Energy and Wasted Energy	Understand
29	B	13.4 Useful Energy and Wasted Energy	Understand
30	C	13.4 Useful Energy and Wasted Energy	Remember
31	C	13.4 Useful Energy and Wasted Energy	Evaluate
32	A	13.5 Conserving Energy for the Future	Remember
33	B	13.5 Conserving Energy for the Future	Understand
34	B	13.5 Conserving Energy for the Future	Understand
35	C	13.5 Conserving Energy for the Future	Analyse
36	D	13.5 Conserving Energy for the Future	Analyse
37	A	13.5 Conserving Energy for the Future	Remember

Q	Ans	Curriculum Area	Cognitive Level
38	D	13.5 Conserving Energy for the Future	Understand
39	B	13.5 Conserving Energy for the Future	Understand
40	A	13.5 Conserving Energy for the Future	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 3 (7%) | Evaluate: 2 (5%) | Remember: 11 (27%) | Understand: 19 (47%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [4 marks]

- P - **chemical energy** (1 mark).
- Q - **electrical energy** (1 mark).
- R - **light energy** (1 mark).
- S - **heat (thermal) energy** (1 mark).
- **Marking note: 'Battery energy' or 'current' earns no mark.**

(b) [3 marks]

- **Energy supplied = 60 J** (1 mark).
- **Energy given out = 15 J + 45 J = 60 J** (1 mark).
- **The two totals are equal, so the energy has only been changed in form and none has been destroyed** (1 mark).
- **Marking note: The sum must be shown; a bare statement of the law earns the third mark only.**

(c) [3 marks]

- (i) **R, the light energy** (1 mark); **because the torch is used in order to give light** (1 mark).
- (ii) **The torch feels warm (the cells or the bulb heat up)** (1 mark).
- **Marking note: Reason mark only if R is chosen.**

(d) [2 marks]

- **In a closed system no energy enters or leaves without being counted** (1 mark).
- **So all the energy given out can be added up and shown to equal the energy supplied** (1 mark).
- **Marking note: 'It is covered' or 'it is in a box' alone earns no mark.**

Question 2 [Total: 10 marks]

(a) [4 marks]

- A - **solar (light) energy to electrical energy** (1 mark).
- B - **electrical energy to heat energy** (1 mark).
- C - **electrical energy to kinetic energy** (1 mark).
- D - **chemical energy to electrical energy to sound energy** (1 mark).
- **Marking note: Accept 'electrical to sound' for D.**

(b) [3 marks]

- **Useful: kinetic energy of the blades** (1 mark).
- **Wasted: sound energy and heat energy** (1 mark).
- **The wasted energy spreads into the air and surroundings and cannot be used again** (1 mark).
- **Marking note: 'It is destroyed' earns no mark.**

(c) [3 marks]

- Any THREE of: **sunlight is free and renewable, so no fuel is bought**; **no smoke or carbon dioxide is given off**; **it makes no noise**; **it needs little maintenance** (1 mark each).
- **Marking note: 'It is modern' is not a scientific reason.**

Question 3 [Total: 10 marks]**(a)** [2 marks]

- **Energy can neither be created nor destroyed** (1 mark).
- **It can only be changed from one form to another (the total remains the same)** (1 mark).
- **Marking note: Both halves are needed for the full 2 marks.**

(b) [3 marks]

- **Potential energy of the stored water shown first** (1 mark).
- **Kinetic energy of the falling water and of the turbine shown next** (1 mark).
- **Electrical energy shown last, with the boxes joined by arrows in the correct order** (1 mark).
- **Marking note: A written chain with arrows is accepted in place of boxes.**

(c) [3 marks]

- (i) **1,000 J - 850 J** (1 mark) **= 150 J** (1 mark, unit required).
- (ii) **Heat energy (accept sound energy)** (1 mark).
- **Marking note: A correct answer with no unit scores 1 of the 2 marks.**

(d) [2 marks]

- **The 150 J is not destroyed; it is changed into heat and sound** (1 mark).
- **Counting every form, the total is still 1,000 J** (1 mark).
- **Marking note: 'The generator is weak' earns no mark.**

Question 4 [Total: 10 marks]**(a)** [4 marks]

- Microphone - **sound to electrical** (1 mark).
- Phone being charged - **electrical to chemical** (1 mark).

- Candle burning - **chemical to heat and light** (1 mark).
- Green plant - **light (solar) to chemical** (1 mark).
- **Marking note:** A change written the wrong way round earns no mark.*

(b) [3 marks]

- **Plants change light energy into chemical energy stored in food (photosynthesis)** (1 mark).
- **People and animals eat the food and obtain this energy** (1 mark).
- **The energy is released in respiration for growth, movement and warmth (accept: wood and charcoal fuels come from plants)** (1 mark).
- **Marking note:** 'Plants give us oxygen' is true but does not answer the question on energy.*

(c) [3 marks]

- **Materials named: a small motor, dry cells, wires, switch and a paper blade or disc** (1 mark).
- **Method: join them in a complete circuit and close the switch** (1 mark).
- **Observation: the blade spins, showing electrical energy changed into kinetic energy** (1 mark).
- **Marking note:** Any other workable activity, such as a battery-driven toy car, earns full marks.*

Question 5 [Total: 10 marks]**(a)** [2 marks]

- **Using energy carefully and avoiding waste** (1 mark).
- **So that the supply lasts longer and is available in the future** (1 mark).
- **Marking note:** Do not confuse this with the LAW of conservation of energy; a statement of the law earns no mark here.*

(b) [4 marks]

- Any FOUR of: **switch off lights, fans and the television in empty rooms**; **use LED lamps in place of filament bulbs**; **dry clothes in the sun**; **iron all the clothes at one time**; **cover cooking pots and open the fridge less often**; **unplug chargers that are not in use** (1 mark each).
- **Marking note:** Two answers that are really the same point count once.*

(c) [4 marks]

- (i) **A smaller electricity bill; appliances and cells last longer** (1 mark).
- (ii) **Less electricity has to be generated, so less fuel is burnt and there are fewer power cuts** (1 mark); **the money saved can be used for other national needs** (1 mark).
- (iii) **Less smoke and carbon dioxide, so cleaner air, less global warming and fewer habitats destroyed** (1 mark).
- **Marking note:** Award the marks under the correct headings.*

Question

(a) [3 marks]

- **Renewable:** replaced naturally and does not run out (1 mark);
- non-renewable:** exists in a fixed amount and is used up for ever (1 mark).
- (ii) **One correct example of each, for example sunlight and crude oil** (1 mark).
- **Marking note:** Both examples are needed for the single mark in (ii).*

(b) [3 marks]

- **Agree that energy is never destroyed** (1 mark).
- **But it is changed into heat and sound that spread out and can no longer be used** (1 mark).
- **The fuels used to supply useable energy are limited and burning them harms the environment, so saving is still necessary** (1 mark).
- **Marking note:** A reply that only says 'he is wrong' earns no mark.*

(c) [2 marks]

- **Two slogans, each naming a clear energy-saving action or giving a sound reason** (1 mark each).
- **Marking note:** A slogan with no action and no reason, such as 'Save energy', earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 3: Conversion and Conservation of Energy | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.4.3.1** - Demonstrate an understanding of the principle of conservation and conversion of energy and their application in real life situations.

> Source: NaCCA Science Common Core Programme Curriculum (B7/J

Crop Production

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	5.1 Plant Nutrients and the Needs of Crops	Remember
2	A	5.1 Plant Nutrients and the Needs of Crops	Remember
3	D	5.1 Plant Nutrients and the Needs of Crops	Understand
4	A	5.1 Plant Nutrients and the Needs of Crops	Remember
5	A	5.1 Plant Nutrients and the Needs of Crops	Analyse
6	B	5.1 Plant Nutrients and the Needs of Crops	Understand
7	C	5.2 Organic Sources of Plant Nutrients	Remember
8	A	5.2 Organic Sources of Plant Nutrients	Remember
9	D	5.2 Organic Sources of Plant Nutrients	Remember
10	B	5.2 Organic Sources of Plant Nutrients	Remember
11	A	5.2 Organic Sources of Plant Nutrients	Understand
12	C	5.2 Organic Sources of Plant Nutrients	Understand
13	C	5.2 Organic Sources of Plant Nutrients	Analyse
14	D	5.3 Inorganic Sources of Plant Nutrients	Remember
15	B	5.3 Inorganic Sources of Plant Nutrients	Remember
16	A	5.3 Inorganic Sources of Plant Nutrients	Remember
17	C	5.3 Inorganic Sources of Plant Nutrients	Remember
18	D	5.3 Inorganic Sources of Plant Nutrients	Understand
19	A	5.3 Inorganic Sources of Plant Nutrients	Understand
20	B	5.3 Inorganic Sources of Plant Nutrients	Analyse
21	A	5.4 Comparing Organic and Inorganic Sources	Understand
22	C	5.4 Comparing Organic and Inorganic Sources	Understand
23	D	5.4 Comparing Organic and Inorganic Sources	Apply
24	D	5.4 Comparing Organic and Inorganic Sources	Understand
25	C	5.4 Comparing Organic and Inorganic Sources	Evaluate
26	B	5.4 Comparing Organic and Inorganic Sources	Analyse
27	D	5.5 Physical Characteristics of the Nutrie...	Remember
28	C	5.5 Physical Characteristics of the Nutrie...	Remember
29	B	5.5 Physical Characteristics of the Nutrie...	Remember
30	D	5.5 Physical Characteristics of the Nutrie...	Understand
31	C	5.5 Physical Characteristics of the Nutrie...	Understand
32	C	5.5 Physical Characteristics of the Nutrie...	Analyse
33	D	5.5 Physical Characteristics of the Nutrie...	Analyse
34	A	5.6 Applying Plant Nutrients in the Field ...	Remember
35	B	5.6 Applying Plant Nutrients in the Field ...	Remember
36	B	5.6 Applying Plant Nutrients in the Field ...	Understand
37	A	5.6 Applying Plant Nutrients in the Field ...	Remember

Q	Ans	Curriculum Area	Cognitive Level
38	B	5.6 Applying Plant Nutrients in the Field ...	Understand
39	C	5.6 Applying Plant Nutrients in the Field ...	Apply
40	D	5.6 Applying Plant Nutrients in the Field ...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 2 (5%) | Evaluate: 2 (5%) | Remember: 17 (42%) | Understand: 13 (32%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [4 marks]

- W - **organic** (1 mark).
- X - **inorganic** (1 mark).
- Y - **organic** (1 mark).
- Z - **inorganic** (1 mark).
- *Marking note: 'Natural' and 'artificial' are accepted for organic and inorganic respectively.*

(b) [4 marks]

- W - any TWO of: **dark brown or black**; **crumbly or loose**; **bulky and heavy**; **moist**; **earthy-smelling** (1 mark each).
- X - any TWO of: **small hard granules (or crystals)**; **dry**; **light in colour** (white, grey or pinkish); **sold in a sealed sack**; **concentrated, so little is needed** (1 mark each).
- *Marking note: A characteristic that is not physical, such as 'it is cheap', earns no mark.*

(c) [2 marks]

- **W (the compost)** (1 mark); accept Y (poultry manure) with the same reason.
- **Because organic manures are bulky and contain only a small proportion of nutrients, while inorganic fertilizers are concentrated** (1 mark).
- *Marking note: The reason must mention the low concentration or the bulkiness of the organic material.*

Question 2 [Total: 12 marks]

(a) [4 marks]

- Any TWO organic: **compost**; **farmyard (cattle, sheep or goat) manure**; **poultry droppings**; **green manure**; **household or market waste** (1 mark each).
- Any TWO inorganic: **NPK**; **sulphate of ammonia**; **urea**; **superphosphate**; **muriate of potash**; **liquid (foliar) fertilizer** (1 mark each).
- *Marking note: 'Fertilizer' written alone earns no mark; a named type is required.*

(b) [4 marks]

- Source - **remains of plants and animals / manufactured from minerals in a factory** (1 mark).
- Quantity - **bulky, large quantities needed / concentrated, small quantities needed** (1 mark).
- Speed of action - **slow release over a long time / quick action** (1 mark).
- Effect on the soil - **adds humus and improves soil structure / adds no humus and may spoil the soil if used alone** (1 mark).
- **Marking note: Both halves of a row must be filled in correctly before the mark for that row is given.**

(c) [2 marks]

- **No, it would not feed the plot as well** (1 mark).
- **The manure is bulky and contains a much smaller proportion of nutrients, so a far larger quantity is needed** (1 mark).
- **Marking note: No mark for the reason if the candidate answers 'yes'.**

(d) [2 marks]

- Organic - any ONE of: **cheap or free**; **made from waste on the compound**; **adds humus and improves the soil**; **releases nutrients slowly** (1 mark).
- Inorganic - any ONE of: **concentrated and easy to carry**; **acts quickly**; **supplies known amounts of named nutrients** (1 mark).
- **Marking note: An advantage written twice in different words earns 1 mark only.**

Question 3 [Total: 10 marks]**(a)** [3 marks]

- P - **broadcasting** (1 mark).
- Q - **ring application** (accept: circle or spot application) (1 mark).
- R - **applying liquid fertilizer by watering at the base of the plant** (accept: liquid or soil drench application) (1 mark).
- **Marking note: 'Foliar spraying' is not accepted for R; the liquid is poured at the base, not sprayed on the leaves.**

(b) [2 marks]

- **P (broadcasting)** (1 mark).
- **The plants are many and close together, so the whole bed must be covered evenly; ringing each plant would be slow and would disturb them** (1 mark).
- **Marking note: No mark for the reason if Q is chosen.**

(c) [2 marks]

- **Fertilizer against the stem would scorch (burn) the stem and roots** (1 mark).
- **At a short distance it lies over the feeding roots, which can still take up the nutrients** (1 mark).
- **Marking note: 'It will spoil the plant' alone earns 1 mark.**

(d) [3 marks]

- Any THREE of: **use only the measured quantity**; **apply when the soil is moist, not in a dry, hot afternoon**; **do not broadcast on a windy day**; **keep the fertilizer off the leaves and stems**; **use gloves or a scoop and wash the hands afterwards**; **keep fertilizer away from food and drinking water** (1 mark each).
- **Marking note: 'Be careful' earns no mark.**

Question 4 [Total: 10 marks]**(a)** [4 marks]

- **Clear a shaded space or dig a shallow pit** (1 mark).
- **Heap the materials in layers, mixing dry and fresh material, with a little soil between the layers** (1 mark).
- **Water the heap to keep it damp and cover it** (1 mark).
- **Turn the heap regularly (every one to two weeks) until it is well rotted** (1 mark).
- **Marking note: The steps must be in a workable order; a bare list of materials earns no mark.**

(b) [3 marks]

- Any THREE of: **dark brown or black in colour**; **crumbly, with the original materials no longer recognisable**; **earthy smell, not a foul smell**; **the heap has cooled down and shrunk in size** (1 mark each).
- **Marking note: 'It looks like soil' is accepted once, for the colour or the crumbly texture, but not for both.**

(c) [3 marks]

- Any THREE of: **it costs nothing and uses waste from the compound**; **it adds humus and improves the soil**; **it releases nutrients slowly and is not easily washed away**; **there is no risk of scorching the crop**; **it keeps the compound clean** (1 mark each).
- **Marking note: Candidates who argue that NPK should also be used in small amounts should not be penalised, provided three reasons for compost are given.**

Question**(a)** [3 marks]

- (i) **Bed 3** (1 mark).
- (ii) **7 kg** (1 mark).
- (iii) **Bed 1** (1 mark).
- **Marking note:** Accept 7 without the unit in (ii).

(b) [3 marks]

- **Too much fertilizer was applied** (1 mark).
- **The excess made the soil solution too strong and scorched (burnt) the roots and leaves** (1 mark).
- **The damaged plants grew poorly, so the yield fell** (1 mark).
- **Marking note:** Accept 'the plants were burnt by the fertilizer' for 2 marks if the idea of excess is also stated.

(c) [2 marks]

- Any TWO of: **the same size and type of bed and soil**; **the same crop**; **the same number of plants**; **the same watering and weeding**; **the same length of time** (1 mark each).
- **Marking note:** The treatment itself is the variable being tested and must not be given as an answer.

(d) [2 marks]

- **Use poultry manure worked into the soil together with the correct measured amount of NPK** (accept: use the measured amount of NPK only) (1 mark).
- **Reason drawn from the table: Bed 3 (correct amount) gave the best yield and Bed 4 shows that extra fertilizer lowers the yield; the manure also improves the soil** (1 mark).
- **Marking note:** The reason must refer to the results in the table.

Question 6 [Total: 8 marks]**(a)** [3 marks]

- Any THREE, each with its reason: **store in a dry place** - **it absorbs moisture and cakes**; **keep the bag sealed and off the floor** - **to keep out moisture**; **store away from food, water and children** - **the chemicals are poisonous**; **keep it out of the sun and away from fire** - **heat spoils it and some fertilizers are a fire risk** (1 mark each).
- **Marking note:** A rule given without its reason earns no mark.

(b) [3 marks]

- ****In dry soil the fertilizer cannot dissolve, so the plant cannot take it up**** (1 mark).
- ****The plants may be scorched, and some of the nutrient is lost in the heat**** (1 mark).
- ****Apply when the soil is moist - after a light rain or watering, in the cool of the morning or evening**** (1 mark).
- ***Marking note: The better time must be stated for the third mark.***

(c) [2 marks]

- Any TWO of: ****read the label and dilute it with the stated volume of water****; ****apply in the cool of the morning or evening****; ****spread it evenly and avoid spraying into the wind****; ****wash the sprayer and his hands after use****; ****do not apply just before harvesting**** (1 mark each).
- ***Marking note: 'Pour it on the plants' earns no mark; the idea of diluting or of measuring must appear in at least one answer.***

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 2: Cycles | ****Sub-strand:** 3: Crop Production | ****Class:** Basic 7 (JHS 1)******

****Content Standard:** ****B7/JHS1.2.3.1**** - Demonstrat**

Ecosystem

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	9.1 The Concept of an Ecosystem	Remember
2	C	9.1 The Concept of an Ecosystem	Understand
3	B	9.1 The Concept of an Ecosystem	Remember
4	D	9.1 The Concept of an Ecosystem	Remember
5	A	9.1 The Concept of an Ecosystem	Remember
6	C	9.1 The Concept of an Ecosystem	Analyse
7	A	9.1 The Concept of an Ecosystem	Understand
8	B	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Remember
9	D	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Remember
10	B	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Remember
11	B	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Remember
12	A	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Understand
13	B	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Understand
14	D	9.2 Terrestrial, Aquatic and Arboreal Ecos...	Understand
15	D	9.3 Biotic Components of an Ecosystem	Remember
16	B	9.3 Biotic Components of an Ecosystem	Remember
17	D	9.3 Biotic Components of an Ecosystem	Understand
18	C	9.3 Biotic Components of an Ecosystem	Understand
19	A	9.3 Biotic Components of an Ecosystem	Understand
20	D	9.3 Biotic Components of an Ecosystem	Analyse
21	C	9.4 Abiotic Components of an Ecosystem	Remember
22	C	9.4 Abiotic Components of an Ecosystem	Remember
23	B	9.4 Abiotic Components of an Ecosystem	Understand
24	D	9.4 Abiotic Components of an Ecosystem	Understand
25	D	9.4 Abiotic Components of an Ecosystem	Understand
26	A	9.4 Abiotic Components of an Ecosystem	Analyse
27	C	9.5 Organisms of Different Ecosystems	Understand
28	B	9.5 Organisms of Different Ecosystems	Understand
29	C	9.5 Organisms of Different Ecosystems	Understand
30	D	9.5 Organisms of Different Ecosystems	Analyse
31	A	9.5 Organisms of Different Ecosystems	Remember
32	A	9.5 Organisms of Different Ecosystems	Analyse
33	B	9.5 Organisms of Different Ecosystems	Analyse
34	A	9.6 Interactions and Interdependence in an...	Remember
35	A	9.6 Interactions and Interdependence in an...	Understand
36	D	9.6 Interactions and Interdependence in an...	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	C	9.6 Interactions and Interdependence in an...	Remember
38	A	9.6 Interactions and Interdependence in an...	Understand
39	C	9.6 Interactions and Interdependence in an...	Analyse
40	B	9.6 Interactions and Interdependence in an...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 7 (17%) | Evaluate: 1 (2%) | Remember: 15 (37%) | Understand: 17 (42%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [5 marks]

- A - **the Sun (sunlight)** (1 mark).
- B - **a water plant** (accept: reed, water weed) (1 mark).
- C - **a fish** (1 mark).
- D - **the mud and stones at the bottom of the pond** (accept: soil, the pond bed) (1 mark).
- E - **a bird** (accept: a wading bird such as a heron or an egret) (1 mark).
- **Marking note:** A general answer such as 'plant' for B and 'animal' for C or E is accepted at this level.*

(b) [3 marks]

- Any TWO of **B**, **C** and **E** as biotic (1 mark each).
- Either **A** or **D** as abiotic (1 mark).
- **Marking note:** Letters alone are enough; names need not be repeated.*

(c) [2 marks]

- **Water plant to fish to bird** (1 mark).
- **Arrows drawn in the correct direction, from the eaten to the eater** (1 mark).
- **Marking note:** A chain written the wrong way round earns the first mark only if the three organisms are correct.*

Question 2 [Total: 12 marks]**(a)** [3 marks]

- P - **terrestrial** (accept: land ecosystem) (1 mark).
- Q - **aquatic** (accept: water ecosystem) (1 mark).
- R - **arboreal** (accept: tree ecosystem) (1 mark).
- **Marking note:** 'Forest' written for R earns no mark; the category wanted is arboreal.*

(b) [3 marks]

- Terrestrial - any ONE named Ghanaian land area, for example **the northern savanna grassland** (1 mark).
- Aquatic - any ONE named Ghanaian water body, for example **the Volta Lake** (1 mark).
- Arboreal - **the life on the branches of a named tall tree, such as a silk-cotton tree** (1 mark).
- **Marking note:** An example placed under the wrong category earns no mark.*

(c) [4 marks]

- Q - a named water organism, for example **tilapia** (1 mark), with **gills** for breathing in water, fins for swimming or a streamlined body (1 mark).
- R - a named tree-living organism, for example **a monkey, a squirrel or a bird** (1 mark), with **limbs or a tail** for gripping branches, claws for climbing or wings for flying (1 mark).
- **Marking note:** The feature must match the organism named; a feature with no organism earns no mark.*

(d) [2 marks]

- **The fish breathes through gills, which work only in water** (1 mark).
- **On land the gills dry up, so the fish cannot take in oxygen and it suffocates** (1 mark).
- **Marking note:** 'There is no water' alone earns 1 mark; the gills must be mentioned for the second.*

Question 3 [Total: 10 marks]**(a) [4 marks]**

- (i) **A living part of the ecosystem** (1 mark) **plus a correct example** (1 mark).
- (ii) **A non-living part of the ecosystem** (1 mark) **plus a correct example** (1 mark).
- **Marking note:** An example that does not match the definition earns no mark.*

(b) [3 marks]

- Grass - **producer, makes its own food using sunlight** (1 mark).
- Grasshopper - **consumer, feeds on plants** (1 mark).
- Fungus - **decomposer, breaks down dead material and returns nutrients to the soil** (1 mark).
- **Marking note:** The name of the role alone is enough for the mark.*

(c) [3 marks]

- **The grass makes its own food from sunlight, so food is produced inside the ecosystem** (1 mark).
- **The grasshopper and other consumers feed on the grass** (1 mark).
- **The fungus breaks down the dead material and returns the nutrients to the soil for the grass to use again** (1 mark).
- **Marking note:** The return of nutrients is essential for the third mark.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- **The waste killed the water plants** (1 mark).
- **The fish lost their food (and the polluted water also harmed them), so their numbers fell** (1 mark).
- **The birds, which feed on the fish, had no food and left** (1 mark).
- *Marking note: The chain must run plants to fish to birds; a single sentence naming pollution earns 1 mark only.*

(b) [2 marks]

- **The components are linked and depend on one another (interdependence)** (1 mark).
- **A change in one part affects the other parts, including people** (1 mark).
- *Marking note: 'Pollution is bad' earns no mark.*

(c) [3 marks]

- Any THREE of: **stop the untreated waste or require the factory to treat it**; **enforce by-laws and punish offenders**; **stop the dumping of refuse at the lagoon**; **replant the plants around the lagoon**; **close the lagoon to fishing for a season**; **educate the community** (1 mark each).
- *Marking note: 'Pray for the lagoon' and other non-scientific measures earn no mark.*

(d) [2 marks]

- Any ONE of: **competition**; **pollination**; **shelter or nesting**; **seed dispersal** (1 mark).
- **A correct example of the interaction named** (1 mark).
- *Marking note: Predation is a feeding relationship and earns no mark here.*

Question 5 [Total: 10 marks]**(a) [4 marks]**

- Any FOUR, each with a correct reason: **water - the medium in which the organisms live**; **sand or gravel - holds the roots of the plants**; **water plants - make food and give out oxygen**; **small fish or snails - the consumers**; **light - needed by the plants to make food**; **stones - shelter for the animals** (1 mark each).
- *Marking note: An item named with no reason earns no mark.*

(b) [2 marks]

- **Biotic items correctly grouped** (1 mark).
- **Abiotic items correctly grouped** (1 mark).
- *Marking note: One item in the wrong group loses the mark for that group only.*

(c) [2 marks]

- Any TWO of: **remove dead material and uneaten food**; **top up or partly change the water**; **feed the fish lightly**; **keep the tank in the light but out of the hot midday Sun**; **check the plants** (1 mark each).
- **Marking note:** 'Clean it' alone is too vague.*

(d) [2 marks]

- A clear position, for example **I agree, the tank is a model rather than a true ecosystem** (1 mark).
- **Reason:** a true ecosystem is self-sustaining, while the tank is too small and needs food to be added and waste removed (1 mark).
- **Marking note:** Either position may be taken; the mark is for the reason, which must mention self-sustaining or outside help.*

Question 6 [Total: 8 marks]**(a)** [3 marks]

- Any THREE of: **animals lose shelter and food and move away or die**; **the soil is eroded and loses fertility**; **the area becomes hotter and drier**; **streams dry up**; **shade-loving plants die** (1 mark each).
- **Marking note:** 'The forest is finished' is a restatement and earns no mark.*

(b) [3 marks]

- Any THREE of: **timber, firewood, fruits, bush meat or herbs**; **clean air and oxygen**; **protection of the soil from erosion**; **water and rainfall**; **shade and a cooler climate**; **a place for study and tourism** (1 mark each).
- **Marking note:** Two answers about wood and firewood count as one.*

(c) [2 marks]

- **A workable activity, for example raising and planting seedlings of local trees at the start of the rains** (1 mark).
- **A way of judging success, for example counting the surviving seedlings or noting the return of grasses, insects and birds** (1 mark).
- **Marking note:** An activity with no way of judging success earns 1 mark only.*

CURRICULUM ASSESSMENT BLUEPRINT**Strand:** 3: Systems | **Sub-strand:** 3: Ecosystem | **Class:** Basic 7 (JHS 1)**Content Standard:** **B7/JHS1.3.3.1** - Recognise the components of and interdependences in an ecosystem, and appreciate their inte

Electricity and Elect

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	12.1 Generation of Thermal and Nuclear Ele...	Remember
2	C	12.1 Generation of Thermal and Nuclear Ele...	Understand
3	C	12.1 Generation of Thermal and Nuclear Ele...	Remember
4	D	12.1 Generation of Thermal and Nuclear Ele...	Remember
5	C	12.1 Generation of Thermal and Nuclear Ele...	Understand
6	A	12.1 Generation of Thermal and Nuclear Ele...	Remember
7	B	12.1 Generation of Thermal and Nuclear Ele...	Understand
8	B	12.1 Generation of Thermal and Nuclear Ele...	Analyse
9	C	12.2 Effects of Electricity Generation on ...	Remember
10	A	12.2 Effects of Electricity Generation on ...	Understand
11	B	12.2 Effects of Electricity Generation on ...	Understand
12	A	12.2 Effects of Electricity Generation on ...	Remember
13	B	12.2 Effects of Electricity Generation on ...	Understand
14	D	12.2 Effects of Electricity Generation on ...	Evaluate
15	B	12.2 Effects of Electricity Generation on ...	Understand
16	B	12.3 Basic Electronic Components and Their...	Remember
17	A	12.3 Basic Electronic Components and Their...	Remember
18	A	12.3 Basic Electronic Components and Their...	Understand
19	D	12.3 Basic Electronic Components and Their...	Remember
20	C	12.3 Basic Electronic Components and Their...	Remember
21	A	12.3 Basic Electronic Components and Their...	Remember
22	C	12.3 Basic Electronic Components and Their...	Remember
23	D	12.3 Basic Electronic Components and Their...	Understand
24	A	12.3 Basic Electronic Components and Their...	Apply
25	C	12.4 The P-N Junction Diode	Remember
26	B	12.4 The P-N Junction Diode	Understand
27	D	12.4 The P-N Junction Diode	Remember
28	C	12.4 The P-N Junction Diode	Analyse
29	B	12.4 The P-N Junction Diode	Remember
30	C	12.5 Assembling a Simple Electronic Circuit	Understand
31	B	12.5 Assembling a Simple Electronic Circuit	Understand
32	A	12.5 Assembling a Simple Electronic Circuit	Remember
33	A	12.5 Assembling a Simple Electronic Circuit	Understand
34	D	12.5 Assembling a Simple Electronic Circuit	Apply
35	D	12.5 Assembling a Simple Electronic Circuit	Understand
36	C	12.6 How the Components Affect One Another	Analyse

Q	Ans	Curriculum Area	Cognitive Level
37	D	12.6 How the Components Affect One Another	Analyse
38	D	12.6 How the Components Affect One Another	Analyse
39	A	12.6 How the Components Affect One Another	Understand
40	B	12.6 How the Components Affect One Another	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 2 (5%) | Evaluate: 1 (2%) | Remember: 16 (40%) | Understand: 16 (40%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [4 marks]

- W - **battery (two dry cells in series)** (1 mark).
- X - **switch** (1 mark).
- Y - **resistor** (1 mark).
- Z - **LED (light-emitting diode)** (1 mark).
- **Marking note: 'Bulb' or 'lamp' written for Z earns no mark.***

(b) [4 marks]

- W - **supplies the electrical energy (drives the current)** (1 mark).
- X - **completes or breaks the circuit, switching the current on and off** (1 mark).
- Y - **limits the current and so protects the LED** (1 mark).
- Z - **gives out light when current passes through it** (1 mark).
- **Marking note: 'It carries current' is true of every part and earns no mark.***

(c) [2 marks]

- **No, the LED is not lit** (1 mark).
- **The switch is open, so the circuit is incomplete and no current flows** (1 mark).
- **Marking note: No mark for the reason if the candidate says the LED is lit.***

(d) [2 marks]

- **Each dry cell supplies 1.5 V** (1 mark).
- **Two cells joined in series add up: $1.5 + 1.5 = 3 \text{ V}$** (1 mark).
- **Marking note: The word 'series' or a correct description of it is needed for the second mark.***

Question 2 [Total: 10 marks]

(a) [4 marks]

- A - **furnace or boiler** (1 mark).
- B - **turbine** (1 mark).
- C - **generator** (1 mark).
- D - **transmission pylon (power lines)** (1 mark).
- **Marking note: 'Chimney' written for A earns no mark; the chimney only carries away the waste gases.***

(b) [3 marks]

- **A: chemical energy of the fuel to heat energy** (1 mark).
- **B: heat energy of the steam to kinetic energy of the turbine** (1 mark).
- **C: kinetic energy to electrical energy** (1 mark).
- **Marking note:** The three changes must be in the right order.*

(c) [3 marks]

- Any THREE of: **the heat comes from splitting uranium nuclei in a reactor, not from burning fuel**; **no smoke or carbon dioxide is given off**; **a very small mass of fuel gives a great deal of energy**; **it produces radioactive waste**; **it is far more costly to build** (1 mark each).
- **Marking note:** A point that is true of BOTH stations, such as 'it uses a turbine', earns no mark.*

Question 3 [Total: 10 marks]**(a) [4 marks]**

- Any FOUR of: **carbon dioxide adds to global warming**; **smoke and soot cause breathing diseases**; **sulphur dioxide causes acid rain**; **hot water kills fish in the river**; **oil spills pollute land and water**; **noise pollution**; **a non-renewable fuel is used up** (1 mark each).
- **Marking note:** Two answers that both say 'air pollution' count as one.*

(b) [3 marks]

- Any THREE of: **fit filters or scrubbers to the chimneys**; **use cleaner fuel such as natural gas**; **use more solar, wind and hydro power**; **cool the water before returning it**; **site stations away from homes and plant trees**; **save electricity so that less is generated** (1 mark each).
- **Marking note:** 'Stop making electricity' is not a practical measure and earns no mark.*

(c) [3 marks]

- **Agree that it gives off no smoke and no carbon dioxide** (1 mark).
- **But it produces radioactive waste that stays dangerous for a very long time** (1 mark).
- **And an accident or leak would release harmful radiation (accept: warm water still harms aquatic life)** (1 mark).
- **Marking note:** A reply that merely says 'he is wrong' earns no mark.*

Question 4 [Total: 10 marks]**(a) [4 marks]**

- **Correct symbol for the two cells in series** (1 mark).
- **Correct symbols for the switch and the resistor** (1 mark).

- **Correct symbol for the LED (triangle and bar)** (1 mark).
- **A complete single loop with all four components in series and labelled** (1 mark).
- **Marking note:** A circuit with a gap anywhere, other than at the switch, loses the completeness mark.*

(b) [3 marks]

- **The longer leg (positive leg or anode) is joined towards the positive terminal of the battery** (1 mark).
- **Connected this way the LED is forward biased and lights** (1 mark).
- **Turned round it is reverse biased, no current flows and it does not light** (1 mark).
- **Marking note:** 'It will burn out' is wrong for a reversed LED at this voltage and earns no mark.*

(c) [3 marks]

- (i) **The circuit is complete, current flows and the LED lights** (1 mark).
- (ii) **The path is broken, the current stops and the LED goes out** (1 mark).
- **The resistor limits the current and protects the LED** (1 mark).
- **Marking note:** 'The electricity rests' earns no mark.*

Question 5 [Total: 10 marks]**(a) [4 marks]**

- Resistor - **limits or controls the current** (1 mark).
- Capacitor - **stores charge and releases it later** (1 mark).
- Diode - **allows current in one direction only** (1 mark).
- Inductor - **opposes a change in the current (stores energy in a magnetic field)** (1 mark).
- **Marking note:** A function written against the wrong component earns no mark.*

(b) [3 marks]

- (i) **A component made by joining a P region and an N region of semiconductor, which passes current one way only** (1 mark).
- (ii) **The P region and the N region** (1 mark); **the P region is joined to the positive terminal** (1 mark).
- **Marking note:** Accept 'positive region' for P and 'negative region' for N.*

(c) [3 marks]

- Any THREE of: **unplug the gadget and remove the cells first**; **never open a gadget connected to the mains; leave mains parts to the teacher**; **work under supervision with the proper tools**; **keep the parts safely and sketch the arrangement before removing anything**; **handle sharp edges with care and**

wash hands afterwards** (1 mark each).

- *Marking note: 'Be careful' alone is too vague for a mark.*

Question 6 [Total: 8 marks]

(a) [3 marks]

- **The LED becomes dimmer** (1 mark).

- **The total resistance of the circuit increases** (1 mark).

- **So the current falls, and the brightness depends on the current** (1 mark).

- *Marking note: No reason marks if the candidate says the LED becomes brighter.*

(b) [3 marks]

- **The LED did not light** (1 mark).

- **The diode was reverse biased and a diode conducts in one direction only** (1 mark).

- **So no current flowed round the circuit** (1 mark).

- *Marking note: 'The diode was spoilt' earns no mark.*

(c) [2 marks]

- **The components depend on one another; a change in one affects the working of the others** (1 mark).

- **Example: the resistor fixes the current, and the current decides the brightness of the LED (or protects it from damage)** (1 mark).

- *Marking note: A definition with no example earns 1 mark only.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 2: Electricity and Electronics |
 Class: Basic 7 (JHS 1)

Content Standard: **B7/JHS1.4.2.1** - Demonstrate understanding of forms of electricity, its generation and effects on the environment

Energy

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	11.1 Forms of Energy	Remember
2	B	11.1 Forms of Energy	Remember
3	B	11.1 Forms of Energy	Remember
4	D	11.1 Forms of Energy	Remember
5	D	11.1 Forms of Energy	Remember
6	A	11.1 Forms of Energy	Remember
7	B	11.1 Forms of Energy	Remember
8	D	11.1 Forms of Energy	Understand
9	D	11.2 Potential, Kinetic and Mechanical Energy	Remember
10	A	11.2 Potential, Kinetic and Mechanical Energy	Understand
11	C	11.2 Potential, Kinetic and Mechanical Energy	Understand
12	C	11.2 Potential, Kinetic and Mechanical Energy	Understand
13	A	11.2 Potential, Kinetic and Mechanical Energy	Understand
14	C	11.2 Potential, Kinetic and Mechanical Energy	Understand
15	A	11.2 Potential, Kinetic and Mechanical Energy	Understand
16	C	11.3 Daily Applications of Energy	Understand
17	A	11.3 Daily Applications of Energy	Understand
18	A	11.3 Daily Applications of Energy	Understand
19	C	11.3 Daily Applications of Energy	Understand
20	A	11.3 Daily Applications of Energy	Apply
21	B	11.3 Daily Applications of Energy	Analyse
22	B	11.3 Daily Applications of Energy	Understand
23	A	11.4 Calculating Potential and Kinetic Energy	Remember
24	D	11.4 Calculating Potential and Kinetic Energy	Remember
25	C	11.4 Calculating Potential and Kinetic Energy	Apply
26	B	11.4 Calculating Potential and Kinetic Energy	Apply
27	D	11.4 Calculating Potential and Kinetic Energy	Analyse
28	D	11.4 Calculating Potential and Kinetic Energy	Analyse
29	B	11.5 Transfer of Heat	Remember
30	B	11.5 Transfer of Heat	Remember
31	C	11.5 Transfer of Heat	Remember
32	B	11.5 Transfer of Heat	Understand
33	C	11.5 Transfer of Heat	Understand
34	C	11.5 Transfer of Heat	Analyse
35	B	11.6 Characteristics of Light	Understand
36	D	11.6 Characteristics of Light	Apply

Q	Ans	Curriculum Area	Cognitive Level
37	D	11.6 Characteristics of Light	Remember
38	A	11.6 Characteristics of Light	Understand
39	A	11.6 Characteristics of Light	Remember
40	C	11.6 Characteristics of Light	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 4 (10%) | Remember: 15 (37%) | Understand: 16 (40%)

Energy

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 12 marks]****(a)** [2 marks]

- A - **potential energy** (1 mark).
- C - **kinetic energy** (1 mark).
- **Marking note:** 'Gravitational potential energy' is accepted for A.*

(b) [3 marks]

- **The potential energy decreases as the height falls** (1 mark).
- **The kinetic energy increases as the speed rises (at B the ball has both)** (1 mark).
- **The potential energy lost is changed into kinetic energy, so the total mechanical energy stays the same** (1 mark).
- **Marking note:** 'The ball moves faster' alone earns 1 mark only.*

(c) [4 marks]

- (i) **$PE = mgh = 0.5 \times 10 \times 3.2$** (1 mark) **$= 16 \text{ J}$** (1 mark).
- (ii) **$KE = 16 \text{ J}$, so $\frac{1}{2} \times 0.5 \times v^2 = 16$, $v^2 = 64$** (1 mark) **$v = 8 \text{ m/s}$** (1 mark).
- **Marking note:** Award the method marks for a correct formula and correct substitution even if the arithmetic is wrong.*

(d) [3 marks]

- **There is friction between the ball and the slope, and air resistance** (1 mark).
- **Some of the energy is used to do work against friction, so less becomes kinetic energy** (1 mark).
- **The missing energy becomes heat (and a little sound); it is not destroyed** (1 mark).
- **Marking note:** 'The energy is lost' without naming heat or sound earns no mark for the third point.*

Question 2 [Total: 10 marks]**(a)** [2 marks]

- (i) **The ability (capacity) to do work** (1 mark).
- (ii) **The joule (J)** (1 mark).
- **Marking note:** 'Power' or 'watt' earns no mark in (ii).*

(b) [4 marks]

- Any FOUR forms, each with a correct source, for example **heat** - a cooking fire; **light** - a lamp; **sound** - a drum; **solar** - the Sun; **electrical** - a dry cell; **chemical** - food or charcoal; **nuclear** - a nuclear reactor (1 mark each).

- *Marking note: A form named without a source earns no mark.*

(c) [4 marks]

- Electric bulb - **electrical to light (and heat)** (1 mark).

- Dry cell - **chemical to electrical** (1 mark).

- Electric fan - **electrical to kinetic (mechanical)** (1 mark).

- Solar panel - **solar to electrical** (1 mark).

- *Marking note: The change must be written as one form to another; a single word earns no mark.*

Question 3 [Total: 10 marks]**(a)** [2 marks]

- (i) **The sum of the potential energy and the kinetic energy of a body** (1 mark).

- (ii) **$PE = mgh$, with m = mass, g = acceleration due to gravity and h = height** (1 mark).

- *Marking note: The expression must be accompanied by the meanings of the letters.*

(b) [4 marks]

- (i) **$KE = \frac{1}{2} \times 2\,000 \times 10^2$** (1 mark) **$= 100\,000\text{ J}$** (1 mark).

- (ii) **$KE = \frac{1}{2} \times 2\,000 \times 20^2$** (1 mark) **$= 400\,000\text{ J}$** (1 mark).

- *Marking note: Accept 100 kJ and 400 kJ. A missing unit loses the final mark of the part only.*

(c) [2 marks]

- **Doubling the speed makes the kinetic energy four times as large, since KE depends on the square of the speed** (1 mark).

- **The lorry then needs a much longer distance to stop and does far more damage in a crash** (1 mark).

- *Marking note: 'Speeding is bad' without reference to the energy earns no mark.*

(d) [2 marks]

- **The mass of the body** (1 mark).

- **The height above the ground** (accept: the acceleration due to gravity) (1 mark).

- *Marking note: 'Speed' is a factor for kinetic energy and earns no mark here.*

Question**(a) [3 marks]**

- **Pin P** (1 mark).
- **Metal is a good conductor, so heat travels quickly along the rod to the far end** (1 mark).
- **The wax at that end melts first and releases the pin (the other two rods are poor conductors)** (1 mark).
- **Marking note: No reason marks if the wrong pin is named.**

(b) [2 marks]

- (i) **Conduction** (1 mark).
- (ii) **Convection** (1 mark).
- **Marking note: Radiation written for (i) earns no mark.**

(c) [2 marks]

- Any TWO of: **the same length and thickness of rod**; **the same depth in the same hot water (same temperature)**; **the same amount of wax**; **pins of the same size, started at the same time** (1 mark each).
- **Marking note: 'The material' must NOT be given; that is what is being changed.**

(d) [3 marks]

- **Radiation** (1 mark).
- **It needs no material medium; it can travel through empty space (a vacuum)** (1 mark).
- **Example: heat from the Sun reaching the Earth, or warmth felt in front of a coal pot** (1 mark).
- **Marking note: 'It does not need water' is not enough for the second mark.**

Question 5 [Total: 10 marks]**(a) [4 marks]**

- **Stand the three cards one behind the other and line up the holes, using the thread** (1 mark).
- **Place the lighted candle in line at one end and look through the nearest hole** (1 mark).
- **Observation: the flame is seen when the holes are in a straight line** (1 mark).
- **Observation: the flame cannot be seen when one card is moved out of line** (1 mark).
- **Marking note: Both observations are needed; the second is the one that proves the point.**

(b) [4 marks]

- **Reflection: light bounces back when it falls on a surface** (1 mark), **example: an image in a mirror or in still water** (1 mark).
- **Refraction: light bends as it passes from one transparent medium into another** (1 mark), **example: a pencil looking bent in a glass of water** (1 mark).
- **Marking note: An example that belongs to the other effect earns no mark.**

(c) [2 marks]

- (i) **Dispersion** (1 mark).
- (ii) **A glass prism** (1 mark).
- **Marking note: 'A mirror' earns no mark in (ii).**

Question 6 [Total: 8 marks]**(a)** [3 marks]

- **Solar energy to electrical energy** (1 mark).
- Any TWO of: **Ghana has strong sunshine for most of the year**; **sunlight is free, so running costs are low**; **the panels work when the mains supply fails**; **they are clean, quiet and give off no smoke** (1 mark each).
- **Marking note: 'It is modern' earns no mark.**

(b) [3 marks]

- Any THREE of: **switch off lights and fans in empty rooms**; **use energy-saving (LED) bulbs**; **use daylight instead of lights**; **service the appliances**; **appoint class energy monitors**; **cook with a lid on the pot on an efficient stove** (1 mark each).
- **Marking note: Two answers about switching things off count as one.**

(c) [2 marks]

- **The water at the bottom is heated, expands, becomes less dense and rises, while cooler water sinks to take its place** (1 mark).
- **These convection currents carry the heat through the whole pot of water** (1 mark).
- **Marking note: The word convection alone, with no mention of the rising and sinking water, earns 1 mark.**

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 1: Energy | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.4.1.1** - Demonstrate an understanding of

forms of energy and their daily applications. **B7/JHS1.4.1.2*

Farming Systems

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	10.1 Meaning and Types of Farming Systems	Remember
2	A	10.1 Meaning and Types of Farming Systems	Understand
3	D	10.1 Meaning and Types of Farming Systems	Understand
4	B	10.1 Meaning and Types of Farming Systems	Remember
5	A	10.1 Meaning and Types of Farming Systems	Remember
6	D	10.1 Meaning and Types of Farming Systems	Remember
7	D	10.2 Land Rotation	Remember
8	B	10.2 Land Rotation	Remember
9	B	10.2 Land Rotation	Remember
10	D	10.2 Land Rotation	Understand
11	D	10.2 Land Rotation	Understand
12	B	10.2 Land Rotation	Understand
13	C	10.2 Land Rotation	Analyse
14	A	10.3 Crop Rotation	Remember
15	B	10.3 Crop Rotation	Understand
16	B	10.3 Crop Rotation	Remember
17	A	10.3 Crop Rotation	Understand
18	C	10.3 Crop Rotation	Understand
19	B	10.3 Crop Rotation	Analyse
20	A	10.3 Crop Rotation	Apply
21	A	10.4 Mixed Cropping	Apply
22	C	10.4 Mixed Cropping	Understand
23	A	10.4 Mixed Cropping	Understand
24	C	10.4 Mixed Cropping	Understand
25	D	10.4 Mixed Cropping	Analyse
26	A	10.4 Mixed Cropping	Analyse
27	C	10.5 Mixed Farming	Apply
28	B	10.5 Mixed Farming	Understand
29	D	10.5 Mixed Farming	Understand
30	B	10.5 Mixed Farming	Understand
31	B	10.5 Mixed Farming	Understand
32	C	10.5 Mixed Farming	Analyse
33	C	10.6 Organic Farming and the Choice of a S...	Remember
34	D	10.6 Organic Farming and the Choice of a S...	Remember
35	C	10.6 Organic Farming and the Choice of a S...	Understand
36	D	10.6 Organic Farming and the Choice of a S...	Understand
37	A	10.6 Organic Farming and the Choice of a S...	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	C	10.6 Organic Farming and the Choice of a S...	Analyse
39	C	10.6 Organic Farming and the Choice of a S...	Evaluate
40	D	10.6 Organic Farming and the Choice of a S...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 3 (7%) | Evaluate: 2 (5%) | Remember: 11 (27%) | Understand: 18 (45%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [3 marks]**

- A - ****mixed cropping**** (1 mark).
- B - ****mixed farming**** (1 mark).
- C - ****land rotation**** (accept: shifting cultivation) (1 mark).
- *Marking note: Mixed farming written for A, or mixed cropping for B, earns no mark; the two must be told apart.*

(b) [4 marks]

- Any TWO advantages: ****income from two sources****; ****manure for the crops****; ****crop residues as feed****; ****land and labour used all year**** (1 mark each).
- Any TWO difficulties: ****more capital needed****; ****more labour and skill needed****; ****animals may stray into the crops****; ****animal diseases must be controlled**** (1 mark each).
- *Marking note: A general answer such as 'it is good' earns no mark.*

(c) [3 marks]

- ****The plots have lost their fertility after cropping**** (1 mark).
- ****The bush that grows and decays on them returns nutrients and restores the fertility**** (1 mark).
- ****Reason: population growth has reduced the land available, so fallow periods are too short**** (accept: land is now scarce or expensive) (1 mark).
- *Marking note: 'He is lazy' earns no mark.*

Question 2 [Total: 12 marks]**(a) [2 marks]**

- (i) ****Crop rotation**** (1 mark).
- (ii) ****Different crops follow one another in a planned order on the same land, and the plan is repeated**** (1 mark).
- *Marking note: 'Land rotation' earns no mark; the land does not change.*

(b) [3 marks]

- ****Groundnut is a legume**** (1 mark).
- ****Bacteria in the nodules on its roots fix nitrogen from the air**** (1 mark).
- ****The soil is therefore enriched for the crop that follows (and the farmer saves on fertilizer)**** (1 mark).
- *Marking note: 'It is a good food' alone earns no mark in this part.*

(c) [4 marks]

- Any FOUR of: **soil fertility is maintained**; **pests and diseases are controlled**; **weeds are kept down**; **the soil structure is improved by different rooting depths**; **the resting year lets the land recover**; **a variety of food and income** (1 mark each).
- **Marking note:** Two answers that both say 'the soil becomes rich' count as one.*

(d) [3 marks]

- **A plan naming all four in a sensible order over four years** (2 marks; 1 mark if one item is missing or out of place).
- **A sound reason, for example the legume follows the heavy feeder to restore nitrogen, or the resting year comes last before the plan is repeated** (1 mark).
- **Marking note:** Any order is accepted provided the reason given supports it.*

Question 3 [Total: 10 marks]**(a)** [3 marks]

- Opoku - **land rotation** (accept: shifting cultivation) (1 mark).
- Mariama - **crop rotation** (1 mark).
- Yaw - **mixed cropping** (1 mark).
- **Marking note:** The names must match the descriptions exactly.*

(b) [4 marks]

- Opoku - **the piece of land changes (the crop may stay the same)** (1 mark).
- Mariama - **the crop changes, the land stays the same** (1 mark).
- Opoku - **a large area of land is needed** (1 mark).
- Mariama - **only a small area is needed** (1 mark).
- **Marking note:** A row left blank earns no mark; both columns must be completed.*

(c) [3 marks]

- Any TWO of: **several foods from one small plot**; **insurance if one crop fails**; **the soil is covered, so less erosion and fewer weeds**; **the legume adds nitrogen** (1 mark each).
- Any ONE of: **the crops compete for light, water and nutrients**; **machines cannot easily be used**; **the crops ripen at different times, so more labour is needed** (1 mark).
- **Marking note:** A difficulty written as an advantage earns no mark.*

Question 4**(a)** [2 marks]

- **Mixed farming** (1 mark).
- **Reason: she keeps animals and grows crops on the same holding at the same time** (1 mark).
- **Marking note: 'Mixed cropping' earns no mark.**

(b) [4 marks]

- Fowls to crops - any TWO of: **droppings used as manure**; **they eat insect pests**; **they scratch out weed seeds** (1 mark each).
- Crops to fowls - any TWO of: **maize grain and bran as feed**; **green leaves and crop residues as feed**; **stalks used as litter or for the house** (1 mark each).
- **Marking note: The same point written twice in different words counts once.**

(c) [2 marks]

- Any TWO of: **compost or allow the droppings to rot before use**; **dig them into the soil and keep them off the leaves**; **do not apply shortly before harvesting leafy vegetables**; **wash hands and wash the vegetables well** (1 mark each).
- **Marking note: 'Use gloves' is accepted as one answer.**

(d) [2 marks]

- Any TWO of: **stop using artificial fertilizer**; **stop using chemical pesticides or weedicides and use natural methods**; **check that the poultry feed and any water running onto the plot are free of chemicals**; **keep records and obtain proper certification** (1 mark each).
- **Marking note: 'Grow more vegetables' earns no mark.**

Question 5 [Total: 10 marks]**(a)** [2 marks]

- **Farming without artificial fertilizers and chemical pesticides or weedicides** (1 mark).
- **Using compost, manure and natural methods of pest and weed control instead** (1 mark).
- **Marking note: 'Farming without chemicals' alone earns 1 mark.**

(b) [4 marks]

- Any FOUR of: **apply compost or manure**; **practise crop rotation with legumes**; **mulch the beds**; **use neem extract, ash or other natural pest control**; **hand weed and hand pick pests**; **encourage natural enemies**; **use resistant varieties** (1 mark each).
- **Marking note: 'Do not use chemicals' is a definition, not a practice, and earns**

no mark here.*

(c) [4 marks]

- Any TWO merits: ****food free from chemical residues****; ****soil life and structure protected****; ****no pollution of water bodies****; ****less money spent on chemicals****; ****higher price for the produce**** (1 mark each).
- Any TWO demerits: ****lower yields****; ****much more labour****; ****pests and diseases harder to control quickly****; ****certification costs money and takes time**** (1 mark each).
- *Marking note: A merit written in the demerit column earns no mark.*

Question 6 [Total: 8 marks]

(a) [4 marks]

- ****A named system, for example mixed cropping (accept mixed farming or crop rotation, if well defended)**** (1 mark).
- Any THREE reasons that fit the farmer's situation: ****the land is small, so no land can be left fallow for long****; ****several foods from one plot and food spread through the year****; ****insurance if one crop fails****; ****a legume keeps the soil fertile without costly fertilizer****; ****the work can be done by hand**** (1 mark each).
- *Marking note: The reasons must refer to this farmer's land, money or family; general praise of the system earns no mark.*

(b) [2 marks]

- ****Land rotation needs a large area, with several plots lying fallow at a time**** (1 mark).
- ****On two acres the fallow would be too short to restore fertility, and the resting land would feed nobody**** (1 mark).
- *Marking note: 'It is an old system' earns no mark.*

(c) [2 marks]

- Any TWO of: ****a demonstration plot in the school garden shown to the farmers****; ****an open day, durbar or exhibition with posters****; ****simple leaflets or posters in the local language****; ****a talk on local radio**** (1 mark each).
- *Marking note: 'Tell them' is too vague for a mark.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 3: Systems | ****Sub-strand:** 4: Farming Systems | ****Class:** Basic 7 (JHS 1)******

****Content Standard:**** ****B7/JHS1.3.4.1**** - Demonstrate an understanding of the differences among the various farming systems: Land Rotation, Crop Rotation, Mixed Cropping, Mixed Farming



Force and Motion

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	14.1 Forces, Balanced and Unbalanced Force...	Remember
2	A	14.1 Forces, Balanced and Unbalanced Force...	Remember
3	D	14.1 Forces, Balanced and Unbalanced Force...	Understand
4	D	14.1 Forces, Balanced and Unbalanced Force...	Understand
5	A	14.1 Forces, Balanced and Unbalanced Force...	Understand
6	B	14.1 Forces, Balanced and Unbalanced Force...	Apply
7	C	14.1 Forces, Balanced and Unbalanced Force...	Remember
8	D	14.1 Forces, Balanced and Unbalanced Force...	Analyse
9	A	14.2 Newton's First Law of Motion	Remember
10	B	14.2 Newton's First Law of Motion	Remember
11	D	14.2 Newton's First Law of Motion	Understand
12	B	14.2 Newton's First Law of Motion	Understand
13	C	14.2 Newton's First Law of Motion	Understand
14	C	14.2 Newton's First Law of Motion	Analyse
15	B	14.2 Newton's First Law of Motion	Remember
16	A	14.3 Applications of Newton's First Law in...	Understand
17	A	14.3 Applications of Newton's First Law in...	Understand
18	D	14.3 Applications of Newton's First Law in...	Analyse
19	B	14.3 Applications of Newton's First Law in...	Analyse
20	A	14.3 Applications of Newton's First Law in...	Understand
21	D	14.3 Applications of Newton's First Law in...	Analyse
22	B	14.4 Magnets: Their Behaviour and Uses	Remember
23	D	14.4 Magnets: Their Behaviour and Uses	Remember
24	C	14.4 Magnets: Their Behaviour and Uses	Remember
25	C	14.4 Magnets: Their Behaviour and Uses	Remember
26	A	14.4 Magnets: Their Behaviour and Uses	Understand
27	B	14.4 Magnets: Their Behaviour and Uses	Analyse
28	C	14.4 Magnets: Their Behaviour and Uses	Understand
29	A	14.4 Magnets: Their Behaviour and Uses	Understand
30	B	14.5 Simple Machines and the Classes of Le...	Remember
31	D	14.5 Simple Machines and the Classes of Le...	Understand
32	C	14.5 Simple Machines and the Classes of Le...	Remember
33	D	14.5 Simple Machines and the Classes of Le...	Understand
34	A	14.5 Simple Machines and the Classes of Le...	Remember
35	C	14.5 Simple Machines and the Classes of Le...	Apply
36	A	14.6 Work Input, Work Output and Efficiency	Remember

Q	Ans	Curriculum Area	Cognitive Level
37	D	14.6 Work Input, Work Output and Efficiency	Understand
38	B	14.6 Work Input, Work Output and Efficiency	Apply
39	B	14.6 Work Input, Work Output and Efficiency	Understand
40	C	14.6 Work Input, Work Output and Efficiency	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 3 (7%) | Remember: 14 (35%) | Understand: 17 (42%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 10 marks]

(a) [2 marks]

- (i) **A push or a pull acting on a body** (1 mark).
- (ii) **The newton (N)** (1 mark).
- **Marking note: 'Energy used' earns no mark for (i).**

(b) [3 marks]

- (i) **500 N - 420 N** (1 mark) **= 80 N** (1 mark, unit required).
- (ii) **Towards the Blue House team (the direction of the larger force)** (1 mark).
- **Marking note: An answer of 920 N shows that the directions have been ignored; no marks.**

(c) [3 marks]

- **Balanced: equal forces in opposite directions, resultant zero** (1 mark).
- **Unbalanced: the forces do not cancel; a resultant force remains** (1 mark).
- **A body at rest stays at rest under balanced forces but starts to move under unbalanced forces** (1 mark).
- **Marking note: Credit a clear labelled sketch that makes the same points.**

(d) [2 marks]

- **The tendency of a body to resist a change in its state of rest or of motion** (1 mark).
- **It is measured by the mass of the body** (1 mark).
- **Marking note: 'Weight' is not accepted for the second mark.**

Question 2 [Total: 8 marks]

(a) [2 marks]

- **A body continues in its state of rest or of uniform motion in a straight line** (1 mark).
- **unless an unbalanced (external) force acts on it** (1 mark).
- **Marking note: Only one mark if the state of uniform motion is left out.**

(b) [3 marks]

- (i) **The ball was at rest and the push was an unbalanced force that set it moving** (1 mark).
- (ii) **The moving ball would have continued in a straight line** (1 mark); **the block (and friction) exerted an unbalanced force that stopped it** (1 mark).

- *Marking note: 'It became tired' earns no mark.*

(c) [3 marks]

- **The statement is not correct** (1 mark).
- **A body keeps moving at constant velocity when no unbalanced force acts on it** (1 mark).
- **A force is needed on Earth only to overcome friction and air resistance** (1 mark).
- *Marking note: No explanation marks if the candidate agrees with the statement.*

Question 3 [Total: 10 marks]

(a) [3 marks]

- **The passengers were moving forward with the vehicle** (1 mark).
- **The braking force stopped the vehicle** (1 mark).
- **The passengers' bodies continued moving forward because of inertia** (1 mark).
- *Marking note: The word inertia, or a correct description of it, is needed for the third mark.*

(b) [2 marks]

- **It provides the backward force that stops the passenger with the vehicle** (1 mark).
- **So the passenger is held in the seat and is not thrown forward and injured** (1 mark).
- *Marking note: 'It makes the person comfortable' earns no mark.*

(c) [3 marks]

- Any THREE of: **passengers thrown backwards when a vehicle starts off**;
tightening a hoe or cutlass blade by knocking the handle;
beating dust out of a mat;
pulling a cloth from under a glass;
shaking water off an umbrella (1 mark each).
- *Marking note: Three versions of the same braking example count once.*

(d) [2 marks]

- Any TWO of: **it explains the need for seat belts and other safety measures, so preventing injury**;
it guides safe driving, such as keeping a safe distance;
it is used in designing vehicles, machines and tools (1 mark each).
- *Marking note: 'It is in the syllabus' earns no mark.*

Question**(a)** [4 marks]

- (i) **P is a north pole** (1 mark); **Q is a north pole** (1 mark); **because the magnets repelled each other and like poles repel (accept: the other ends are marked S)** (1 mark).
- (ii) **North pole** (1 mark).
- **Marking note: Accept 'north-seeking pole'.**

(b) [3 marks]

- **Like poles repel each other** (1 mark).
- **Unlike poles attract each other (a magnet attracts iron)** (1 mark).
- **A freely suspended magnet comes to rest in the north-south direction** (1 mark).
- **Marking note: 'It is strong' is not a characteristic.**

(c) [2 marks]

- **Bring each bar near a known suspended magnet (or hang each bar freely)** (1 mark).
- **The bar that repels is the magnet, since repulsion is the sure test (accept: the bar that comes to rest north-south is the magnet)** (1 mark).
- **Marking note: An answer based on attraction alone earns no second mark.**

(d) [3 marks]

- Any THREE of: **in a compass for finding direction**; **in loudspeakers, earphones and phones**; **in electric motors and generators**; **in fridge and cupboard door catches**; **in cranes for lifting scrap iron**; **for separating iron from other waste** (1 mark each).
- **Marking note: Two answers naming devices that work in the same way, such as a radio and an earphone, count once.**

Question 5 [Total: 12 marks]**(a)** [3 marks]

- (i) **A device that makes work easier to do** (1 mark).
- (ii) Any TWO of: **inclined plane, pulley, wheel and axle, screw, wedge** (1 mark each).
- **Marking note: A lever named in (ii) earns no mark, as it is excluded by the question.**

(b) [3 marks]

- A - **first-class lever** (1 mark).
- B - **second-class lever** (1 mark).
- C - **third-class lever** (1 mark).

- *Marking note: Accept 'first order', 'second order' and 'third order'.*

(c) [3 marks]

- **Fulcrum: at the wheel (the axle at the front)** (1 mark).
- **Load: in the tray, between the wheel and the handles** (1 mark).
- **Effort: at the handles, where the hands lift** (1 mark).
- *Marking note: Marks are for the positions, not merely for the words.*

(d) [3 marks]

- **The load moves a greater distance (and faster) than the effort** (1 mark).
- **The tool gives a delicate, well-controlled grip or movement** (1 mark).
- **One other example: a broom, a fishing rod, a spade or the human forearm** (1 mark).
- *Marking note: A wheelbarrow or a see-saw named as the example earns no mark.*

Question 6 [Total: 8 marks]

(a) [2 marks]

- **Work input: the work done by the effort on the machine (the energy supplied to it)** (1 mark).
- **Work output: the useful work done by the machine on the load** (1 mark).
- *Marking note: The word 'useful' is expected in the second answer.*

(b) [3 marks]

- **Efficiency = (work output / work input) x 100%** (1 mark).
- **= (1,800 / 2,400) x 100%** (1 mark).
- **= 75%** (1 mark, percentage sign required).
- *Marking note: A correct answer with no working earns 2 of the 3 marks.*

(c) [3 marks]

- (i) **600 J was used in overcoming friction and in lifting the moving parts of the machine** (1 mark); **it appears as heat (and sound), so it is wasted but not destroyed** (1 mark).
- (ii) **Oil the moving parts to reduce friction (accept: use lighter or smoother parts)** (1 mark).
- *Marking note: 'The machine is old' earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 4: Forces and Energy | **Sub-strand:** 4: Force and Motion |
 Class: Basic 7 (JHS 1)

****Content Standard:**** ****B7/JHS1.4.4.1**** - Examine the concept of motion, Newton's first law of motion, magnetic force

Human Body Digest

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	7.1 The Concept of Food and the Need to Eat	Remember
2	B	7.1 The Concept of Food and the Need to Eat	Understand
3	C	7.1 The Concept of Food and the Need to Eat	Remember
4	B	7.1 The Concept of Food and the Need to Eat	Understand
5	D	7.1 The Concept of Food and the Need to Eat	Understand
6	A	7.1 The Concept of Food and the Need to Eat	Understand
7	C	7.1 The Concept of Food and the Need to Eat	Analyse
8	A	7.2 The Parts of the Alimentary Canal	Remember
9	A	7.2 The Parts of the Alimentary Canal	Remember
10	D	7.2 The Parts of the Alimentary Canal	Remember
11	C	7.2 The Parts of the Alimentary Canal	Remember
12	B	7.2 The Parts of the Alimentary Canal	Remember
13	D	7.2 The Parts of the Alimentary Canal	Understand
14	B	7.2 The Parts of the Alimentary Canal	Remember
15	D	7.2 The Parts of the Alimentary Canal	Understand
16	A	7.3 What Happens to Food at Each Stage of ...	Understand
17	D	7.3 What Happens to Food at Each Stage of ...	Remember
18	C	7.3 What Happens to Food at Each Stage of ...	Remember
19	A	7.3 What Happens to Food at Each Stage of ...	Understand
20	A	7.3 What Happens to Food at Each Stage of ...	Analyse
21	D	7.3 What Happens to Food at Each Stage of ...	Remember
22	C	7.3 What Happens to Food at Each Stage of ...	Understand
23	C	7.3 What Happens to Food at Each Stage of ...	Understand
24	B	7.3 What Happens to Food at Each Stage of ...	Analyse
25	A	7.4 End Products of Digestion	Remember
26	B	7.4 End Products of Digestion	Remember
27	C	7.4 End Products of Digestion	Remember
28	C	7.4 End Products of Digestion	Apply
29	A	7.4 End Products of Digestion	Apply
30	B	7.4 End Products of Digestion	Understand
31	B	7.4 End Products of Digestion	Understand
32	D	7.5 Absorption and the Use of Digested Food	Remember
33	B	7.5 Absorption and the Use of Digested Food	Understand
34	B	7.5 Absorption and the Use of Digested Food	Remember
35	D	7.5 Absorption and the Use of Digested Food	Understand
36	C	7.5 Absorption and the Use of Digested Food	Analyse
37	A	7.6 Food Tests	Remember

Q	Ans	Curriculum Area	Cognitive Level
38	D	7.6 Food Tests	Remember
39	A	7.6 Food Tests	Remember
40	D	7.6 Food Tests	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 3 (7%) | Remember: 19 (47%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [6 marks]**

- A - **mouth** (1 mark).
- B - **oesophagus** (accept: gullet) (1 mark).
- C - **stomach** (1 mark).
- D - **small intestine** (1 mark).
- E - **large intestine** (accept: colon) (1 mark).
- F - **anus** (1 mark).
- *Marking note: 'Windpipe' or 'trachea' written for B earns no mark; that tube carries air, not food.**

(b) [2 marks]

- D - **completes digestion and absorbs the digested food into the blood** (1 mark).
- E - **absorbs water from the undigested material (and stores faeces)** (1 mark).
- *Marking note: 'It digests food' alone is too vague for E.**

(c) [2 marks]

- **A, B, C, D, E, F** (2 marks; 1 mark if one letter is out of place or missing).
- *Marking note: No mark if the list begins anywhere other than A.**

Question 2 [Total: 12 marks]**(a) [4 marks]**

- (i) **A substance taken into the body** (1 mark) **which provides energy, builds and repairs the body or protects it from disease** (1 mark).
- (ii) Any **FOUR** of: **carbohydrates**; **proteins**; **fats and oils**; **vitamins**; **mineral salts**; **water** (half a mark each, 2 marks for four correct).
- *Marking note: 'Something we eat' alone earns 1 mark in (i).**

(b) [4 marks]

- Any **FOUR** of: **to supply energy**; **for growth**; **to repair worn-out or damaged parts**; **to protect the body from disease (keep healthy)**; **to keep the body warm**; **to supply water and roughage for the proper working of the body** (1 mark each).
- *Marking note: 'To become fat' earns no mark.**

(c) [4 marks]

- Any THREE differences: **thin, with ribs showing / normal weight**; **dull hair, dry skin, sunken eyes / bright eyes and healthy skin**; **weak and tired / active and playful**; **often ill and heals slowly / resists disease** (1 mark each).
- **The child should be given regular balanced meals and taken to a health centre** (1 mark).
- **Marking note:** Marks are given for the comparison; 'he is sick' without a visible feature earns no mark.*

Question 3 [Total: 10 marks]**(a) [4 marks]**

- Mouth - **chewing by the teeth, moistening with saliva and the start of starch digestion** (1 mark).
- Stomach - **churning with acid and enzymes; protein digestion begins** (1 mark).
- Small intestine - **digestion is completed and the digested food is absorbed into the blood** (1 mark).
- Large intestine - **water is absorbed and the waste is stored as faeces** (1 mark).
- **Marking note:** A row that merely says 'food passes through' earns no mark.*

(b) [3 marks]

- **Physical digestion: food is broken into smaller pieces without a change in the substance** (1 mark).
- **Chemical digestion: the food substances are changed into simpler ones by enzymes** (1 mark).
- **One correct example of each from the breakfast** (1 mark).
- **Marking note:** Both examples are needed for the third mark.*

(c) [3 marks]

- **She cannot chew, so large pieces of food are swallowed** (1 mark).
- **Large pieces give the enzymes a small surface to act on, so digestion is slow and incomplete** (1 mark).
- **Advice: eat soft, mashed or well cooked food, eat slowly, or see a dentist** (1 mark).
- **Marking note:** The idea of surface area or of slower enzyme action is needed for the second mark.*

Question 4 [Total: 10 marks]

(a) [4 marks]

- A continuous tube from mouth to anus, drawn large and in pencil (1 mark).
- Any THREE further parts correctly placed and clearly named, with label lines touching the parts (1 mark each).
- *Marking note: Marks are for correct position and labelling, not for beauty; a label line that does not touch its part earns no mark.*

(b) [3 marks]

- **The statement is not correct** (1 mark).
- **Food is pushed along by waves of muscular squeezing in the wall of the canal** (1 mark).
- **This works in any position, so a person lying down can still swallow (and food travels upwards in part of the large intestine)** (1 mark).
- *Marking note: No mark for the explanation if the candidate agrees with the pupil.*

(c) [3 marks]

- **It is made up of several different organs joined together** (1 mark).
- **Each organ does a special piece of work** (1 mark).
- **The organs work together in order for one purpose, the digestion and absorption of food** (1 mark).
- *Marking note: Accept 'other organs such as the liver help it' for the third mark.*

Question 5 [Total: 10 marks]**(a)** [3 marks]

- (i) **starch to glucose (simple sugar)** (1 mark).
- (ii) **protein to amino acids** (1 mark).
- (iii) **oils and fats to fatty acids and glycerol** (1 mark).
- *Marking note: The arrow or the word 'gives' must be shown; a bare list of products earns 1 mark in total.*

(b) [3 marks]

- (i) **The passing of digested food through the wall of the alimentary canal into the blood** (1 mark).
- (ii) **The small intestine** (1 mark).
- (iii) **By the blood, through the blood vessels** (1 mark).
- *Marking note: 'Digestion' written for (i) earns no mark.*

(c) [2 marks]

- **They increase the surface area of the inner wall** (1 mark).
- **So more digested food is absorbed, and absorbed more quickly** (1 mark).
- *Marking note: 'They catch the food' earns no mark.*

(d) [2 marks]

- (i) **To release energy in the cells during respiration** (1 mark).
- (ii) **To build new tissue for growth and for the repair of worn-out parts** (1 mark).
- **Marking note: 'For strength' is accepted for (i) only if energy is mentioned.**

Question 6 [Total: 8 marks]**(a)** [4 marks]

- P - **starch** (1 mark).
- Q - **glucose (a simple or reducing sugar)** (1 mark).
- R - **protein** (1 mark).
- S - **oil or fat** (1 mark).
- **Marking note: 'Carbohydrate' is accepted for P and Q only if the candidate also writes starch or sugar.**

(b) [2 marks]

- P - any ONE of: **boiled yam**; **cassava**; **plantain**; **cocoyam**; **rice**; **kenkey or banku**; **bread** (1 mark).
- R - any ONE of: **egg white**; **meat**; **fish**; **beans**; **groundnut**; **milk** (1 mark).
- **Marking note: A food named for the wrong test earns no mark.**

(c) [2 marks]

- **The enzyme in saliva changed some of the starch into sugar** (1 mark).
- **Some starch was left, so the iodine test was still positive but weaker** (1 mark).
- **Marking note: 'The bread was wet' earns no mark; the action of saliva must be given.**

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 3: Systems | **Sub-strand:** 1: The Human Body System | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.3.1.1** - Demonstrate understanding of the concept of food, the process of digestion and its importance in humans.

> Source: NaCCA Science Common Core Programme Curriculum (B7/

Human Health

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	17.1 Food Nutrients and Their Uses in the ...	Remember
2	B	17.1 Food Nutrients and Their Uses in the ...	Understand
3	D	17.1 Food Nutrients and Their Uses in the ...	Remember
4	A	17.1 Food Nutrients and Their Uses in the ...	Understand
5	C	17.1 Food Nutrients and Their Uses in the ...	Understand
6	B	17.1 Food Nutrients and Their Uses in the ...	Remember
7	C	17.1 Food Nutrients and Their Uses in the ...	Remember
8	A	17.1 Food Nutrients and Their Uses in the ...	Remember
9	A	17.2 Deficiency Diseases and the Nutrients...	Remember
10	B	17.2 Deficiency Diseases and the Nutrients...	Remember
11	D	17.2 Deficiency Diseases and the Nutrients...	Remember
12	B	17.2 Deficiency Diseases and the Nutrients...	Remember
13	D	17.2 Deficiency Diseases and the Nutrients...	Remember
14	C	17.2 Deficiency Diseases and the Nutrients...	Remember
15	A	17.2 Deficiency Diseases and the Nutrients...	Remember
16	D	17.2 Deficiency Diseases and the Nutrients...	Analyse
17	C	17.3 Symptoms, Effects and Prevention of D...	Remember
18	B	17.3 Symptoms, Effects and Prevention of D...	Remember
19	A	17.3 Symptoms, Effects and Prevention of D...	Understand
20	C	17.3 Symptoms, Effects and Prevention of D...	Apply
21	D	17.3 Symptoms, Effects and Prevention of D...	Apply
22	B	17.3 Symptoms, Effects and Prevention of D...	Apply
23	A	17.3 Symptoms, Effects and Prevention of D...	Evaluate
24	D	17.4 The Nature and Transmission of Viral ...	Remember
25	A	17.4 The Nature and Transmission of Viral ...	Remember
26	B	17.4 The Nature and Transmission of Viral ...	Understand
27	C	17.4 The Nature and Transmission of Viral ...	Analyse
28	B	17.4 The Nature and Transmission of Viral ...	Understand
29	A	17.4 The Nature and Transmission of Viral ...	Understand
30	C	17.4 The Nature and Transmission of Viral ...	Remember
31	D	17.4 The Nature and Transmission of Viral ...	Analyse
32	C	17.5 Symptoms, Effects and Prevention of C...	Remember
33	B	17.5 Symptoms, Effects and Prevention of C...	Apply
34	A	17.5 Symptoms, Effects and Prevention of C...	Understand
35	B	17.5 Symptoms, Effects and Prevention of C...	Understand
36	D	17.5 Symptoms, Effects and Prevention of C...	Analyse
37	C	17.6 Roles of the Individual, the Communit...	Evaluate

Q	Ans	Curriculum Area	Cognitive Level
38	D	17.6 Roles of the Individual, the Communit...	Apply
39	C	17.6 Roles of the Individual, the Communit...	Remember
40	D	17.6 Roles of the Individual, the Communit...	Create

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 4 (10%) | Apply: 5 (12%) | Create: 1 (2%) | Evaluate: 2 (5%) | Remember: 19 (47%) | Understand: 9

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [3 marks]

- A - **protein** (1 mark).
- B - **vitamin C (ascorbic acid)** (1 mark).
- C - **vitamin A (accept: fats and oils)** (1 mark).
 - *Marking note: 'Vitamins' alone for B or C earns no mark; the particular vitamin is required.*

(b) [3 marks]

- **Protein - growth and repair of worn-out tissues** (1 mark).
- **Vitamin C - keeps the gums, skin and blood vessels healthy; helps wounds to heal** (1 mark).
- **Vitamin A - needed for vision, especially in dim light, and for healthy skin** (1 mark).
- *Marking note: 'It makes the body strong' is too vague for a mark.*

(c) [3 marks]

- A - **kwashiorkor** (1 mark).
- B - **scurvy** (1 mark).
- C - **night blindness** (1 mark).
- *Marking note: Accept 'protein-energy malnutrition' for A.*

(d) [3 marks]

- **Carbohydrates - one local example such as yam or kenkey** (1 mark).
- **Fats and oils - one local example such as palm oil or groundnut** (1 mark).
- **Vitamins and mineral salts - one local example such as kontomire or oranges (accept water and roughage)** (1 mark).
- *Marking note: A class named without an example earns no mark.*

Question 2 [Total: 10 marks]

(a) [3 marks]

- First child - **kwashiorkor** (1 mark).
- Second child - **rickets** (1 mark).
- Third child - **anaemia** (1 mark).
- *Marking note: Mark each diagnosis separately.*

(b) [3 marks]

- First child - **protein** (1 mark).
- Second child - **vitamin D (and calcium)** (1 mark).
- Third child - **iron** (1 mark).
- *Marking note: The nutrient must match the disease named in (a).*

(c) [4 marks]

- **Two protein-rich foods, such as fish, beans, eggs, groundnuts or milk** (1 mark each).
- **Two further effects, such as stunted growth, weakness, low resistance to infection, poor mental development or even death** (1 mark each).
- *Marking note: A food that is chiefly a carbohydrate, such as kenkey, earns no mark.*

Question 3 [Total: 8 marks]**(a)** [4 marks]

- Scurvy - **symptom: bleeding, swollen gums (accept slow healing of wounds)** (1 mark); **prevention: eat fresh fruits and vegetables rich in vitamin C** (1 mark).
- Goitre - **symptom: swelling in front of the neck** (1 mark); **prevention: use iodated salt; eat sea fish** (1 mark).
- *Marking note: A cause written in the symptom column earns no mark.*

(b) [2 marks]

- **Sunlight on the skin helps the body to make vitamin D** (1 mark).
- **Vitamin D helps the body to absorb calcium, which hardens the bones** (1 mark).
- *Marking note: 'The sun is good for the body' earns no mark.*

(c) [2 marks]

- Any TWO of: **the foods grown are mostly starchy staples, so meals are not balanced**; **the protein foods and vegetables are sold instead of eaten**; **ignorance of what each food supplies**; **poor cooking habits destroy the vitamins**; **customs that deny children certain foods** (1 mark each).
- *Marking note: 'They are poor' alone earns no mark without an explanation.*

Question 4 [Total: 10 marks]**(a)** [3 marks]

- (i) **A very small agent that can multiply only inside the living cells of a host** (1 mark).
- (ii) Any TWO of: **they spread easily, often before symptoms appear**;

****antibiotics do not cure them****; ****the viruses change into new forms****; ****there is often no specific medicine**** (1 mark each).

- ***Marking note: 'It is a germ' alone is too vague for (i).***

(b) [4 marks]

- (i) ****In droplets from coughing, sneezing or talking, breathed in or picked up from surfaces on the hands**** (1 mark).

- (ii) ****Infected people travel to markets, funerals and other gatherings in nearby communities**** (1 mark).

- (iii) ****Travellers carry it by air, road or sea, often before showing symptoms**** (1 mark).

- ****One correct symptom named, such as fever, dry cough or loss of taste and smell**** (1 mark).

- ***Marking note: 'By air' alone for (iii) is not enough; the travel of infected people must be clear.***

(c) [3 marks]

- (i) ****It spread to very many countries and continents at the same time**** (1 mark); ****and affected a very large number of people, beyond local control**** (1 mark).

- (ii) ****Ebola spreads by direct contact with body fluids, not chiefly by droplets in the air**** (1 mark).

- ***Marking note: 'Many people died' alone does not define a pandemic.***

Question 5 [Total: 12 marks]

(a) [4 marks]

- A - ****washing the hands with soap under running water**** (1 mark).

- B - ****wearing a nose mask**** (1 mark).

- C - ****keeping a social (physical) distance**** (1 mark).

- D - ****vaccination or immunisation**** (1 mark).

- ***Marking note: 'Injection' is accepted for D only if vaccination is clearly meant.***

(b) [4 marks]

- A - ****removes and destroys virus particles on the hands**** (1 mark).

- B - ****traps the droplets breathed out and keeps out those from others**** (1 mark).

- C - ****droplets fall to the ground before reaching another person**** (1 mark).

- D - ****the body makes antibodies in advance, so the person is protected**** (1 mark).

- ***Marking note: A repeat of the name of the measure, with no explanation, earns no mark.***

(c) [2 marks]

- Any TWO of: **cover the nose and mouth when coughing or sneezing**; **avoid crowded places**; **use hand sanitiser**; **isolate or quarantine infected persons**; **avoid handshaking and touching the face**; **keep rooms well ventilated** (1 mark each).

- *Marking note: A measure already shown in Fig. 2 earns no mark.*

(d) [2 marks]

- Any TWO of: **schools close and learning is lost**; **businesses close, people lose their incomes and the economy suffers**; **hospitals are overwhelmed so other patients suffer**; **travel, trade and social gatherings are restricted** (1 mark each).

- *Marking note: Two answers that both say 'people suffer' count once.*

Question 6 [Total: 8 marks]**(a)** [3 marks]

- (i) **Observe the safety measures and report symptoms early** (1 mark).
- (ii) **Educate and support one another, avoid stigma and help the health workers** (1 mark).
- (iii) **Provide testing, treatment, vaccines and accurate information, and enforce regulations** (1 mark).

- *Marking note: A role placed at the wrong level earns no mark.*

(b) [2 marks]

- **People hide their illness instead of reporting it, so cases are found late and spread unchecked** (1 mark).
- **The family is left without food and support and must move about to find help, spreading the disease** (1 mark).

- *Marking note: 'It is unkind' alone earns no mark; the effect on the spread of disease is required.*

(c) [3 marks]

- Any THREE of: **a clear heading naming the disease**; **the main symptoms**; **how the disease is spread**; **the preventive measures, with simple pictures**; **what to do and where to report, with a contact number**; **large, simple lettering in a language the people read** (1 mark each).

- *Marking note: 'Make it beautiful' earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans and the Environment | **Sub-strand:** 2: Human Health |

****Class:**** Basic 7 (JHS 1)

****Content Standard:**** ****B7/JHS1.5.2.1**** - Demonstrate knowledge of common deficiency diseases of humans,

|

Life Cycle Housefly

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	4.1 Life Cycles and Metamorphosis	Remember
2	C	4.1 Life Cycles and Metamorphosis	Remember
3	D	4.1 Life Cycles and Metamorphosis	Understand
4	C	4.1 Life Cycles and Metamorphosis	Understand
5	A	4.1 Life Cycles and Metamorphosis	Analyse
6	C	4.2 The Stages of the Life Cycle of the Housefly	Remember
7	B	4.2 The Stages of the Life Cycle of the Housefly	Remember
8	D	4.2 The Stages of the Life Cycle of the Housefly	Remember
9	B	4.2 The Stages of the Life Cycle of the Housefly	Remember
10	B	4.2 The Stages of the Life Cycle of the Housefly	Remember
11	C	4.2 The Stages of the Life Cycle of the Housefly	Understand
12	B	4.2 The Stages of the Life Cycle of the Housefly	Remember
13	D	4.2 The Stages of the Life Cycle of the Housefly	Remember
14	C	4.2 The Stages of the Life Cycle of the Housefly	Analyse
15	D	4.3 Features of Each Stage of the Housefly	Remember
16	D	4.3 Features of Each Stage of the Housefly	Understand
17	B	4.3 Features of Each Stage of the Housefly	Understand
18	A	4.3 Features of Each Stage of the Housefly	Understand
19	D	4.3 Features of Each Stage of the Housefly	Analyse
20	A	4.3 Features of Each Stage of the Housefly	Analyse
21	A	4.4 Feeding Habits and Breeding Places of the Housefly	Remember
22	C	4.4 Feeding Habits and Breeding Places of the Housefly	Understand
23	B	4.4 Feeding Habits and Breeding Places of the Housefly	Remember
24	D	4.4 Feeding Habits and Breeding Places of the Housefly	Understand
25	C	4.4 Feeding Habits and Breeding Places of the Housefly	Remember
26	D	4.4 Feeding Habits and Breeding Places of the Housefly	Understand
27	B	4.4 Feeding Habits and Breeding Places of the Housefly	Analyse
28	B	4.5 The Housefly as a Menace to Humans	Remember
29	A	4.5 The Housefly as a Menace to Humans	Remember
30	B	4.5 The Housefly as a Menace to Humans	Remember
31	C	4.5 The Housefly as a Menace to Humans	Understand
32	D	4.5 The Housefly as a Menace to Humans	Analyse
33	D	4.5 The Housefly as a Menace to Humans	Evaluate
34	A	4.5 The Housefly as a Menace to Humans	Understand
35	C	4.6 Reducing the Housefly Menace in the Community	Understand
36	A	4.6 Reducing the Housefly Menace in the Community	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	A	4.6 Reducing the Housefly Menace in the Co...	Understand
38	C	4.6 Reducing the Housefly Menace in the Co...	Analyse
39	B	4.6 Reducing the Housefly Menace in the Co...	Evaluate
40	A	4.6 Reducing the Housefly Menace in the Co...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 7 (17%) | Evaluate: 3 (7%) | Remember: 16 (40%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [4 marks]**

- A - **larva** (accept: maggot) (1 mark).
- B - **adult** (adult housefly) (1 mark).
- C - **egg** (eggs) (1 mark).
- D - **pupa** (1 mark).
- *Marking note: 'Caterpillar' or 'nymph' written for A earns no mark.*

(b) [2 marks]

- **C, A, D, B** (2 marks; 1 mark if only one letter is out of place).
- *Marking note: Accept the same order written in a circle, for example A, D, B, C, provided the sequence is unbroken.*

(c) [2 marks]

- A - **the larva feeds on the rotting matter and grows quickly** (1 mark).
- D - **the insect rests inside a hard case, does not feed, and changes into the adult** (1 mark).
- *Marking note: 'It sleeps' alone is not accepted for D; the change into the adult must be shown.*

(d) [2 marks]

- **The adult lays eggs, which start the same stages again** (1 mark).
- **The same stages follow one another in the same order over and over again** (1 mark).
- *Marking note: A candidate who writes only 'because it is round' earns no mark.*

Question 2 [Total: 12 marks]**(a) [3 marks]**

- J - **food left uncovered (open) on a table, with flies on it** (1 mark).
- K - **a heap of refuse (rubbish) with flies around it** (1 mark).
- L - **food covered with a net cover, with no flies on it** (1 mark).
- *Marking note: The idea of 'uncovered' at J and 'covered' at L must be clear.*

(b) [3 marks]

- **K provides food and a breeding place for the flies** (1 mark).
- **J is open and easy to reach, and the smell attracts the flies** (1 mark).
- **The net at L keeps the flies off the food although they can still smell it** (1 mark).

mark).

- *Marking note: Accept 'the flies cannot pass through the holes of the net' for the third mark.*

(c) [3 marks]

- Any THREE of: **they transfer disease germs to the food and cause diseases such as dysentery, diarrhoea, cholera or typhoid fever**; **they contaminate the food and cause food poisoning**; **they are a nuisance in the home and environment**; **they spoil the food so that it must be thrown away** (1 mark each).

- *Marking note: Malaria earns no mark; it is spread by the mosquito.*

(d) [3 marks]

- Any THREE of: **keep all food covered until it is eaten**; **clear the refuse heap and use a bin with a tight lid**; **dispose of refuse at the proper dumping site regularly**; **sweep the compound and keep gutters clean and dry**; **clear away food remains and wash plates at once** (1 mark each).

- *Marking note: 'Kill all the flies' is not accepted as a lasting change.*

Question 3 [Total: 10 marks]

(a) [4 marks]

- Egg - **in rotting rubbish, manure, faeces or decaying food** (1 mark).
 - Larva - **in the same rotting matter, where it feeds** (1 mark).
 - Pupa - **in the drier part of the rubbish or in the soil beneath it** (1 mark).
 - Adult - **flying about the compound, on refuse, faeces and uncovered food** (1 mark).

- *Marking note: 'In dirty places' may be credited once only.*

(b) [3 marks]

- **The warm, damp conditions suit the eggs and larvae** (1 mark).
 - **Rotting rubbish and waste, which are food and breeding places, are plentiful** (1 mark).
 - **The life cycle is completed in only one to three weeks and many eggs are laid each time** (1 mark).

- *Marking note: Accept 'the flies breed faster because it is warm and wet' for 1 mark only.*

(c) [3 marks]

- Any THREE of: **the eggs and larvae cannot move away, so they are easy to destroy**; **killing adults does not stop the ones already breeding in the refuse from hatching**; **removing the breeding places prevents new flies altogether, so the control lasts**; **adults can fly in from other compounds**; **cleaning needs no chemicals, so it is cheap and safe** (1 mark each).

- *Marking note: A candidate who merely repeats 'it is better' without a reason

earns no mark.*

Question 4 [Total: 10 marks]

(a) [3 marks]

- **It settles on the food and pours out saliva onto it** (1 mark).
- **The saliva dissolves or softens the food** (1 mark).
- **The fly sucks up the liquid food through its sponge-like mouth parts** (1 mark).
- *Marking note: 'It bites the food' earns no mark; the housefly has no biting mouth parts.*

(b) [3 marks]

- **Germs from the gutter stick to the hairy legs, body and mouth parts of the fly** (1 mark).
- **The fly lands on the kenkey and leaves the germs on it (it may also vomit or pass waste on the food)** (1 mark).
- **The customer eats the contaminated food and takes in the germs** (1 mark).
- *Marking note: The three steps must be in a sensible order for full marks.*

(c) [2 marks]

- Any TWO of: **dysentery**; **diarrhoea**; **cholera**; **typhoid fever**; **food poisoning** (1 mark each).
- *Marking note: Malaria earns no mark.*

(d) [2 marks]

- Any TWO of: **insecticide on or near food can poison the customers**; **it kills only the flies present at the time**; **the gutter keeps breeding new flies, so they return at once** (1 mark each, maximum 1 mark if no alternative is given).
- **She should clear the gutter and keep the food in a covered glass case or under a net cover** (1 mark).
- *Marking note: Full marks need both a reason and the better practice.*

Question 5 [Total: 10 marks]

(a) [3 marks]

- Any THREE of: **a clean-up of refuse heaps and choked gutters**; **refuse containers with tight lids, emptied regularly**; **keeping the school toilets clean and covered**; **asking vendors to keep food covered**; **keeping the compound swept and dry** (1 mark each).
- *Marking note: A bare list is enough; a short description is not required for the mark but should be encouraged.*

(b) [3 marks]

- (i) Any TWO materials, for example **a plastic bottle** and **bait** such as rotting fruit, fish or sugar water (1 mark for both).
- (ii) **The smell of the bait attracts the flies into the bottle through a narrow opening** (1 mark); **the flies cannot find their way out and are trapped** (1 mark).
 - **Marking note:** Accept any workable trap, for example a bowl of soapy water with bait covered by a perforated lid.*

(c) [2 marks]

- **Count the flies (in a trap or at a fixed place) before the intervention and again some weeks after it** (1 mark).
- **Compare the two counts; fewer flies afterwards shows that the intervention worked** (accept: check whether fewer pupils report sick with diarrhoea) (1 mark).
 - **Marking note:** A candidate who writes only 'look at the place' earns no mark; some form of comparison is needed.*

(d) [2 marks]

- Any TWO short, correct messages about covering food, clearing refuse, keeping gutters clean, using and covering latrines or washing hands (1 mark each).
 - **Marking note:** The message must be about reducing flies or protecting food; a general greeting earns no mark.*

Question 6 [Total: 8 marks]**(a)** [2 marks]

- (i) **Home W** (1 mark).
- (ii) **39** (1 mark).
- **Marking note:** Accept 39 flies written with or without the word 'flies'.*

(b) [3 marks]

- **Homes with open refuse (W and Y) caught many more flies than homes with covered refuse (X and Z)** (1 mark).
- **Open refuse gives the flies food and a breeding place** (1 mark).
- **Emptying often also matters: X, emptied daily, caught fewer flies than Z, emptied weekly, because refuse left standing allows the eggs to hatch** (1 mark).
 - **Marking note:** Candidates must quote at least one pair of figures from the table for the first mark.*

(c) [3 marks]

- (i) **Keep refuse in a covered bin and empty it daily** (1 mark).
- (ii) Any TWO of: **only four homes were tested**; **the count was made on one day only**; **other differences between the homes, such as cleanliness or a nearby dump, were not controlled**; **the traps may not catch the same share of**

the flies in every home** (1 mark each).

- *Marking note: 'They did not count well' earns no mark without a reason.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 2: Cycles | ****Sub-strand:**** 2: Life Cycle of Organisms | ****Class:****
Basic 7 (JHS 1)

****Content Standard:**** ****B7/JHS1.2.2.1***

Living Cells

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	B	2.1 Cells as the Units of Living Things	Remember
2	A	2.1 Cells as the Units of Living Things	Remember
3	D	2.1 Cells as the Units of Living Things	Apply
4	B	2.1 Cells as the Units of Living Things	Understand
5	B	2.1 Cells as the Units of Living Things	Remember
6	D	2.1 Cells as the Units of Living Things	Remember
7	A	2.2 The Structure of the Animal Cell	Remember
8	C	2.2 The Structure of the Animal Cell	Remember
9	D	2.2 The Structure of the Animal Cell	Remember
10	C	2.2 The Structure of the Animal Cell	Remember
11	B	2.2 The Structure of the Animal Cell	Understand
12	C	2.2 The Structure of the Animal Cell	Remember
13	A	2.2 The Structure of the Animal Cell	Analyse
14	C	2.2 The Structure of the Animal Cell	Apply
15	A	2.3 Functions of the Organelles of the Ani...	Remember
16	C	2.3 Functions of the Organelles of the Ani...	Understand
17	A	2.3 Functions of the Organelles of the Ani...	Remember
18	B	2.3 Functions of the Organelles of the Ani...	Understand
19	D	2.3 Functions of the Organelles of the Ani...	Remember
20	C	2.3 Functions of the Organelles of the Ani...	Analyse
21	C	2.3 Functions of the Organelles of the Ani...	Analyse
22	D	2.4 The Structure of the Plant Cell	Remember
23	A	2.4 The Structure of the Plant Cell	Remember
24	D	2.4 The Structure of the Plant Cell	Remember
25	D	2.4 The Structure of the Plant Cell	Remember
26	C	2.4 The Structure of the Plant Cell	Remember
27	D	2.4 The Structure of the Plant Cell	Understand
28	B	2.4 The Structure of the Plant Cell	Analyse
29	C	2.5 Functions of the Organelles of the Pla...	Understand
30	A	2.5 Functions of the Organelles of the Pla...	Remember
31	B	2.5 Functions of the Organelles of the Pla...	Understand
32	A	2.5 Functions of the Organelles of the Pla...	Analyse
33	A	2.5 Functions of the Organelles of the Pla...	Understand
34	B	2.5 Functions of the Organelles of the Pla...	Apply
35	D	2.6 Comparing Plant and Animal Cells	Remember
36	B	2.6 Comparing Plant and Animal Cells	Understand
37	B	2.6 Comparing Plant and Animal Cells	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	A	2.6 Comparing Plant and Animal Cells	Analyse
39	D	2.6 Comparing Plant and Animal Cells	Create
40	C	2.6 Comparing Plant and Animal Cells	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 4 (10%) | Create: 1 (2%) | Evaluate: 1 (2%) | Remember: 19 (47%) | Understand: 9

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [5 marks]**

- A - **cell membrane** (1 mark).
- B - **cytoplasm** (1 mark).
- C - **nucleus** (1 mark).
- D - **mitochondrion** (1 mark).
- E - **vacuole** (1 mark).
- **Marking note:** 'Cell wall' written for A earns no mark; an animal cell has no cell wall.*

(b) [3 marks]

- A - **controls what enters and leaves the cell** (accept: holds the cell together) (1 mark).
- C - **controls the activities of the cell** (1 mark).
- D - **releases energy from food (respiration)** (1 mark).
- **Marking note:** 'It protects the cell' alone is not accepted for C.*

(c) [2 marks]

- Any TWO of: **there is no cell wall and the shape is irregular**; **there are no chloroplasts**; **there is no large central vacuole**; **the nucleus is not pushed to one side** (1 mark each).
- **Marking note:** The reason must be visible in the figure.*

Question 2 [Total: 12 marks]**(a) [4 marks]**

- (i) **The smallest unit of a living organism** (1 mark) **that can carry out the activities of life / from which all living things are built** (1 mark).
- (ii) Any TWO of: **both plants and animals are made of cells**; **the cell is the basic unit of all living things**; **cells are too small to be seen without a microscope** (1 mark each).
- **Marking note:** 'A small thing in the body' earns no mark in (i).*

(b) [4 marks]

- Cytoplasm - **where the organelles lie and most reactions of the cell take place** (1 mark).
- Nucleus - **controls the activities of the cell** (1 mark).
- Mitochondrion - **releases energy from food** (1 mark).
- Ribosome - **makes proteins** (1 mark).
- **Marking note:** The table must be copied; one correct function each is

enough.*

(c) [4 marks]

- **A rounded cell with an irregular outline and no cell wall** (1 mark).
- **Four correct parts shown and named** - cell membrane, cytoplasm, nucleus, mitochondrion (accept vacuole or ribosome) (2 marks; 1 mark if only two or three are correct).
- **Each label written outside the drawing with a straight line touching the part named** (1 mark).
- **Marking note:** A drawing that includes a cell wall or chloroplasts loses the first mark.*

Question 3 [Total: 10 marks]

(a) [5 marks]

- P - **cell wall** (1 mark).
- Q - **cell membrane** (1 mark).
- R - **nucleus** (1 mark).
- S - **chloroplast** (1 mark).
- T - **vacuole (large central vacuole)** (1 mark).
- **Marking note:** P and Q must not be interchanged; the cell wall is the outer line.*

(b) [3 marks]

- P - **gives the cell a fixed shape and gives support and protection** (1 mark).
- S - **makes food for the plant by photosynthesis / holds the chlorophyll** (1 mark).
- T - **stores cell sap and keeps the cell firm** (1 mark).
- **Marking note:** 'It protects the cell' alone is not enough for P; the idea of shape or support is needed.*

(c) [2 marks]

- (i) **The root is in the soil where there is no light, so no food can be made there** (1 mark).
- (ii) **Food made in the leaves is carried down to the root through the stem** (1 mark).
- **Marking note:** 'Because it is not green' is accepted for (i) only if the absence of light is also stated.*

Question 4 [Total: 10 marks]

(a) [4 marks]

- Cell wall - **present / absent** (1 mark).
- Chloroplast - **present / absent** (1 mark).
- Vacuole - **present in both, large and central in the plant cell but small in the animal cell** (1 mark).
- Nucleus - **present / present** (1 mark).
- **Marking note:** For the vacuole row, both 'present' entries and the difference are needed for the mark.*

(b) [3 marks]

- **The plant cell has a firm cell wall made of cellulose** (1 mark).
- **The wall is rigid and holds the cell in a fixed shape** (1 mark).
- **The animal cell has only a thin flexible membrane, so its shape is irregular** (1 mark).
- **Marking note:** A candidate who writes only 'the plant cell has a cell wall' earns 1 mark.*

(c) [3 marks]

- **The pupil is not correct; a nucleus and cytoplasm occur in both kinds of cell** (1 mark).
- Any TWO of: **a cell wall**; **chloroplasts**; **a large central vacuole**; **a regular box-like shape** (1 mark each).
- **Marking note:** No mark for the features if the candidate says the pupil is correct.*

Question 5 [Total: 10 marks]**(a)** [4 marks]

- Cell membrane - **rubber bag; thin and flexible and holds everything in** (1 mark).
- Cytoplasm - **starch paste; jelly-like and fills the cell** (1 mark).
- Nucleus - **one stone or large seed; rounded, firm and single** (1 mark).
- Mitochondrion - **maize grains; small, oval and several** (1 mark).
- **Marking note:** A material named without a reason earns no mark. Accept any sensible alternative use of the materials listed.*

(b) [4 marks]

- (i) **The wooden box** (1 mark); **it is firm, straight-sided and keeps a fixed box-like shape** (1 mark).
- (ii) Any TWO of: **chloroplasts (green leaf pieces)**; **a large central vacuole**; **the nucleus pushed to one side** (1 mark each).
- **Marking note:** Naming the same part twice in (ii) scores once.*

(c) [2 marks]

- Any TWO of: ****the model is far larger than the real cell and gives a wrong idea of size****; ****the materials are not alive and carry out none of the activities of a cell****; ****some organelles are too small or too many to be shown****; ****the model is fixed while the contents of a living cell move**** (1 mark each).
- ***Marking note: 'It is not real' alone earns no mark; the limitation must be explained.***

Question 6 [Total: 8 marks]**(a) [4 marks]**

- (i) ****The chloroplasts could make no food in the dark**** (1 mark); ****the chlorophyll broke down and was not replaced, so the green colour faded**** (1 mark).
- (ii) ****The vacuoles lost water (cell sap)**** (1 mark); ****so the cells were no longer firm against their cell walls and the plant drooped**** (1 mark).
- ***Marking note: An answer in terms of the whole plant only, with no mention of cells, earns at most 1 mark in each part.***

(b) [2 marks]

- ****Water the plant (and return it to the light)**** (1 mark).
- ****The vacuoles fill with water again and the cells become firm**** (1 mark).
- ***Marking note: The second mark needs the vacuole or the cell, not just 'the plant drinks water'.***

(c) [2 marks]

- ****The animal cell has no cell wall to resist the water entering**** (1 mark).
- ****So it swells and may burst instead of becoming firm, while the plant cell is held by its wall**** (1 mark).
- ***Marking note: Accept 'the membrane is thin and flexible' for the first mark.***

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 1: Diversity of Matter | ****Sub-strand:** 2: Living Cells | ****Class:** Basic 7 (JHS 1)******

****Content Standard:** ****B7/JHS1.1.2.1**** - Demonstrate understanding of the structure of organisms and functions of cells in living systems.**

> Source: NaCCA Science Common Core Programme C

Materials

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	1.1 Materials and Their Properties	Remember
2	D	1.1 Materials and Their Properties	Remember
3	B	1.1 Materials and Their Properties	Apply
4	B	1.1 Materials and Their Properties	Understand
5	A	1.1 Materials and Their Properties	Understand
6	C	1.1 Materials and Their Properties	Remember
7	B	1.1 Materials and Their Properties	Understand
8	A	1.2 Solids, Liquids and Gases Compared	Remember
9	D	1.2 Solids, Liquids and Gases Compared	Understand
10	B	1.2 Solids, Liquids and Gases Compared	Analyse
11	D	1.2 Solids, Liquids and Gases Compared	Remember
12	A	1.2 Solids, Liquids and Gases Compared	Understand
13	B	1.2 Solids, Liquids and Gases Compared	Analyse
14	B	1.2 Solids, Liquids and Gases Compared	Analyse
15	C	1.3 The Importance of Liquids to Life	Remember
16	C	1.3 The Importance of Liquids to Life	Understand
17	A	1.3 The Importance of Liquids to Life	Understand
18	A	1.3 The Importance of Liquids to Life	Apply
19	C	1.3 The Importance of Liquids to Life	Evaluate
20	B	1.3 The Importance of Liquids to Life	Understand
21	A	1.3 The Importance of Liquids to Life	Evaluate
22	D	1.4 The Importance of Solids to Life	Remember
23	D	1.4 The Importance of Solids to Life	Remember
24	A	1.4 The Importance of Solids to Life	Understand
25	C	1.4 The Importance of Solids to Life	Remember
26	C	1.4 The Importance of Solids to Life	Understand
27	B	1.4 The Importance of Solids to Life	Evaluate
28	A	1.5 The First Twenty Elements and Their Sy...	Remember
29	D	1.5 The First Twenty Elements and Their Sy...	Remember
30	A	1.5 The First Twenty Elements and Their Sy...	Remember
31	D	1.5 The First Twenty Elements and Their Sy...	Remember
32	C	1.5 The First Twenty Elements and Their Sy...	Remember
33	A	1.5 The First Twenty Elements and Their Sy...	Understand
34	C	1.5 The First Twenty Elements and Their Sy...	Apply
35	D	1.6 Metals, Non-Metals and Noble Gases in ...	Understand
36	B	1.6 Metals, Non-Metals and Noble Gases in ...	Remember
37	D	1.6 Metals, Non-Metals and Noble Gases in ...	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	B	1.6 Metals, Non-Metals and Noble Gases in ...	Understand
39	C	1.6 Metals, Non-Metals and Noble Gases in ...	Analyse
40	D	1.6 Metals, Non-Metals and Noble Gases in ...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 3 (7%) | Evaluate: 3 (7%) | Remember: 15 (37%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [3 marks]

- P - **solid** (1 mark).
- Q - **liquid** (1 mark).
- R - **gas** (1 mark).
- **Marking note:** One mark for each correct state; no half marks.*

(b) [3 marks]

- P - **the particles are closely packed in a regular pattern and the material keeps its own shape** (1 mark).
- Q - **the particles are close together but irregular and there is a flat surface; the material takes the shape of the bottom of the jar** (1 mark).
- R - **the particles are far apart, moving, and spread through the whole jar** (1 mark).
- **Marking note:** The reason must come from the figure; 'because it is a solid' earns no mark.*

(c) [2 marks]

- (i) **It escapes and spreads out into the room** (1 mark).
- (ii) **The particles of a gas move freely in all directions, so a gas spreads to fill the space available** (1 mark).
- **Marking note:** 'It finishes' alone earns no mark in (i).*

(d) [2 marks]

- P - any solid, e.g. **salt, sugar, charcoal, stone, cooking pot** (1 mark).
- Q - any liquid, e.g. **water, palm oil, kerosene, milk** (1 mark).
- **Marking note:** The example must be found in a home; 'iron ore' is not accepted.*

Question 2 [Total: 12 marks]**(a)** [6 marks]

- Solids - **chalk, river sand** (1 mark each).
- Liquids - **borehole water, groundnut oil** (1 mark each).
- Gases - **cooking gas, air in the balloon** (1 mark each).
- **Marking note:** A material entered under the wrong heading earns no mark; the table must be copied.*

(b) [3 marks]

- (i) **Fixed shape and fixed volume** (1 mark).
- (ii) **No fixed shape, but fixed volume** (1 mark).
- (iii) **Neither fixed shape nor fixed volume** (1 mark).
 - *Marking note: Both halves of the answer are needed for the mark in each case.*

(c) [3 marks]

- **Push an inverted empty glass (or blow up a balloon) into a bucket of water** (1 mark).
- **Observation: the water does not fill the glass and the inside remains dry (the balloon swells)** (1 mark).
- **Conclusion: the air (gas) in the glass occupies the space** (1 mark).
 - *Marking note: An activity with no stated observation earns 1 mark only.*

Question 3 [Total: 10 marks]**(a)** [4 marks]

- Any FOUR of: **drinking**; **cooking**; **bathing or washing the body**; **washing clothes and utensils**; **cleaning the house and compound**; **watering plants**; **giving to livestock** (1 mark each).
 - *Marking note: Two answers that mean the same thing, such as 'washing' and 'washing clothes', score once.*

(b) [2 marks]

- Any TWO of: **helps in digestion**; **carries digested food and oxygen round the body in the blood**; **removes waste as urine and sweat**; **controls body temperature** (1 mark each).
 - *Marking note: Uses outside the body earn no mark here.*

(c) [2 marks]

- Any TWO of: **water is scarce, especially in the dry season**; **it costs time, labour and money to fetch**; **it cannot be replaced at once when wasted**; **uncovered or wasted water breeds mosquitoes and causes disease**; **all life in the village depends on it** (1 mark each).
 - *Marking note: Two distinct reasons are required.*

(d) [2 marks]

- Liquid: **palm oil, groundnut oil, kerosene, petrol or diesel** (1 mark).
- Use: **cooking** / **lighting a lamp or stove** / **fuel for vehicles and machines** (1 mark).
 - *Marking note: The use must match the liquid named.*

Question 4

(a) [4 marks]

- Wood - ****furniture, doors or roofing timber**** (1 mark).
- Cement blocks - ****building walls or foundations**** (1 mark).
- Iron - ****cutlasses, hoes, nails or burglar bars**** (1 mark).
- Glass - ****windows, louvre blades, bottles or mirrors**** (1 mark).
- ***Marking note: The table must be copied; one correct use per material is enough.***

(b) [4 marks]

- (i) ****Iron is hard and strong**** (1 mark); ****so it takes and keeps a sharp edge without breaking or bending**** (1 mark).
- (ii) ****Wood is light but strong**** (1 mark); ****and it is easily cut, shaped and smoothed with simple tools**** (1 mark).
- ***Marking note: A property without the reason earns 1 mark only.***

(c) [2 marks]

- Any TWO of: ****measure carefully and avoid wastage****; ****keep and use off-cuts****; ****repair and re-use old articles****; ****send scrap iron and glass for recycling****; ****plant trees to replace the timber used**** (1 mark each).
- ***Marking note: A measure that does not preserve material, such as 'work faster', earns no mark.***

Question 5 [Total: 10 marks]**(a)** [4 marks]

- W - ****magnesium**** (1 mark).
- X - ****sulphur**** (1 mark).
- Y - ****argon**** (1 mark).
- Z - ****potassium**** (1 mark).
- ***Marking note: Accept the correct chemical symbol written in place of the name.***

(b) [2 marks]

- (i) ****Y**** (1 mark).
- (ii) ****X**** (1 mark).
- ***Marking note: The letter alone is enough; no mark for naming a different element.***

(c) [2 marks]

- (i) ****Beryllium (Be)**** or ****calcium (Ca)**** (1 mark).
- (ii) ****They have similar chemical properties**** (accept: the same number of outermost electrons) (1 mark).
- ***Marking note: Only one element need be named in (i).***

(d) [2 marks]

- Rule: **in order of increasing atomic number** (1 mark).
- Evidence: **the numbers in the boxes increase by one from each element to the next across a period and on to the next period** (1 mark).
- **Marking note:** 'In order of their mass' earns no mark.*

Question 6 [Total: 8 marks]**(a)** [4 marks]

- Carbon - **C** (1 mark).
- Sodium - **Na** (1 mark).
- Aluminium - **Al** (1 mark).
- Helium - **He** (1 mark).
- **Marking note:** A symbol written wholly in capitals, such as NA or AL, earns no mark.*

(b) [2 marks]

- **Place the sample in a simple circuit with a cell and a bulb** (accept: hammer it gently, or rub it) (1 mark).
- **If the bulb lights, the sample conducts electricity and is a metal** (accept: it flattens into a sheet instead of breaking; it is shiny) (1 mark).
- **Marking note:** A test with no stated result earns 1 mark only.*

(c) [2 marks]

- Any TWO differences, each stated for both sides: **metals conduct heat and electricity, non-metals generally do not**; **metals are shiny, non-metals are dull**; **metals can be beaten into sheets and drawn into wires, non-metals are brittle**; **most metals are solids with high melting points, non-metals may be solids, liquids or gases** (1 mark each).
- **Marking note:** A statement about metals only, with nothing said about non-metals, earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 1: Diversity of Matter | **Sub-strand:** 1: Materials | **Class:** Basic 7 (JHS 1)

**

Science and Industry

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	18.1 Careers in Science and What Each Work...	Remember
2	C	18.1 Careers in Science and What Each Work...	Remember
3	D	18.1 Careers in Science and What Each Work...	Remember
4	B	18.1 Careers in Science and What Each Work...	Remember
5	A	18.1 Careers in Science and What Each Work...	Remember
6	B	18.1 Careers in Science and What Each Work...	Remember
7	B	18.1 Careers in Science and What Each Work...	Remember
8	D	18.1 Careers in Science and What Each Work...	Understand
9	D	18.1 Careers in Science and What Each Work...	Remember
10	C	18.2 Ghanaian Scientists and Science Educa...	Remember
11	A	18.2 Ghanaian Scientists and Science Educa...	Remember
12	D	18.2 Ghanaian Scientists and Science Educa...	Remember
13	A	18.2 Ghanaian Scientists and Science Educa...	Remember
14	C	18.2 Ghanaian Scientists and Science Educa...	Understand
15	B	18.2 Ghanaian Scientists and Science Educa...	Evaluate
16	C	18.2 Ghanaian Scientists and Science Educa...	Understand
17	D	18.2 Ghanaian Scientists and Science Educa...	Remember
18	C	18.3 Internationally Recognised Scientists...	Remember
19	A	18.3 Internationally Recognised Scientists...	Understand
20	B	18.3 Internationally Recognised Scientists...	Remember
21	B	18.3 Internationally Recognised Scientists...	Remember
22	C	18.3 Internationally Recognised Scientists...	Remember
23	B	18.3 Internationally Recognised Scientists...	Understand
24	D	18.3 Internationally Recognised Scientists...	Evaluate
25	A	18.3 Internationally Recognised Scientists...	Apply
26	B	18.4 The Impact of Science, Technology and...	Remember
27	C	18.4 The Impact of Science, Technology and...	Understand
28	D	18.4 The Impact of Science, Technology and...	Understand
29	C	18.4 The Impact of Science, Technology and...	Analyse
30	A	18.4 The Impact of Science, Technology and...	Understand
31	C	18.4 The Impact of Science, Technology and...	Understand
32	B	18.4 The Impact of Science, Technology and...	Analyse
33	D	18.4 The Impact of Science, Technology and...	Remember
34	C	18.5 Science and Technology Careers Ghana ...	Evaluate
35	A	18.5 Science and Technology Careers Ghana ...	Understand
36	B	18.5 Science and Technology Careers Ghana ...	Evaluate
37	A	18.5 Science and Technology Careers Ghana ...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	A	18.5 Science and Technology Careers Ghana ...	Apply
39	D	18.5 Science and Technology Careers Ghana ...	Apply
40	A	18.5 Science and Technology Careers Ghana ...	Understand

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 2 (5%) | Apply: 4 (10%) | Evaluate: 4 (10%) | Remember: 19 (47%) | Understand: 11 (27%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [4 marks]

- A - ****laboratory scientist or technician (accept biologist, chemist)**** (1 mark).
- B - ****doctor or nurse (accept any named health worker)**** (1 mark).
- C - ****agricultural scientist (accept agronomist, soil scientist)**** (1 mark).
- D - ****civil engineer (accept architect, surveyor)**** (1 mark).
- ***Marking note: 'He works in a laboratory' is not the name of a career.***

(b) [4 marks]

- A - ****tests samples so that disease is detected and food and water are safe**** (1 mark).
- B - ****treats the sick and immunises people, saving lives**** (1 mark).
- C - ****improves crops and methods, so more food is produced**** (1 mark).
- D - ****designs safe roads, bridges, dams and buildings**** (1 mark).
- ***Marking note: 'He earns money' is not a benefit to the people.***

(c) [2 marks]

- ****Integrated science and mathematics (accept English for one)**** (1 mark).
- ****Reason: science supplies the knowledge and mathematics the measurement and calculation that all these careers require**** (1 mark).
- ***Marking note: Subjects named with no reason earn 1 mark only.***

(d) [2 marks]

- ****A scientist carries out research and makes discoveries in a field of science**** (1 mark).
- ****A science educator studies and improves the teaching and learning of science (trains teachers, writes books, develops the curriculum)**** (1 mark).
- ***Marking note: 'One is cleverer than the other' earns no mark.***

Question 2 [Total: 10 marks]

(a) [3 marks]

- (i) ****Alexander Fleming**** (1 mark).
- (ii) ****Penicillin**** (1 mark).
- (iii) ****A substance from the mould had killed or stopped the growth of the bacteria**** (1 mark).
- ***Marking note: 'Antibiotic' alone is accepted for (ii) only if penicillin is named.***

(b) [3 marks]

- Q - **the substance is tested and studied in the laboratory and the results recorded** (1 mark).
- R - **the medicine is manufactured in a factory and packed** (1 mark).
- S - **the patient is treated and recovers** (1 mark).
- **Marking note: Credit any clear wording of each stage.**

(c) [2 marks]

- **Science makes the discovery and tests it** (1 mark).
- **Industry manufactures it in large quantities so that it reaches the people** (1 mark).
- **Marking note: A general statement with no reference to the stages earns 1 mark at most.**

(d) [2 marks]

- Any TWO of: **keen observation**; **curiosity and an open mind**; **patience and careful follow-up work**; **careful recording and reporting of findings** (1 mark each).
- **Marking note: 'He was clever' is too vague for a mark.**

Question 3 [Total: 10 marks]**(a)** [4 marks]

- Allotey - **mathematics and physics** (1 mark).
- Oduro - **food science (food and crop processing)** (1 mark).
- Addy - **biochemistry and the biological sciences** (1 mark).
- Anamuah-Mensah - **science education** (1 mark).
- **Marking note: A field written against the wrong person earns no mark.**

(b) [3 marks]

- **Prof. Theophilus Ossei-Anto** (1 mark); **Prof. Christian Anthony-Krueger** (1 mark).
- **They improve the teaching and learning of science, so that more pupils understand it and take up science careers** (1 mark).
- **Marking note: Any other correctly named Ghanaian science educator is accepted.**

(c) [3 marks]

- Any THREE of: **it shows that Ghanaians can succeed in science**; **it gives models the pupils can relate to**; **it shows science applied to Ghana's own problems**; **it builds national pride and encourages pupils to serve their country** (1 mark each).
- **Marking note: Two answers that both say 'to encourage us' count once.**

Question 4**(a)** [3 marks]

- (i) **One sound home example, such as a refrigerator or a mobile phone** (1 mark).
- (ii) **One sound school example, such as computers or laboratory apparatus** (1 mark).
- (iii) **One sound community example, such as treated water, electricity or mobile money** (1 mark).
- **Marking note: The same example may not be used twice.**

(b) [3 marks]

- Any THREE of: **people communicate quickly over long distances**; **farmers and traders check market prices before travelling**; **money is sent and received by mobile money**; **emergencies can be reported at once**; **information for learning becomes available** (1 mark each).
- **Marking note: 'It is nice to have' earns no mark.**

(c) [2 marks]

- Any TWO of: **air and water pollution from factories and vehicles**; **plastic and electronic waste**; **loss of jobs to machines**; **too much screen time harming study and health**; **destruction of land and rivers by mining machinery** (1 mark each).
- **Marking note: A harmful effect must be stated, not merely a device named.**

(d) [2 marks]

- **Putting a new or improved idea into practical use to solve a problem** (1 mark).
- **One sound Ghanaian example, such as mobile money, delivery drones or solar lamps** (1 mark).
- **Marking note: A copy of an existing article is not an innovation.**

Question 5 [Total: 10 marks]**(a)** [4 marks]

- **First career named** (1 mark) **with a sound reason** (1 mark).
- **Second career named** (1 mark) **with a sound reason** (1 mark).
- **Marking note: A career named with no reason earns 1 mark only.**

(b) [3 marks]

- Any THREE of: **study science and mathematics seriously and ask questions**; **take part in practical work and the science club**; **read about science and about scientists**; **keep a portfolio of findings**; **visit places of work and talk to scientists** (1 mark each).
- **Marking note: 'Pray to become a scientist' earns no mark.**

(c) [3 marks]

- **The belief is wrong; science needs observation and hard work, which do not depend on sex** (1 mark).
- **Evidence: named Ghanaian women scientists, such as Prof. Ibok Oduro or Prof. Ewurama Addy (accept women doctors, engineers and so on)** (1 mark).
- **Excluding girls wastes half the nation's talent, so girls should be encouraged equally** (1 mark).
- **Marking note: A reply that agrees with the belief earns no mark.**

Question 6 [Total: 8 marks]**(a) [2 marks]**

- **An organised collection of one's own work and evidence on a topic** (1 mark).
- **One reason: it keeps the evidence in one place for presentation and shows how the work has developed** (1 mark).
- **Marking note: 'A bag for books' earns no mark.**

(b) [4 marks]

- Any FOUR of: **notes from books and the internet**; **photographs or drawings of technology in use**; **records of interviews with people at work**; **newspaper cuttings and leaflets**; **tables or charts of data the class collected**; **a written report with conclusions** (1 mark each).
- **Marking note: 'Everything about science' earns no mark.**

(c) [2 marks]

- Any TWO of: **speak clearly and in an orderly sequence, keeping to time**; **show charts, pictures and specimens as evidence**; **share the work among the members**; **practise beforehand and be ready for questions** (1 mark each).
- **Marking note: 'Wear fine clothes' earns no mark.**

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans and the Environment | **Sub-strand:** 3: Science and Industry | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.5.3.1** - Realise how careers in science can improve human life, and research about Ghanaian and internationally recognised

Solar System

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	8.1 The Solar System and the Sun	Remember
2	A	8.1 The Solar System and the Sun	Remember
3	B	8.1 The Solar System and the Sun	Remember
4	D	8.1 The Solar System and the Sun	Understand
5	D	8.1 The Solar System and the Sun	Understand
6	C	8.1 The Solar System and the Sun	Understand
7	B	8.1 The Solar System and the Sun	Analyse
8	D	8.2 The Inner Planets and Their Properties	Remember
9	B	8.2 The Inner Planets and Their Properties	Remember
10	B	8.2 The Inner Planets and Their Properties	Remember
11	C	8.2 The Inner Planets and Their Properties	Understand
12	D	8.2 The Inner Planets and Their Properties	Remember
13	D	8.2 The Inner Planets and Their Properties	Remember
14	A	8.2 The Inner Planets and Their Properties	Remember
15	C	8.2 The Inner Planets and Their Properties	Understand
16	C	8.2 The Inner Planets and Their Properties	Understand
17	D	8.3 The Galaxy and the Milky Way	Remember
18	A	8.3 The Galaxy and the Milky Way	Remember
19	A	8.3 The Galaxy and the Milky Way	Understand
20	C	8.3 The Galaxy and the Milky Way	Understand
21	B	8.3 The Galaxy and the Milky Way	Analyse
22	C	8.3 The Galaxy and the Milky Way	Analyse
23	A	8.4 The Paths and Movements of the Planets	Remember
24	B	8.4 The Paths and Movements of the Planets	Remember
25	A	8.4 The Paths and Movements of the Planets	Remember
26	D	8.4 The Paths and Movements of the Planets	Remember
27	C	8.4 The Paths and Movements of the Planets	Understand
28	A	8.4 The Paths and Movements of the Planets	Analyse
29	B	8.4 The Paths and Movements of the Planets	Remember
30	A	8.5 The Planets Mercury and Venus	Understand
31	D	8.5 The Planets Mercury and Venus	Analyse
32	B	8.5 The Planets Mercury and Venus	Apply
33	B	8.5 The Planets Mercury and Venus	Understand
34	C	8.5 The Planets Mercury and Venus	Understand
35	A	8.5 The Planets Mercury and Venus	Understand
36	D	8.5 The Planets Mercury and Venus	Analyse
37	A	8.6 Making a Model of the Solar System	Understand

Q	Ans	Curriculum Area	Cognitive Level
38	C	8.6 Making a Model of the Solar System	Apply
39	C	8.6 Making a Model of the Solar System	Evaluate
40	B	8.6 Making a Model of the Solar System	Apply

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 3 (7%) | Evaluate: 1 (2%) | Remember: 16 (40%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a)** [4 marks]

- W - **Mercury** (1 mark).
- X - **Venus** (1 mark).
- Y - **Earth** (1 mark).
- Z - **Mars** (1 mark).
- *Marking note: The four names must be in the correct order from the Sun; a correct name in the wrong position earns no mark.*

(b) [2 marks]

- **The inner planets** (accept: the rocky or terrestrial planets) (1 mark).
- Any ONE of: **they have solid rocky surfaces**; **they are small**; **they are close to the Sun**; **they have few or no moons** (1 mark).
- *Marking note: 'They are hot' alone is too vague.*

(c) [2 marks]

- **Y, the Earth** (1 mark).
- Any ONE of: **it is neither too hot nor too cold**; **it has liquid water**; **it has air (oxygen) to breathe** (1 mark).
- *Marking note: 'Because people live there' is a restatement and earns no mark for the reason.*

(d) [2 marks]

- **Z is farther from the Sun, so its path (orbit) is much longer** (1 mark).
- **It also travels more slowly than W, so it takes much more time to complete one round** (1 mark).
- *Marking note: The idea of a longer path is essential for the first mark.*

Question 2 [Total: 10 marks]**(a)** [3 marks]

- (i) **A very large group (collection) of stars, dust and gas** (1 mark) **held together by gravity** (1 mark).
- (ii) **The Milky Way** (1 mark).
- *Marking note: 'Many stars' alone earns 1 mark in (i).*

(b) [3 marks]

- **The light of the Milky Way is faint** (1 mark).
- **The bright lights of the town brighten the sky and hide it, while the village is dark** (1 mark).

- **Suggestion: travel to a dark place outside the town, or switch off the compound lights and observe on a moonless night** (1 mark).

- **Marking note: 'Too many buildings in Accra' earns no mark.**

(c) [2 marks]

- **Planet, solar system, galaxy** (1 mark).

- **Reason: the planet is one member of the solar system, and the solar system is one small part of the galaxy** (1 mark).

- **Marking note: No mark for the reason if the order is wrong.**

(d) [2 marks]

- Any TWO of: **a star gives out its own light and heat, a planet reflects light; a star is far larger and hotter; stars twinkle, planets shine steadily; a planet moves round a star** (1 mark each).

- **Marking note: 'Stars are far away and planets are near' is too weak on its own.**

Question 3 [Total: 10 marks]

(a) [4 marks]

- P - **the axis (of the planet)** (1 mark).

- Q - **rotation** (accept: the planet turning or spinning on its own axis) (1 mark).

- R - **revolution** (accept: the planet moving round the Sun) (1 mark).

- S - **the orbit** (accept: the elliptical path of the planet) (1 mark).

- **Marking note: Rotation written for R, or revolution for Q, earns no mark.**

(b) [2 marks]

- Q - **day and night** (1 mark).

- R - **the year** (accept: the seasons) (1 mark).

- **Marking note: Day and night written for R earns no mark.**

(c) [2 marks]

- **Elliptical (an ellipse)** (1 mark).

- **The distance between the planet and the Sun changes as the planet moves; it is nearer at some times and farther at others** (1 mark).

- **Marking note: 'The planet is always the same distance from the Sun' is wrong and earns no mark.**

(d) [2 marks]

- **The gravitational pull of the Sun holds the planet and stops it flying away** (1 mark).

- **The planet is also moving forward fast along its path, so it keeps going round instead of falling in** (1 mark).

- *Marking note: Gravity alone earns 1 mark; the forward motion is needed for the second.*

Question 4 [Total: 12 marks]

(a) [4 marks]

- Mercury - **first (nearest) planet from the Sun** (1 mark).
- Venus - **second planet from the Sun** (1 mark).
- Mercury - any ONE of: **the smallest inner planet**; **has almost no atmosphere**; **very hot by day and very cold by night**; **has the shortest year** (1 mark).
- Venus - any ONE of: **the hottest planet**; **has a very thick atmosphere**; **nearly the same size as the Earth**; **seen as the morning or evening star** (1 mark).
- *Marking note: A property given for the wrong planet earns no mark.*

(b) [3 marks]

- **Venus has a very thick atmosphere** (1 mark).
- **The atmosphere traps the heat so that it cannot escape, and the heat builds up** (1 mark).
- **Mercury has almost no atmosphere, so its heat escapes quickly into space** (1 mark).
- *Marking note: The trapping of heat must be stated; 'Venus is bigger' earns no mark.*

(c) [3 marks]

- (i) ** $365 / 88$ ** (1 mark) ** $= 4.1$, which is about 4 times** (1 mark).
- (ii) **A year on Mercury is much shorter than an Earth year (about a quarter as long)** (1 mark).
- *Marking note: Award the second mark for 4 even if the division is not shown, but the working mark is then lost.*

(d) [2 marks]

- **It is seen shining brightly just before sunrise or just after sunset** (1 mark).
- **It is not a star but a planet, which shines by reflecting the light of the Sun** (1 mark).
- *Marking note: The second mark requires the idea of reflected light or of Venus being a planet.*

Question 5 [Total: 10 marks]

(a) [3 marks]

- Any THREE materials, each with a correct use, for example: **a large ball of clay for the Sun**; **smaller balls, beads or seeds for the planets**; **loops of

wire or lines on cardboard for the paths**; **thread or sticks to hold the planets** (1 mark each).

- *Marking note: A material named with no use, or a use with no material, earns no mark.*

(b) [4 marks]

- Any FOUR steps in a workable order: **fix the largest ball at the centre as the Sun**; **make four planets of different sizes**; **make four oval paths of increasing size round the Sun**; **place each planet on its own path in the correct order**; **label the bodies and state that the model is not to scale** (1 mark each).

- *Marking note: Steps given out of a workable order lose 1 mark overall.*

(c) [3 marks]

- Any TWO of: **the order of the planets from the Sun**; **that each planet has its own path**; **the oval shape of the paths**; **that the Sun is at the centre** (1 mark each).

- Any ONE of: **the true sizes of the bodies**; **the true distances between them**; **the real speeds of the planets**; **that the paths are not solid rings** (1 mark).

- *Marking note: 'It is not real' is too vague for the third mark.*

Question 6 [Total: 8 marks]

(a) [3 marks]

- **The pupil is not correct** (1 mark).

- **The Earth rotates on its own axis once in about 24 hours** (1 mark).

- **The half facing the Sun has day and the half turned away has night, so the Sun only appears to move** (1 mark).

- *Marking note: A candidate who agrees with the pupil can earn no mark in this part.*

(b) [3 marks]

- Any THREE of: **it supplies light for seeing and for photosynthesis**; **it supplies heat and keeps the Earth warm**; **it drives the water cycle and the winds**; **its gravity holds the planets in their paths**; **it supplies energy for solar panels** (1 mark each).

- *Marking note: 'It makes the day bright' and 'it gives light' count as one answer only.*

(c) [2 marks]

- **Shine a torch (the Sun) on a globe or ball marked to show Ghana, in a darkened room** (1 mark).

- **Keep the torch still and turn the globe slowly; the lit half has day, the dark

half night, and the mark passes from one to the other** (1 mark).

- *Marking note: No mark for an activity in which the torch is moved round the globe.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 3: Systems | **Sub-strand:** 2: The Solar System | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.3.2.1** - Demonstrate knowledge of the inner planets of the solar system and understand their mov

Understanding the Environment

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	20.1 Land Forms in Ghana and the Living Th...	Remember
2	D	20.1 Land Forms in Ghana and the Living Th...	Remember
3	A	20.1 Land Forms in Ghana and the Living Th...	Remember
4	D	20.1 Land Forms in Ghana and the Living Th...	Remember
5	A	20.1 Land Forms in Ghana and the Living Th...	Remember
6	B	20.1 Land Forms in Ghana and the Living Th...	Understand
7	B	20.1 Land Forms in Ghana and the Living Th...	Understand
8	D	20.1 Land Forms in Ghana and the Living Th...	Understand
9	C	20.2 Characteristics that Enable Animals t...	Remember
10	A	20.2 Characteristics that Enable Animals t...	Understand
11	B	20.2 Characteristics that Enable Animals t...	Understand
12	C	20.2 Characteristics that Enable Animals t...	Understand
13	D	20.2 Characteristics that Enable Animals t...	Understand
14	A	20.2 Characteristics that Enable Animals t...	Understand
15	C	20.2 Characteristics that Enable Animals t...	Analyse
16	D	20.2 Characteristics that Enable Animals t...	Analyse
17	D	20.3 Characteristics that Enable Plants to...	Understand
18	B	20.3 Characteristics that Enable Plants to...	Understand
19	B	20.3 Characteristics that Enable Plants to...	Remember
20	D	20.3 Characteristics that Enable Plants to...	Understand
21	B	20.3 Characteristics that Enable Plants to...	Understand
22	C	20.3 Characteristics that Enable Plants to...	Analyse
23	B	20.3 Characteristics that Enable Plants to...	Analyse
24	A	20.3 Characteristics that Enable Plants to...	Apply
25	D	20.4 Associations among Plants and Animals	Remember
26	C	20.4 Associations among Plants and Animals	Understand
27	D	20.4 Associations among Plants and Animals	Remember
28	B	20.4 Associations among Plants and Animals	Apply
29	D	20.4 Associations among Plants and Animals	Apply
30	B	20.4 Associations among Plants and Animals	Apply
31	C	20.4 Associations among Plants and Animals	Apply
32	C	20.4 Associations among Plants and Animals	Analyse
33	A	20.4 Associations among Plants and Animals	Remember
34	A	20.5 Mechanisms for Survival, Field Study ...	Remember
35	B	20.5 Mechanisms for Survival, Field Study ...	Understand
36	A	20.5 Mechanisms for Survival, Field Study ...	Remember

Q	Ans	Curriculum Area	Cognitive Level
37	A	20.5 Mechanisms for Survival, Field Study ...	Remember
38	C	20.5 Mechanisms for Survival, Field Study ...	Apply
39	A	20.5 Mechanisms for Survival, Field Study ...	Evaluate
40	C	20.5 Mechanisms for Survival, Field Study ...	Analyse

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 6 (15%) | Evaluate: 1 (2%) | Remember: 13 (32%) | Understand: 14 (35%)

SECTION B: THEORY — MARKING SCHEME

Question 1 [Total: 12 marks]

(a) [4 marks]

- A - **mountain** (accept mountain peak) (1 mark).
- B - **valley** (1 mark).
- C - **plateau** (accept table land) (1 mark).
- D - **plain** (1 mark).
- *Marking note: 'Hill' is accepted for A only if the candidate describes it as high land rising to a peak.*

(b) [2 marks]

- C - **the Kwahu Plateau** (accept any correctly named Ghanaian plateau) (1 mark).
- D - **the Accra Plains or the Afram Plains** (accept any correctly named Ghanaian plain) (1 mark).
- *Marking note: A named place that is not of the land form asked for earns no mark.*

(c) [4 marks]

- Any TWO plants of the plains: **grasses** (elephant grass, spear grass); **baobab**; **shea tree**; **dawadawa**; **acacia** (1 mark each).
- Any TWO animals of the plains: **cattle, sheep or goats**; **antelope**; **grasscutter**; **guinea fowl**; **grasshopper**; **egret** (1 mark each).
- *Marking note: A forest tree such as the odum is not accepted as a plant of the open plain.*

(d) [2 marks]

- Any TWO of: **the soil on the slopes is thin, stony and poor, so few plants can grow**; **it is colder and windier high up**; **water drains away quickly from the steep slope**; **with few plants there is little food and shelter for animals**; **the steep rocky ground is difficult to move about on** (1 mark each).
- *Marking note: 'Because it is a mountain' earns no mark.*

Question 2 [Total: 10 marks]

(a) [2 marks]

- **A feature of the body, or a habit, that helps a living thing to survive where it lives** (1 mark).
- **One correct example** (1 mark).
- *Marking note: An example alone, without the meaning, earns 1 mark.*

(b) [6 marks]

- Goat - **strong legs with hard split hooves** (1 mark) - **grip the rock, so it does not slip on the steep slope** (1 mark).
- Egret - **long thin legs with a long neck and beak** (1 mark) - **wades in shallow water and catches food without wetting its body** (1 mark).
- Fish - **streamlined body with strong fins and tail** (1 mark) - **water flows past it easily, so it can hold its place in the current** (1 mark).
- **Marking note: A feature entered without an advantage earns the feature mark only.**

(c) [2 marks]

- **Its features (gills, fins, moist skin) work only in water, and there is no water on the hill top** (1 mark).
- **It would not find its usual food or any shelter, and would dry up in the heat** (1 mark).
- **Marking note: 'Because it is a fish' earns no mark without the reason.**

Question 3 [Total: 11 marks]**(a)** [3 marks]

- P - **mutualism** (1 mark).
- Q - **commensalism** (1 mark).
- R - **parasitism** (1 mark).
- **Marking note: 'Symbiosis' alone is not accepted for P.**

(b) [6 marks]

- P - **the bee gets nectar for food** (1 mark); **the flower is pollinated, so it can form seeds** (1 mark).
- Q - **the bird gets the insects to eat** (1 mark); **the cow is neither helped nor harmed** (1 mark).
- R - **the plant on the branch gets water and food from the tree** (1 mark); **the tree is harmed and grows weak** (1 mark).
- **Marking note: Marks are for the benefit and the effect, not for naming the association again.**

(c) [2 marks]

- **He should not remove the bees (he should manage them instead)** (1 mark).
- **Reason: the bees pollinate the flowers, and without them the trees would set far less fruit** (1 mark).
- **Marking note: Advice with no reason earns 1 mark only.**

Question**(a) [4 marks]**

- Any FOUR of: **long or widely spreading roots**; **small, narrow or needle-like leaves**; **thick waxy or hairy leaf surfaces**; **fleshy stems, roots or tubers that store water**; **thick fire-resistant bark**; **shedding of leaves in the dry season**; **thorns in place of leaves** (1 mark each).

- **Marking note: 'It grows well' is not a feature.**

(b) [3 marks]

- Any THREE of: **stilt or prop roots that support it in the soft mud**; **breathing roots that grow up out of the mud for air**; **thick leathery leaves that cope with the salt**; **the ability to get rid of or store excess salt**; **seedlings that germinate on the parent plant** (1 mark each).

- **Marking note: Credit any clear description even if the technical name is not used.**

(c) [2 marks]

- **Tall growth lets the tree reach the sunlight above the crowded forest** (1 mark).

- **Broad leaves catch as much light as possible, and the plentiful water makes the water loss affordable** (1 mark).

- **Marking note: 'To look beautiful' earns no mark.**

Question 5 [Total: 10 marks]**(a) [3 marks]**

- Any THREE items, each with its correct use, for example: **notebook and pencil - to record observations**; **camera or phone - to photograph what cannot be collected**; **hand lens - to examine small specimens**; **tape or rule - to measure**; **bags or envelopes - to carry leaf specimens**; **field guide or chart - to help name the organisms** (1 mark each).

- **Marking note: An item named without its use earns no mark.**

(b) [3 marks]

- Any THREE of: **the name of the organism (English and local name)**; **the land form and place where it was found**; **the date it was seen**; **the features noticed and the adaptation each gives**; **the name of the recorder** (1 mark each).

- **Marking note: 'A nice drawing' is not a record.**

(c) [2 marks]

- Any TWO of: ****the valley is damper and the water collects there, while the plateau drains quickly****; ****the valley soil is deeper and richer, the plateau soil shallow and stony****; ****the plateau is more exposed to sun and wind****; ****temperatures differ between the high and the low ground****; ****farming and grazing differ in the two places**** (1 mark each).

- *Marking note: 'One is up and one is down' alone earns no mark.*

(d) [2 marks]

- Any TWO of: ****observe, draw or photograph instead of uprooting and killing****; ****collect only small specimens where necessary****; ****do not disturb nests, eggs or young animals****; ****keep to the paths and avoid trampling****; ****light no fires****; ****carry all rubbish away**** (1 mark each).

- *Marking note: 'Behave well' is too vague for a mark.*

Question 6 [Total: 8 marks]**(a)** [2 marks]

- ****The natural place in which a plant or animal lives**** (1 mark).
 - ****It supplies the food, water, shelter and breeding place that the organism needs**** (1 mark).

- *Marking note: 'Where animals sleep' is accepted only with the idea of food or shelter.*

(b) [4 marks]

- Any FOUR of: ****animals lose shelter and nesting places and move away or die****; ****food plants are destroyed, so food becomes scarce and competition increases****; ****the shade is lost, so the ground is hot and dry and shade-loving plants die****; ****the bare soil is washed away by rain and the streams are silted****; ****associations such as pollination are broken****; ****some kinds of plants and animals disappear from the area**** (1 mark each).

- *Marking note: Two answers that both say 'animals die' count once.*

(c) [2 marks]

- Any TWO of: ****replant trees and protect the seedlings from fire and animals****; ****set part of the slope aside as a protected area****; ****stop charcoal burning and use improved stoves or gas****; ****farm the flat land instead of the slope****; ****agree a closed season for hunting**** (1 mark each).

- *Marking note: 'Report them to the police' alone earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:** 5: Humans and the Environment | ****Sub-strand:** 5: Understanding****

the Environment | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.5.5.1** - Demonstrate understanding of diffe

Waste Management

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	C	16.1 Types and Sources of Waste	Remember
2	C	16.1 Types and Sources of Waste	Remember
3	D	16.1 Types and Sources of Waste	Remember
4	D	16.1 Types and Sources of Waste	Remember
5	C	16.1 Types and Sources of Waste	Remember
6	D	16.1 Types and Sources of Waste	Remember
7	A	16.1 Types and Sources of Waste	Understand
8	B	16.1 Types and Sources of Waste	Remember
9	C	16.2 Biodegradable and Non-Biodegradable W...	Remember
10	B	16.2 Biodegradable and Non-Biodegradable W...	Understand
11	B	16.2 Biodegradable and Non-Biodegradable W...	Remember
12	D	16.2 Biodegradable and Non-Biodegradable W...	Analyse
13	A	16.2 Biodegradable and Non-Biodegradable W...	Understand
14	D	16.2 Biodegradable and Non-Biodegradable W...	Analyse
15	B	16.2 Biodegradable and Non-Biodegradable W...	Analyse
16	C	16.2 Biodegradable and Non-Biodegradable W...	Understand
17	B	16.3 Good Waste Management Practices	Remember
18	C	16.3 Good Waste Management Practices	Understand
19	D	16.3 Good Waste Management Practices	Understand
20	A	16.3 Good Waste Management Practices	Understand
21	D	16.3 Good Waste Management Practices	Remember
22	A	16.3 Good Waste Management Practices	Analyse
23	A	16.3 Good Waste Management Practices	Understand
24	B	16.3 Good Waste Management Practices	Evaluate
25	A	16.3 Good Waste Management Practices	Apply
26	C	16.4 Composting and the Science Behind It	Remember
27	B	16.4 Composting and the Science Behind It	Understand
28	C	16.4 Composting and the Science Behind It	Understand
29	A	16.4 Composting and the Science Behind It	Understand
30	B	16.4 Composting and the Science Behind It	Analyse
31	C	16.4 Composting and the Science Behind It	Understand
32	D	16.4 Composting and the Science Behind It	Understand
33	A	16.5 Effects of Poor Waste Management and ...	Understand
34	B	16.5 Effects of Poor Waste Management and ...	Remember
35	A	16.5 Effects of Poor Waste Management and ...	Understand
36	A	16.5 Effects of Poor Waste Management and ...	Apply
37	B	16.5 Effects of Poor Waste Management and ...	Apply

Q	Ans	Curriculum Area	Cognitive Level
38	D	16.5 Effects of Poor Waste Management and ...	Apply
39	C	16.5 Effects of Poor Waste Management and ...	Evaluate
40	D	16.5 Effects of Poor Waste Management and ...	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 5 (12%) | Apply: 4 (10%) | Evaluate: 3 (7%) | Remember: 13 (32%) | Understand: 15 (37%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 12 marks]****(a)** [4 marks]

- A - **organic (food or kitchen) waste** (1 mark).
- B - **plastic waste** (1 mark).
- C - **glass and metal waste** (1 mark).
- D - **paper waste** (1 mark).
- **Marking note:** Accept 'biodegradable waste' for A and D only if the material is also named.*

(b) [4 marks]

- A - **biodegradable** (1 mark).
- B - **non-biodegradable** (1 mark).
- C - **non-biodegradable** (1 mark).
- D - **biodegradable** (1 mark).
- **Marking note:** Mark each entry separately.*

(c) [2 marks]

- **Biodegradable waste is broken down by micro-organisms into simple substances** (1 mark).
- **Non-biodegradable waste is not broken down and stays in the environment for a very long time** (1 mark).
- **Marking note:** 'One rots and the other does not' is accepted for the first mark only if decay is clearly meant.*

(d) [2 marks]

- **Sort it into a separate container and send it for recycling (accept: bury it safely in a pit)** (1 mark).
- **Because glass and metal do not decay; broken glass causes injury and tins collect water in which mosquitoes breed** (1 mark).
- **Marking note:** 'Burn it' earns no mark.*

Question 2 [Total: 8 marks]**(a)** [3 marks]

- Any THREE of: **flies spread typhoid, cholera or diarrhoea**; **mosquitoes breeding in the tins spread malaria**; **rats and cockroaches spread disease**; **the smell and smoke affect breathing**; **sharp objects cause injury** (1 mark each).
- **Marking note:** A disease named without its link to the heap still earns the mark if the link is obvious.*

(b) [3 marks]

- **The organic waste is decaying, being broken down by bacteria and fungi** (1 mark).
- **Gases such as ammonia and hydrogen sulphide are released, and these smell bad** (1 mark).
- **Mosquitoes lay eggs in the still water and breed there, spreading malaria** (1 mark).
- **Marking note: Accept 'rotting' for decay.**

(c) [2 marks]

- Any TWO of: **burning plastics and rubber releases poisonous smoke and gases that pollute the air and harm the lungs**; **the fire may spread and destroy property**; **useful material that could be composted or recycled is destroyed** (1 mark each).
- **Marking note: 'It is against the law' earns no mark without a scientific reason.**

Question 3 [Total: 10 marks]**(a)** [3 marks]

- **Reduce: produce less waste by using less and avoiding unnecessary items** (1 mark).
- **Reuse: use an article again as it is, instead of throwing it away** (1 mark).
- **Recycle: process used material into new articles** (1 mark).
- **Marking note: An example alone, with no explanation, earns no mark.**

(b) [3 marks]

- **Reduce: one sound example, such as a refillable bottle instead of daily sachets** (1 mark).
- **Reuse: one sound example, such as tins used for seedlings** (1 mark).
- **Recycle: one sound example, such as collecting plastic bottles for a recycling centre** (1 mark).
- **Marking note: An example placed under the wrong term earns no mark.**

(c) [2 marks]

- **Separating waste into different containers according to its kind** (1 mark).
- **So that each kind can be composted, reused or recycled instead of being dumped** (1 mark).
- **Marking note: Accept 'sorting' for segregation.**

(d) [2 marks]

- **Three sensible labels, such as organic waste, plastics and paper** (1 mark).
- **A correct item named for each bin** (1 mark).
- **Marking note: Labels such as 'rubbish 1, rubbish 2' earn no mark.**

Question 4**(a)** [4 marks]

- P - **coarse dry material: maize stalks, twigs or sticks** (1 mark).
- Q - **green plant material and kitchen waste: grass, weeds, peels** (1 mark).
- R - **topsoil or animal droppings (manure)** (1 mark).
- S - **a cover of palm fronds, dry grass or an old sack** (1 mark).
- **Marking note: Plastics or tins named in any layer earn no mark.**

(b) [3 marks]

- (i) **Water keeps the heap moist** (1 mark); **the micro-organisms need moisture to live and act on the waste** (1 mark).
- (ii) **The micro-organisms respire as they feed, and respiration gives out heat** (1 mark).
- **Marking note: 'The Sun heats it' earns no mark for (ii).**

(c) [3 marks]

- **Turning lets air (oxygen) into the heap** (1 mark).
- **The decomposing micro-organisms are aerobic and need oxygen; turning also mixes the layers for even decay** (1 mark).
- **Without turning, decay becomes slow and anaerobic and the heap smells foul** (1 mark).
- **Marking note: Accept 'the heap would rot slowly and smell' for the third mark.**

(d) [2 marks]

- Any TWO of: **the waste is turned into manure that feeds the garden and improves the soil**; **money is saved on fertiliser**; **less waste goes to the dump, so fewer flies, less smell and less pollution** (1 mark each).
- **Marking note: 'It is cheaper' alone, with no explanation, earns no mark.**

Question 5 [Total: 10 marks]**(a)** [3 marks]

- Any THREE of: **research good waste management practices**; **obtain permission from the authorities and inform the traders**; **fix a date and share the members into groups with clear duties**; **collect the tools and protective items needed**; **arrange for the waste collected to be carted away** (1 mark each).
- **Marking note: 'Tell everybody' is too vague for a mark.**

(b) [3 marks]

- **Hand gloves - to protect the hands from cuts and germs** (1 mark).
- **Nose mask - to keep out dust, smell and germs** (1 mark).
- **Boots or closed shoes - to protect the feet from sharp objects and dirty**

water** (1 mark).

- *Marking note: No mark for an item named without a reason.*

(c) [4 marks]

- **Sort (segregate) the waste as it is collected** (1 mark).

- **Compost the organic waste** (1 mark).

- **Send the plastics, paper and metal for recycling or reuse** (1 mark).

- **Hand over only the remaining unusable waste for proper disposal, with no open burning** (1 mark).

- *Marking note: A plan that ends with everything being dumped or burnt scores no more than 1 mark.*

Question 6 [Total: 8 marks]

(a) [4 marks]

- Any FOUR of: **title, date and place**; **aim of the project**; **materials and method used**; **findings or observations**; **conclusion**; **recommendations** (1 mark each, the heading with a correct phrase of content).

- *Marking note: A heading with nothing said about its content earns half of nothing - award the mark only when both are given.*

(b) [2 marks]

- Any TWO of: **provide enough covered, labelled bins and empty them regularly**; **fix a weekly clean-up day**; **assign someone to sweep daily**; **educate the traders with posters and announcements**; **enforce a by-law against dumping in gutters** (1 mark each).

- *Marking note: 'They should keep it clean' is too vague for a mark.*

(c) [2 marks]

- **It spreads the knowledge to the whole school, so that others change their habits** (1 mark).

- **It makes the findings and recommendations known so that they can be acted upon (accept: it encourages others to take part)** (1 mark).

- *Marking note: 'Because the teacher asked for it' earns no mark.*

CURRICULUM ASSESSMENT BLUEPRINT

Strand: 5: Humans and the Environment | **Sub-strand:** 1: Waste Management | **Class:** Basic 7 (JHS 1)

Content Standard: **B7/JHS1.5.1.1** - Exhibit knowledge and skill of

scientific basis for management practices of types of waste in the en



Water Cycle

ANSWER KEY

SECTION A: OBJECTIVE TEST

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	D	3.1 Natural Sources of Water	Remember
2	A	3.1 Natural Sources of Water	Remember
3	C	3.1 Natural Sources of Water	Remember
4	D	3.1 Natural Sources of Water	Understand
5	A	3.1 Natural Sources of Water	Understand
6	B	3.2 Evaporation and Transpiration	Remember
7	D	3.2 Evaporation and Transpiration	Remember
8	C	3.2 Evaporation and Transpiration	Understand
9	B	3.2 Evaporation and Transpiration	Remember
10	C	3.2 Evaporation and Transpiration	Understand
11	A	3.2 Evaporation and Transpiration	Analyse
12	A	3.2 Evaporation and Transpiration	Understand
13	D	3.2 Evaporation and Transpiration	Apply
14	D	3.3 Condensation and Cloud Formation	Remember
15	C	3.3 Condensation and Cloud Formation	Understand
16	B	3.3 Condensation and Cloud Formation	Remember
17	D	3.3 Condensation and Cloud Formation	Understand
18	A	3.3 Condensation and Cloud Formation	Understand
19	B	3.3 Condensation and Cloud Formation	Understand
20	A	3.3 Condensation and Cloud Formation	Understand
21	B	3.4 Precipitation, Run-off and Collection	Remember
22	A	3.4 Precipitation, Run-off and Collection	Understand
23	B	3.4 Precipitation, Run-off and Collection	Remember
24	C	3.4 Precipitation, Run-off and Collection	Remember
25	C	3.4 Precipitation, Run-off and Collection	Remember
26	D	3.4 Precipitation, Run-off and Collection	Apply
27	A	3.4 Precipitation, Run-off and Collection	Analyse
28	B	3.5 The Water Cycle as a Repeated Pattern	Understand
29	D	3.5 The Water Cycle as a Repeated Pattern	Understand
30	B	3.5 The Water Cycle as a Repeated Pattern	Understand
31	A	3.5 The Water Cycle as a Repeated Pattern	Understand
32	C	3.5 The Water Cycle as a Repeated Pattern	Analyse
33	C	3.5 The Water Cycle as a Repeated Pattern	Analyse
34	B	3.6 The Importance of the Water Cycle	Understand
35	C	3.6 The Importance of the Water Cycle	Understand
36	A	3.6 The Importance of the Water Cycle	Understand

Q	Ans	Curriculum Area	Cognitive Level
37	D	3.6 The Importance of the Water Cycle	Understand
38	B	3.6 The Importance of the Water Cycle	Analyse
39	D	3.6 The Importance of the Water Cycle	Analyse
40	C	3.6 The Importance of the Water Cycle	Evaluate

Answer Distribution:

A: 10 (25%) | B: 10 (25%) | C: 10 (25%) | D: 10 (25%)

Cognitive Level Distribution:

Analyse: 6 (15%) | Apply: 2 (5%) | Evaluate: 1 (2%) | Remember: 12 (30%) | Understand: 19 (47%)

SECTION B: THEORY — MARKING SCHEME**Question 1 [Total: 10 marks]****(a) [5 marks]**

- A - **evaporation** (1 mark).
- B - **transpiration** (1 mark).
- C - **condensation** (1 mark).
- D - **precipitation** (accept: rainfall) (1 mark).
- E - **run-off** (accept: surface run-off, collection of water in rivers returning to the sea) (1 mark).
 - *Marking note: 'Evaporation' written for B earns no mark; water lost from the leaves of a plant is transpiration.*

(b) [3 marks]

- (i) **The sun** (1 mark).
- (ii) **It is left behind in the sea; only pure water vapour rises** (1 mark).
- (iii) **The air is cooler high up, so the vapour condenses into tiny water drops** (1 mark).
 - *Marking note: 'It is hot' is not accepted for (i); the sun must be named.*

(c) [2 marks]

- **The water returns to the sea, the place where it started, so the same water is used again** (1 mark).
- **The same stages follow one another in the same order without ending** (1 mark).
 - *Marking note: A candidate who writes only 'because it happens every year' earns no mark.*

Question 2 [Total: 12 marks]**(a) [4 marks]**

- All four stages written in the correct order (2 marks; 1 mark if one stage is out of place).
- Arrows drawn between the stages to show the direction of the change (1 mark).
- An arrow from **collection back to evaporation**, closing the cycle (1 mark).
 - *Marking note: A chart that stops at precipitation loses the last mark, however neat it is.*

(b) [5 marks]

- Any FIVE of: **a source of energy - energy released during condensation warms the environment**; **a carrier of nutrients**; **it improves (raises) the water table**; **it regulates the weather pattern**; **it provides clean, fresh water**; **it supplies water for crops, animals and people** (1 mark each).
- **Marking note:** 'It gives us water' may be credited once only, as the provision of clean water.*

(c) [3 marks]

- Any THREE of: **rain falling on the town, the farms or the hills**; **the river or the wells being refilled (the water table rising)**; **people using the water for drinking, washing, farming or watering animals**; **water returning to the air by evaporation from the river and transpiration from trees** (1 mark each).
- **Marking note:** Marks are for what the diagram would show, not for drawing it; a candidate who draws it correctly still earns the marks.*

Question 3 [Total: 10 marks]**(a)** [3 marks]

- P - **steam / water vapour rising from the water** (1 mark).
- Q - **drops of water (condensed water) on the underside of the plate** (1 mark).
- R - **the water collected in the cup** (1 mark).
- **Marking note:** 'Smoke' written for P earns no mark.*

(b) [3 marks]

- P - **evaporation** (1 mark).
- Q - **condensation** (1 mark).
- R - **precipitation and collection** (accept either) (1 mark).
- **Marking note:** Accept 'rainfall' for R.*

(c) [2 marks]

- Burner - **the sun** (1 mark).
- Cold metal plate - **the cool air high up in the atmosphere (where clouds form)** (1 mark).
- **Marking note:** 'The sky' alone is not accepted for the plate; the idea of cool air or cloud level is needed.*

(d) [2 marks]

- Any TWO of: **in nature the sun heats the water slowly and the water does not boil**; **the model leaves out transpiration by plants**; **the model leaves out run-off and water soaking into the ground**; **in nature the wind carries the clouds far away before the rain falls**; **the model is very much smaller and lasts only a few minutes** (1 mark each).
- **Marking note:** 'It is not real water' earns no mark.*

Question**(a)** [3 marks]

- **Trees release water vapour into the air by transpiration** (1 mark).
- **With fewer trees, less water vapour enters the air** (1 mark).
- **Less vapour means fewer clouds form by condensation, so less rain falls** (1 mark).
- **Marking note:** The word 'transpiration' is not compulsory if the idea of water vapour from the leaves is clear.*

(b) [3 marks]

- (i) **The level below the ground at which the soil or rock is filled (saturated) with water** (1 mark).
- (ii) **Rain water soaks into the soil (infiltration)** (1 mark) **and collects below the surface, raising the stored water so that wells do not dry up** (1 mark).
- **Marking note:** 'Water under the ground' alone earns the mark in (i) only if the idea of a level is given.*

(c) [4 marks]

- Any TWO actions (1 mark each) with a matching reason (1 mark each): **plant trees and grasses** - **roots slow the water and help it soak into the soil**; **leave open ground, dig trenches or infiltration pits** - **the water is held until it soaks in**; **harvest rain water from roofs into tanks** - **less water runs over the ground and water is stored for the dry season**; **clear and maintain the gutters** - **run-off is carried away without flooding the streets**.
- **Marking note:** An action without a reason earns 1 mark only.*

Question 5 [Total: 10 marks]**(a)** [3 marks]

- Dish 1 - **48 cm³** (1 mark).
- Dish 2 - **22 cm³** (1 mark).
- Dish 3 - **6 cm³** (1 mark).
- **Marking note:** Accept answers without the unit, but a wrong unit (for example g) loses the mark for that dish.*

(b) [3 marks]

- **The dish in the sun received more heat energy from the sun** (1 mark).
- **More heat makes water evaporate faster** (1 mark).
- **The cupboard was cool and the still, closed air soon held as much vapour as it could, so evaporation was slow** (1 mark).
- **Marking note:** Accept 'there was no wind in the cupboard to carry the vapour away' for the third mark.*

(c) [2 marks]

- Any TWO of: ****the dishes were identical, so the exposed surface was the same****; ****the same volume of water was used in each dish****; ****all the dishes were left for the same time (six hours)**** (1 mark each).
- ***Marking note: 'They used water' is not a fair-test measure and earns no mark.***

(d) [2 marks]

- ****The pupil is not correct**** (1 mark).
- ****The water has changed into water vapour and entered the air; it is not destroyed and will return as rain**** (1 mark).
- ***Marking note: No mark for the explanation if the candidate agrees with the pupil.***

Question 6 [Total: 8 marks]**(a)** [3 marks]

- ****Evaporation leaves salts and dirt behind, so the rain that falls is fairly pure**** (1 mark).
- ****Falling rain picks up dust and smoke from the air**** (1 mark).
- ****The roof and gutters add dust, leaves, bird droppings and germs to the water**** (1 mark).
- ***Marking note: A candidate who writes only 'the tank may be dirty' earns 1 mark.***

(b) [2 marks]

- Any TWO of: ****boil the water before drinking****; ****filter it through a clean cloth or filter****; ****treat it with an approved water-treatment agent (for example chlorine tablets)****; ****keep the tank covered and clean****; ****clean the roof and gutters and let the first dirty water run to waste**** (1 mark each).
- ***Marking note: Accept any safe, practical method used in Ghanaian homes.***

(c) [3 marks]

- Any THREE of: ****the water in the tank is still part of the cycle; when it is used it evaporates or soaks into the ground****; ****the total amount of water on the earth does not change, it is only used again and again****; ****the tank holds only a very small part of the rain that falls****; ****collecting roof water reduces run-off and flooding**** (1 mark each).
- ***Marking note: A candidate who merely writes 'the neighbour is wrong' without a reason earns no mark.***

CURRICULUM ASSESSMENT BLUEPRINT

****Strand:**** 2: Cycles | ****Sub-strand:**** 1: Earth Science | ****Class:**** Basic 7 (JHS 1)

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